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An Algorithmic Approach to Managing Supply Chain Data Security: The Differentially Private Newsvendor

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Data is now unanimously considered a key firm asset for enabling better operational decisions. However, data-driven decisions can inadvertently expose private data, leaving firms vulnerable to unforeseen danger. How to manage data security risks by protecting data from being inferred from observable decisions thus becomes an important question. In this paper, we focus on data security in supply chains due to their data-intensive nature. Specifically, we examine a data-driven contextual newsvendor problem. To quantify and ensure data security, we adopt the notion of differential privacy, a mathematically rigorous measure of data security that limits an attacker's inference accuracy. Employing convolution smoothing and noise injection, we propose several differentially private algorithms that provably guarantee both data security and asymptotic optimality with (near) optimal rates. In the non-asymptotic regime, we further identify three drivers of the cost of data security; namely, dataset size, context, and number of products. This finding suggests that gathering more data, collecting detailed context, and pooling data from multiple products can lower data security cost. Lastly, we examine the impact of a newsvendor's private algorithms on supply chain partners. We discover additional distortion to the demand signaling process and lower profit share for an upstream supplier.

Key words: data security, differential privacy, data-driven contextual newsvendor, private learning

1. Introduction

Digitalization has spawned a dramatic growth in the amount of data available to retail firms. Meanwhile, recent developments in artificial intelligence have unleashed the power of data in reducing stock levels, identifying demand trends, and making better operational decisions. It is therefore not surprising that it is proprietary data that builds informational advantages for a firm and holds the key to its long-term success. However, when a firm's proprietary data falls into the wrong hands, the consequences could be dire and far-reaching. Such a data breach incident could cause immediate tangible financial losses from disruption to business operations, and worse, may breed intangible threats to the firm's competitive edge that may endanger business in the future.

When a firm's proprietary data (such as sales data) is acquired by horizontal competitors, the latter can exploit the data to better understand the firm's sales campaigns, inventory plans and market trends. Subsequently, they will be able to devise plans to gain market advantage over the firm. An example is the

case of Newbury Comics in the music industry (Anand and Goyal 2009). Sales data of Newbury Comics was collected by a third-party data broker SoundScan, who then passed on the data to Newbury's competitors such as Handleman Co. Unbeknown to Newbury, Handleman was able to use this set of precise sales data to strategically devise inventory plans, encroaching on Newbury's market share from the blind side.

Meanwhile, when proprietary data is made available to vertical partners, the firm risks being eliminated from the market through disintermediation. Consider a vendor-retailer channel wherein a joint program is launched, such as Vendor Managed Inventory (VMI) and Collaborative Planning, Forecasting, and Replenishment (CPFR). Through such programs, the retailer's proprietary data, including stock levels, storage capacity and demand, are shared with the upstream vendor with hopes of benefiting both parties in terms of better end consumer service levels and reduced inventory-related costs. However, these shared data also provide the vendor with knowledge of the market, previously private and exclusive to the retailer. The vendor can then sell products to consumers directly and potentially disintermediate the retailer. Dukes et al. (2017) observe the same phenomenon and point out that the upstream vendor generally "benefits (from vertical information sharing) at the loss of downstream retailer".

Aside from risks from horizontal competitors and vertical vendors, data security-related crises may also arise from a trusted third-party partner because shared data may be unwittingly leaked due to the partner's cyber vulnerabilities. According to IBM's estimation (IBM 2023), nearly one-fifth of data incidents in 2022 happened at business partners. The leaked data could be acquired by unintended recipients, thereby engendering undetectable risks. Because supply chains are inherently interconnected networks, a cyber attack at a vulnerable node may quickly cascade into a ripple effect across several other firms. This explains recent reports that the supply chain industry recorded the highest financial loss once a data incident occurs (RiskLens 2023), ahead of the healthcare and the finance industries.

In light of the serious data security issues above, it is imperative for any retail firm to properly manage supply chain data security with a unified, versatile and security-by-design approach. While improving information technology (IT) infrastructure security can prevent unauthorized direct access to data by attackers, it does not protect data from being indirectly inferred from operational decisions. For example, inventory ordering decisions in supply chains are usually obtained from a machine learning model trained on demand datasets (Ban and Rudin 2019, Qi et al. 2023). If the model or algorithm is unaware of data security, ordering decisions may be exploited by anyone observing them, potentially revealing demand data on which the model is trained. The following example demonstrates such a data breach risk.

Example 1 (Inadvertent Data Breach through Data-Driven Decisions). *A retailer is using a data-driven newsvendor model to help her replenishment decision making, i.e. construct the empirical Cumulative Distribution Function (CDF)* $\widehat{F}_n(a) = \sum_{i=1}^{n} \mathbb{1}\{y_i \le a\}$ *with* n *historical demand samples; then choose* $\widehat{q}_{n+1} = \widehat{F}_n^{-1}(r) := \inf_{y\in\mathbb{R}}\{y : \widehat{F}_n(y) \ge r\}$ *as the order quantity for day* $n+1$*, where the critical ratio* $r = 0.5$*. Suppose*

Table 1 Data Breach in a Data-Driven Newsvendor

	Day	1	2	3	4	5	6	7
Newsvendor	Demand y_i	0	0	1	1	1	0	-
	Order Quantity $\widehat{q}_i$			-			1	0
Attacker	Inference			-			-	$y_6 = 0$

demand is a binary random variable taking value 0 or 1. Further suppose she has run the business for five days and has observed five historical demands from Day 1 to Day 5 as shown in Table 1.

Following the decision-making process, her order quantity for Day 6 is $\widehat{q}_6 = 1$. If the realized demand for Day 6 is 0, then her order quantity for Day 7 will be $\widehat{q}_7 = 0$. Observing the change in order quantity from $\widehat{q}_6 = 1$ to $\widehat{q}_7 = 0$, anyone can accurately infer that the realized demand for Day 6 is $y_6 = 0$.

The above example demonstrates the inadvertent data breach risk in a security-unaware data-driven algorithm. Intuitively, data is reversely inferred from the nuance in decisions that originates from the nuance in datasets. In the worst scenario, the entire demand dataset could be reverse-engineered.

To address this type of inadvertent breach, we focus our study on securing data by designing security-aware data-driven algorithms that can limit an attacker's inference accuracy. Such an approach warrants data security from the very beginning, and thus is a security-by-design way to defend against data breach risks. While important, theoretical explorations on securing *data* are relatively sporadic. Existing game-theoretical models (Li 2002, Kong et al. 2013, Kong et al. 2017) and empirical studies (Özer et al. 2011, Croson et al. 2014, Cui et al. 2015) have identified *information* leakage issues in supply chains and proposed various managerial insights to mitigate them. However, operational guidelines and implementable algorithms for securing *data* remain lacking. We are not the only ones who notice the absence of attention to data security and the lack of implementable algorithms. Recent calls for research on digital confidentiality in supply chains (Massimino et al. 2018, Kumar and Mallipeddi 2022, Kumar and Tang 2022) have expressed similar concerns. However, there are some potential obstacles that may hinder theory and algorithm development such as how to measure data security and how to quantify an algorithm's data protection capability.

To facilitate better security quantification, we borrow the idea of *Differential Privacy* (DP), a mathematically rigorous measure of data security proposed in the field of computational privacy (Dwork and Roth 2014). A differentially private algorithm guarantees data security by ensuring that the output from a dataset containing some individual data point is almost as likely to come from a dataset without that data point. Hence, an attacker can hardly tell whether that data point is included in the dataset and cannot make any accurate inference about that data point. The concept of differential privacy is formally introduced below.

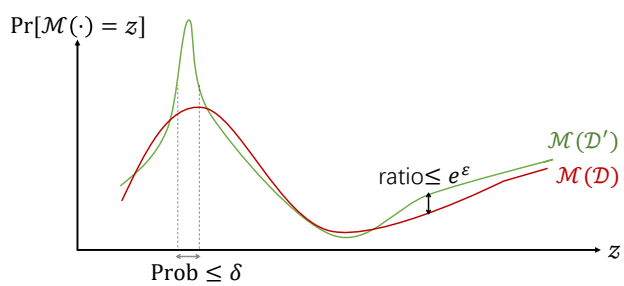
Definition 1 ((ε, δ)-Differential Privacy, Dwork and Roth 2014). *Let $\mathcal{X} \times \mathcal{Y}$ be the data point space, and $\mathcal{Z}$ be the output space. For any given $\varepsilon \in (0, \infty)$, $\delta \in [0, 1]$, a (randomized) algorithm $\mathcal{M}\colon \mathcal{X}^n \times \mathcal{Y}^n \to \mathcal{Z}$*

is said to be (ε, δ)-Differentially Private if for any pair of neighboring datasets $\mathcal{D} \sim \mathcal{D}' \in \mathcal{X}^n \times \mathcal{Y}^n$ that differ in only one data point, the distributions of outputs $\mathcal{M}(\mathcal{D})$ and $\mathcal{M}(\mathcal{D}')$ satisfy the following inequality,

$$\Pr\left[\mathcal{M}(\mathcal{D}) = z\right] \leq e^{\varepsilon} \cdot \Pr\left[\mathcal{M}(\mathcal{D}') = z\right] + \delta, \quad \forall z \in \mathcal{Z}.$$

From a high-level perspective, the definition describes probabilistic indistinguishability between outputs obtained from applying the same algorithm $\mathcal{M}$ to two *neighboring datasets* $\mathcal{D} \sim \mathcal{D}'$ (which differ in one data point only). The extent of indistinguishability is controlled by two DP parameters $(\varepsilon, \delta) \in (0, \infty) \times [0, 1]$, where ε is known as *security loss budget*. To better understand the definition, let us temporarily assume $\delta = 0$ and let ε take a small positive value. According to the definition, the probability ratio $\Pr\left[\mathcal{M}(\mathcal{D}) = z\right] / \Pr\left[\mathcal{M}(\mathcal{D}') = z\right]$ is uniformly upper bounded by $e^{\varepsilon} \approx 1 + \varepsilon$, which implies that $\mathcal{M}(\mathcal{D})$ and $\mathcal{M}(\mathcal{D}')$ are close to each other. Next, we let δ take a positive but tiny value. The positive value of δ allows violations of security loss budget where $\Pr\left[\mathcal{M}(\mathcal{D}) = z\right] / \Pr\left[\mathcal{M}(\mathcal{D}') = z\right] > e^{\varepsilon}$, but requires the probability of such occurrence to be smaller than δ.Therefore, an attacker attempting to distinguish $\mathcal{D}$ from $\mathcal{D}'$ will come to the following situation. With probability at least $1 - \delta$, the attacker will favor $\mathcal{D}$ over $\mathcal{D}'$ since $\mathcal{D}$ has a higher likelihood of being the input of observed output $\mathcal{M}(\cdot)$, but such preference of likelihood will not exceed e^{ε}. This explanation is visualized in Figure 1. For effective protection, both parameters (ε, δ)

Figure 1 **Graphical Illustration on (ε, δ)-Differential Privacy.**



should take sufficiently small values, making distinguishing $\mathcal{D}$ from $\mathcal{D}'$ nearly impossible. Moreover, the protection should work well for all pairs of neighboring datasets. With these guarantees, we can then say that the dataset at hand is adequately protected.

Differential privacy has been widely adopted as the data privacy and security standard by government agencies and leading technology companies, such as the US Census Bureau (Long 2022), the Swiss Federal Statistical Office (Raphaël et al. 2023), Apple (Greenberg 2016), Google (Erlingsson et al. 2014, Florian and Kairouz 2023), LinkedIn (Rogers et al. 2021), Meta (Herdağdelen et al. 2020) and Microsoft (Ding et al. 2017). However, the application of differential privacy in supply chain management remains unstudied, to our best knowledge. To show how DP works in our context, we revisit Example 1 below.

Example 2 (Differential Privacy Prevents Data Breach). *Consider the same retailer using a data-driven newsvendor model as in Example 1. Suppose she adopts an $(\varepsilon, 0)$-DP decision making algorithm: with n historical demand samples, the order quantity for Day $n+1$ is decided by the following randomized policy,* $\Pr\left[\widehat{q}_{n+1} = i\right] \propto \exp\left(\varepsilon \cdot \mathbb{1}\left\{\widehat{F}_n^{-1}(r) = i\right\}\right),\ \forall i = 0, 1$. *Observing five historical demand points $\boldsymbol{y} = (0, 0, 1, 1, 1)$, her order quantity decision for Day 6 becomes*

$$\widehat{q}_6(\boldsymbol{y}) = \begin{cases} 0, & \textit{with prob. } \frac{1}{1+\exp(\varepsilon)}; \\ 1, & \textit{with prob. } \frac{\exp(\varepsilon)}{1+\exp(\varepsilon)}. \end{cases}$$

Now, consider an arbitrary neighboring dataset $\boldsymbol{y}' = (0, 0, 1, 1, 0)$. If she had $\boldsymbol{y}'$, her decision would be

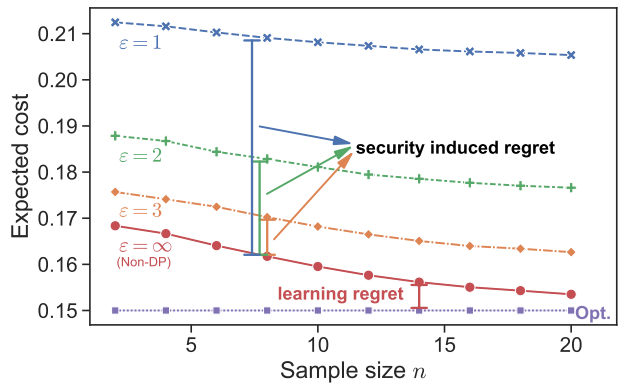
$$\widehat{q}_6(\boldsymbol{y}') = \begin{cases} 0, & \textit{with prob. } \frac{\exp(\varepsilon)}{1+\exp(\varepsilon)}; \\ 1, & \textit{with prob. } \frac{1}{1+\exp(\varepsilon)}. \end{cases}$$

It is easy to check that the likelihood ratio of taking decision $i \in \{0, 1\}$ is $\Pr\left[\widehat{q}_6(\boldsymbol{y}) = i\right] / \Pr\left[\widehat{q}_6(\boldsymbol{y}') = i\right] = (1/\exp(\varepsilon))^{1-2i}$, *which is close to 1 if ε takes a small value. The ratio implies that $\widehat{q}_6$ almost equally likely comes from $\boldsymbol{y}$ and $\boldsymbol{y}'$, regardless of the actual value it takes (which could be either 0 or 1). Therefore, an attacker cannot accurately distinguish $\boldsymbol{y}$ from $\boldsymbol{y}'$, even if the attacker has the retailer's full operational knowledge. The above argument can be applied to any arbitrary neighboring dataset; hence, the dataset at hand $\boldsymbol{y}$ is sufficiently protected.*

As shown in Example 2, differentially private algorithms can limit reverse inference accuracy at any specified level, and thus guarantee data security. Moreover, these guarantees hold for any dataset and preclude any attacker from accurate inferences, making DP a robust and versatile approach to managing data security risks. Despite the security advantages, Example 2 also reveals a decision sub-optimality downside, i.e. with a non-trivial probability, the decision maker must place a sub-optimal order quantity. This deviation from better decisions is necessary to meet DP requirements and obfuscate attackers, but in turn, it compromises optimality. As shown in Figure 2, imposing a DP requirement incurs higher expected costs, and the more stringent the requirement (i.e. smaller ε), the higher the costs will be. This observation naturally sparks our interest in the security-optimality trade-off. We henceforth examine this trade-off in an even more general setting of the data driven *contextual* newsvendor, and proceed to characterize the learning regret as well as the security-induced regret, among other measures.

Our study does not mean to challenge the widely accepted wisdom that data and information sharing in supply chains is beneficial. We clearly realize the advantages of data sharing, ranging from facilitating effective decisions and aligning incentives to enhancing coordination and improving traditional operations performance (Cachon and Fisher 2000, Lee and Whang 2000). Nonetheless, our aim is to complement the existing approach with more attention to the critical issue of data security and its associated costs.

Figure 2 **Expected Costs (by simulation) under Various Security Loss Budgets.**



Note. Smaller ε implies stricter security requirement. The bottom purple line, marked as "Opt.", is the optimal cost. Experiment settings follow Example 2. Critical ratio $r = 0.5$, $\Pr[y = 1] = 0.3$, $\delta = 0$.

1.1. Research Questions

Motivated by the severity of data security issues, the inadvertent data breach example in the Newsvendor problem, recent call for research on data security in supply chain management, and the wide adoption of Differential Privacy (DP) in practice, we are interested in keeping data secure under the framework of DP for a data-driven contextual newsvendor, as well as in understanding the security-optimality trade-off. We refer to the loss in optimality due to keeping data secure as the cost of data security. Of particular interest are the following three research questions:

1. Can we design algorithms for a data-driven contextual newsvendor that guarantee data security while minimizing the cost of data security? How much is the cost of data security?
2. What are the drivers of the security cost? How do we mitigate the cost without sacrificing security?
3. What is the impact on supply chain partners if a newsvendor adopts differentially private algorithms?

1.2. Our Contributions

We summarize our contributions in three short answers to the preceding three research questions.

1. In Section 3, we propose several differentially private algorithms (Algorithms 1, 2, and 3) that provably guarantee both data security and asymptotic optimality with (near) optimal rates. The cost of data security is found to be at least $\widetilde{\Omega}\left(\frac{\sqrt{d}}{n\varepsilon}\right)$ (Theorem 7), but no more than $\mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$ (Theorems 2, 4, 6), where n is sample size and d is feature dimension. This result holds for linear demand $y = \langle \boldsymbol{\theta}^\star, \boldsymbol{x}\rangle + \epsilon^\star(\boldsymbol{x})$ with a possibly endogenous error term $\epsilon^\star(\boldsymbol{x})$ in any form.
2. In Section 4, we attribute the cost of security to three aspects of "being small"; namely, small-sized dataset, context-free dataset (Theorem 8) and datasets with a limited number of correlated products (Proposition 1). We refer to this phenomenon as *the curse of being small*, that data security is financially costly for firms possessing only "small" datasets.

3. In Section 5, we show that the adoption of DP algorithms by a newsvendor has negative external effects on upstream suppliers. The demand signaling process will be further distorted, entailing an ambiguous bias (Proposition 2) and a possibly higher variability of order quantity (Proposition 3), which then drive down the supplier's profit share (Proposition 4).

These contributions and insights can help firms understand the costs associated with data security and its impact on supply chain partners. They also lay the foundation for further exploration into more sophisticated DP supply chain management problems.

Our analytical results, albeit not exhaustively summarized in Table 2, are governed by two DP parameters $(\varepsilon, \delta) \in (0, \infty) \times (0, 1]$ that reflect the decision maker's attitude towards data security. Relaxing

Table 2 Main Theoretical Results.

Regret Bounds on Data-driven (DD) Algorithms				Increase in Variability of Order Quantity	
Non-DP Partially DD	Non-DP Fully DD	DP Fully DD		DP Fully DD	
UB†	UB	LB	UB	LB	UB
$\mathcal{O}\left(\frac{1}{\sqrt{n}}\right) + w$	$\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$	$\widetilde{\Omega}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d}}{n\varepsilon}\right)$	$\mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon}\right)$	$\Omega\left(\frac{\ln(1/\delta)}{n^2\varepsilon^2} - \frac{\lambda+h}{n}\right)$	$\mathcal{O}\left(\frac{\ln(1/\delta)}{n^2\varepsilon^2}\right)$
($w > 0$, Theorem 8)	(Remark 2)	(Theorem 7)	(Theorem 2, 4, 6)	(Proposition 3)	(Proposition 3)

Note. "UB" and "LB" are upper and lower bound, respectively. When dataset $\mathcal{D} := \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$ consists of both features and demands, "Partially DD" means only demands are used to train models, while "Fully DD" means both features and demands are used. $n \geq 2$ is sample size and $d \geq 1$ is feature dimension. $(\varepsilon, \delta) \in (0, \infty) \times (0, 1]$ are DP parameters; $\lambda \geq 0$ and $h \geq 0$ are algorithm's parameters that potentially depend on $n, d, \varepsilon, \delta$. The symbol $\widetilde{\Omega}(\cdot)$ hides some logarithmic factors in n or d. † For this upper bound only, we assume features are Gaussian distributed.

either ε to infinity or δ to 1 will recover results in a non-DP setting. In other words, we characterize the security-optimality trade-off between two extremes: no security concerns $(\varepsilon, \delta) \to (\infty, 1)$, and full security $(\varepsilon, \delta) \to (0, 0)$. All proofs are deferred to the Appendix.

1.3. Notation

Symbols (ε, δ) are exclusively reserved for DP, and σ is exclusively reserved for the standard deviation of an injected noise. n represents the sample size while d denotes the feature dimension.

We use $\mathcal{X} \times \mathcal{Y} \subseteq \mathbb{R}^d \times \mathbb{R}$ to denote the data domain, and $\mathcal{B}(B)$ to denote a Euclidean ball with radius B. The r-th newsvendor cost function is $c(u) = ru^+ + (1-r)(-u)^+, \forall u \in \mathbb{R}$ where $u^+ := \max\{0, u\}$, and we use $\ell(\boldsymbol{\theta}; \boldsymbol{x}, y) := c(y - \langle \boldsymbol{\theta}, \boldsymbol{x} \rangle)$ to denote a real-valued loss function induced from $c(\cdot)$. The empirical (i.e., in-sample) risk of any $\boldsymbol{\theta} \in \mathbb{R}^d$ w.r.t. ℓ and dataset $\mathcal{D} := \{\boldsymbol{x}_i, y_i\}_{i=1}^n$ is defined as $\widehat{\mathcal{L}}(\boldsymbol{\theta}; \mathcal{D}) := \frac{1}{n}\sum_{i=1}^n \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)$. The generalization (i.e., out-of-sample) risk of $\boldsymbol{\theta}$ w.r.t. ℓ and distribution $\mathbb{P}$ is defined as $\mathcal{L}(\boldsymbol{\theta}; \mathbb{P}) := \mathbb{E}_{(\boldsymbol{x}, y) \sim \mathbb{P}}[\ell(\boldsymbol{\theta}; \boldsymbol{x}, y)]$. We use $\widehat{\mathcal{L}}(\boldsymbol{\theta})$ and $\mathcal{L}(\boldsymbol{\theta})$ as shorthands for the empirical and generalization risks when the dependence is clear from context. The hat symbol always represents an estimator.

For two sequences of nonnegative values $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$, we use $a_n \lesssim b_n$ to indicate that there exists a constant $C > 0$ independent of n such that $a_n \leq C b_n$ for all n, and $a_n \asymp b_n$ means $a_n \lesssim b_n$ and $b_n \lesssim a_n$ hold at the same time. Big-$\mathcal{O}$ and big-Ω notations are used when the preceding relationships hold only

for sufficiently large n. Stochastic big-$\mathcal{O}_p$ and small-o_p notations are used to describe the limiting behavior of a sequence of random variables $\{X_n\}_n$. Specifically, $X_n = \mathcal{O}_p(a_n)$ means that X_n/a_n is bounded in probability; and $X_n = o_p(a_n)$ means X_n/a_n converges to 0 in probability.

We use "private dataset" to refer to a dataset that needs to be protected. Similarly, "private" algorithm (or estimator) refers to "differentially private" algorithm (or estimator) that can ensure data security in terms of differential privacy. Occasionally, we may use "privacy" and "security" interchangeably since one can roughly regard a firm's proprietary data as its private data.

2. Literature Review and Preliminaries

2.1. Literature Review on Data-Driven Newsvendor Problem

The stream of studies on data-driven newsvendor problems began with operational statistics (Liyanage and Shanthikumar 2005, Chu et al. 2008, Ramamurthy et al. 2012) when only historical demand in the form of samples can be used to make replenishment decisions. The operational statistics framework integrates estimation and optimization, and maps historical samples to decisions directly, but under parametric distribution assumptions. To relax the parametric demand assumptions, Levi et al. (2007) established distribution-free probabilistic bounds on the generalization risk of a decision obtained from solving an Empirical Risk Minimization (ERM) problem. Levi et al. (2015), Cheung and Simchi-Levi (2019), and Lin et al. (2022) further tightened the probabilistic bounds and considered more general settings, such as capacitated models. The analytical results in these works are derived from general bounds in statistical learning literature. To develop specialized bounds, Besbes and Mouchtaki (2023) characterized an ERM-based algorithm's exact worst-case performance over all distributions for arbitrary data sizes.

When contextual information is involved, Ban and Rudin (2019) provided finite sample bounds on the performance of several policies by considering a linear data generation process. Qi et al. (2022) and Zhang et al. (2023) developed closed-form solutions for a distributionally robust problem under Wasserstein ambiguity sets. Besbes et al. (2023a,b) extended their previous work to the contextual case, where they found that the worst-case performance is non-monotone in sample size, and highlighted the importance of high-quality data. Some other works endeavor to combine the contextual newsvendor problem with neural networks (Huber et al. 2019) or deep learning (Oroojlooyjadid et al. 2020, Qi et al. 2023, Han et al. 2023), which resulted in significant performance improvements in practice. Inspired by the above works, we further take data security into consideration.

2.2. Literature Review on Differentially Private Algorithm Design and Applications in Operations Management

The conventional approach to designing a DP algorithm is to artificially create randomness at any step of the "Data-Model-Decision" pipeline. For instance, Warner's Randomized Response introduces randomness by collecting data in a randomized manner where a coin-flipping procedure is involved to hide personal sensitive information, guaranteeing $(\ln 3, 0)$-DP. But noisy data collected is of lower quality and

thus results in poor decisions, as discussed in Chen et al. (2022b). At the other end of the pipeline, injecting Gaussian noise into the final decision is the most common approach. However, this approach could result in inconsistent decisions if the algorithm is not well-designed and the noise magnitude is not well-calibrated. Instead, we design algorithms that inject noise in the Model step. Our algorithms are built upon Objective Perturbation (Kifer et al. 2012) and Gradient Perturbation (Bassily et al. 2014), two common approaches in Computational Privacy literature. However, both approaches cannot be applied to our problem directly because of a technical issue that the newsvendor loss function is nonsmooth. To address the technical issue, we design tailored algorithms in Section 3 and provide detailed discussions therein. We refer interested readers to Dwork and Roth (2014) for a comprehensive discussion on DP algorithm design, and to Selvi et al. (2023) for a distributionally robust optimization perspective.

In recent years, DP operations management has attracted much attention from the management community (Ashley and Rachel 2020, Simson and Claire 2022). Most existing works considered personalized pricing or recommendation problems for online platforms, and focused on users' privacy and data security. For instance, Chen et al. (2022b) considered a dynamic personalized pricing problem with parametric demand learning, while Chen et al. (2022a) extended the previous study to a nonparametric setting. Lei et al. (2023) examined a personalized pricing problem in an offline setting, and further explored a private assortment problem. Fu et al. (2022) studied a personalized recommendation problem, and identified a coarse-grained threshold policy. Another noteworthy work in the literature is Fallah et al. (2023), which considered personal data acquisition from privacy-sensitive users. While DP has been used to protect users' privacy and data, we find no such work on protecting firms' proprietary data in business management research. Hence, our work aims to contribute to this area.

2.3. Preliminaries on DP Algorithm Design

Definition 2 (L-Lipschitz Continuity). *Let $L > 0$. A function $f : \mathbb{R}^d \to \mathbb{R}$ is L-Lipschitz with respect to $\|\cdot\|_2$ over a set $\mathcal{C}$ if for any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{C}$, we have*

$$|f(\boldsymbol{\theta}_1) - f(\boldsymbol{\theta}_2)| \le L \cdot \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 .$$

Definition 3 (β-Smoothness). *Let $\beta > 0$. A differentiable convex function $f : \mathbb{R}^d \to \mathbb{R}$ is β-smooth with respect to $\|\cdot\|_2$ over a set $\mathcal{C}$ if for any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{C}$, we have*

$$f(\boldsymbol{\theta}_1) \le f(\boldsymbol{\theta}_2) + \langle \nabla \ell(\boldsymbol{\theta}_2), \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2 \rangle + \frac{\beta}{2} \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2^2 .$$

Lemma 1 (Gaussian Mechanism). *(Dwork and Roth 2014, Theorem A.1) Let $\mathcal{M} : \mathcal{X}^n \times \mathcal{Y}^n \to \mathcal{Z} \subseteq \mathbb{R}^m$ be a vector-valued function. Define the Global Sensitivity of $\mathcal{M}$ as $\mathrm{GS}_{\mathcal{M}} := \sup_{\mathcal{D} \sim \mathcal{D}'} \|\mathcal{M}(\mathcal{D}) - \mathcal{M}(\mathcal{D}')\|_2$. Then, for any $\varepsilon \in (0, 1)$ and $\delta > 0$, the Gaussian Mechanism that outputs*

$$\mathcal{M}(\mathcal{D}) + \mathcal{N}(0, \sigma^2 \boldsymbol{I}_{m \times m})$$

is an (ε, δ)-DP algorithm if $\sigma \ge \sqrt{2 \ln(1.25/\delta)} \cdot \mathrm{GS}_{\mathcal{M}} / \varepsilon$.

The Gaussian Mechamism Lemma states that it suffices to inject a Gaussian noise with standard deviation $\sigma \geq C \cdot \mathrm{GS}_{\mathcal{M}} \cdot \frac{\sqrt{\ln(1/\delta)}}{\varepsilon}$ for some constant C into the output of an algorithm $\mathcal{M}$ in order to make the algorithm (ε, δ)-DP.

Lemma 2 (Advanced Composition). *(Dwork and Roth 2014, Theorem 3.20) For all $\varepsilon, \delta, \delta' \geq 0$, the class of (ε, δ)-DP mechanisms satisfies $(\varepsilon\sqrt{2k\ln(1/\delta')} + k\varepsilon(e^{\varepsilon} - 1), k\delta + \delta')$-DP under k-fold adaptive composition.*

The Advanced Composition Lemma states that the overall security loss of a sequentially adaptive composition algorithm of k (ε, δ)-DP algorithms that output all intermediate states in the composition process is roughly $(\sqrt{k}\varepsilon, k\delta)$-DP. Adaptive composition means that any subsequent algorithm can take as their inputs intermediate states from upstream algorithms.

Lemma 3 (Post-processing). *(Dwork and Roth 2014, Proposition 2.1) If $\mathcal{M} : \mathcal{X}^n \times \mathcal{Y}^n \to \mathcal{Z}$ is an (ε, δ)-DP algorithm, and $\mathcal{F} : \mathcal{Z} \to \mathcal{U}$ is a (randomized) function of interest. Then the sequentially composited algorithm $\mathcal{F} \circ \mathcal{M}$ is (ε, δ)-DP.*

The Post-processing Lemma states that DP will be preserved under further manipulation as long as the manipulation does not depend on the private dataset. Some typical post-processing operators are projection, shifting and scaling.

3. Differentially Private Data-Driven Contextual Newsvendor Problem

Consider a single-period single-item data-driven contextual newsvendor problem where the decision maker needs to decide the order quantity $q(\boldsymbol{x}) \in \mathbb{R}$ based on observed feature $\boldsymbol{x} \in \mathcal{X} := \mathcal{B}(B_x) \subseteq \mathbb{R}^d$ before a random demand, $y(\boldsymbol{x}) \in \mathcal{Y} \subseteq \mathbb{R}$ almost surely, is realized. Let c_o, c_u represent the unit overage cost and the unit underage cost, and without loss of generality, we assume $c_u + c_o = 1$. Denote critical ratio as $r := c_u/(c_u + c_o)$. Then, after a demand $y(\boldsymbol{x})$ is observed, the cost $c : \mathbb{R} \to \mathbb{R}_+$ at the underage level $u := y(\boldsymbol{x}) - q(\boldsymbol{x})$ is $c(u) = r \cdot u^+ + (1 - r) \cdot (-u)^+$, where $u^+ := \max\{u, 0\}$.

The true demand is assumed to follow a linear model, taking the form of

$$y = \langle \boldsymbol{\theta}^\star, \boldsymbol{x} \rangle + \epsilon^\star(\boldsymbol{x}), \tag{1}$$

where $\boldsymbol{\theta}^\star \in \mathbb{R}^d$ with $d \geq 1$ is the unknown groundtruth coefficient vector and $\epsilon^\star(\boldsymbol{x})$ is a possibly endogenous error term. We emphasize that we allow the error term $\epsilon^\star(\boldsymbol{x})$ to be contingent on $\boldsymbol{x}$, delivering a more flexible model. An oracle with full knowledge of distribution $\mathbb{P}$ of $(\boldsymbol{x}, y)$ can obtain $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta}) := \mathbb{E}_{(\boldsymbol{x}, y) \sim \mathbb{P}}[\ell(\boldsymbol{\theta}; \boldsymbol{x}, y)]$ where loss function $\ell(\boldsymbol{\theta}; \boldsymbol{x}, y) := c(y - \langle \boldsymbol{\theta}, \boldsymbol{x} \rangle)$. Thus, the linear demand model (1) can be reformulated as

$$y = \langle \boldsymbol{\theta}^*, \boldsymbol{x} \rangle + \epsilon(\boldsymbol{x}), \tag{2}$$

where the new error term $\epsilon(\boldsymbol{x})$ has a Cumulative Distribution Function (CDF) $F_{\epsilon|\boldsymbol{x}}$ with $F^{-1}_{\epsilon|\boldsymbol{x}}(r) = 0$, $\forall \boldsymbol{x} \in \mathcal{X}$. This reformulated linear model can be understood as being obtained by shifting the CDF of $\epsilon^\star(\boldsymbol{x})$

leftwards or rightwards so that its r-th quantile is 0, and the shifted part is absorbed into $\boldsymbol{\theta}^*$. Working on the reformulated linear model (2) is common in the literature (Chen et al. 2020, He et al. 2021). Moreover, the following lemma further motivates the adoption of (2) by identifying the closed-form of $\boldsymbol{\theta}^*$ in a special case where the error term is exogenous.

Lemma 4. *Suppose that demand follows the linear form (1), where $\epsilon^\star(\boldsymbol{x}) \equiv \epsilon^\star$ is independent of $\boldsymbol{x}$, and has a CDF $F_{\epsilon^\star}$ being continuous at r-th quantile. We further assume $x_1 \equiv 1$ to incorporate the intercept term. Let $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta})$. Then $\boldsymbol{\theta}^* = \boldsymbol{\theta}^\star + [F_{\epsilon^\star}^{-1}(r), 0, \ldots, 0]^\top$, where $F_{\epsilon^\star}^{-1}(r) := \inf\{y \in \mathbb{R} : F_{\epsilon^\star}(y) \geq r\}$.*

The assumptions of exogeneity and of existence of an intercept term are for the special case only, and are not universally assumed. Throughout the rest of our paper, we will rely on (2), unless otherwise stated.

Facing the DP data-driven contextual newsvendor problem, a decision maker, with access to a dataset $\mathcal{D} := \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n \in \mathcal{X}^n \times \mathcal{Y}^n$ consisting of n data points that are i.i.d. drawn from an unknown distribution $\mathbb{P}$, aims at designing a DP ordering rule $\mathcal{A}$ for any upcoming feature $\boldsymbol{x}$. Inspired by Ban and Rudin (2019) and DP Post-Processing Lemma 3, we can restrict the ordering rule to a linear decision rule taking the form of $\widehat{q}^{\mathcal{A}}(\boldsymbol{x}) := \langle \widehat{\boldsymbol{\theta}}^\pi, \boldsymbol{x}\rangle$ with a DP estimator $\widehat{\boldsymbol{\theta}}^{\mathcal{A}}$ trained on $\mathcal{D}$. The Differentially Private Newsvendor Problem (DP-NV) is formally modelled as

$$\text{(DP-NV)} \qquad \inf_{\mathcal{A}\in\mathcal{F}_{\varepsilon,\delta}} \left\{ \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \left\{ \mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n,\mathcal{A}}[\mathcal{L}(\widehat{\boldsymbol{\theta}}^{\mathcal{A}})] - \min_{\boldsymbol{\theta}\in\mathcal{C}} \mathcal{L}(\boldsymbol{\theta}) \right\} \right\}, \tag{3}$$

where $\mathcal{F}_{\varepsilon,\delta}$ is the family of all (ε,δ)-DP algorithms that map from $\mathcal{X}^n \times \mathcal{Y}^n \to \mathcal{C}$, and $\mathcal{P}(\mathcal{X}\times\mathcal{Y})$ is the class of all probability measures over $\mathcal{X}\times\mathcal{Y}$. The innermost difference in (3), referred to as *regret* and abbreviated as $\mathcal{R}(\mathcal{A};\mathbb{P}) := \mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n,\mathcal{A}}[\mathcal{L}(\widehat{\boldsymbol{\theta}}^{\mathcal{A}})] - \min_{\boldsymbol{\theta}\in\mathcal{C}} \mathcal{L}(\boldsymbol{\theta})$, is the excess generalization risk of algorithm $\mathcal{A}$ over the oracle's minimal risk. We assume $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta})$ is finite and belongs to a Euclidean ball $\mathcal{C} := \mathcal{B}(B_\theta) \subseteq \mathbb{R}^d$. Therefore, the regret of $\mathcal{A}$ under distribution $\mathbb{P}$ can be simplified as $\mathcal{R}(\mathcal{A};\mathbb{P}) := \mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n,\mathcal{A}}[\mathcal{L}(\widehat{\boldsymbol{\theta}}^{\mathcal{A}})] - \mathcal{L}(\boldsymbol{\theta}^*)$. Consequently, the goal of problem DP-NV (3) is to find a best DP algorithm $\mathcal{A}$ that minimizes the regret, uniformly for all distributions.

The DP-NV problem (3) is hard to solve directly since it involves a minimization problem over an infinite dimensional space $\mathcal{F}_{\varepsilon,\delta}$ of (ε,δ)-DP algorithms. It is helpful to ignore the DP constraint for a while, and seek a non-private $\widehat{\boldsymbol{\theta}}$ first. To obtain a non-private $\widehat{\boldsymbol{\theta}}$, Ban and Rudin (2019) proposed to solve an Empirical Risk Minimization (ERM) problem

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \widehat{\mathcal{L}}(\boldsymbol{\theta}) := \frac{1}{n}\sum_{i=1}^n \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i).$$

They have shown that the excess cost $\left|\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}) - \mathcal{L}(\boldsymbol{\theta}^*)\right|$ between objective values of the ERM and the oracle's minimization problem will converge to 0 as sample size n tends to infinity under some regularity conditions. We can borrow their idea as a building block and additionally require $\widehat{\boldsymbol{\theta}}^{\mathcal{A}}$ under our algorithm $\mathcal{A}$ to be differentially private.

The ERM approach and analytical results in Ban and Rudin (2019) are impressive. But when (ε, δ)-DP constraint is imposed, there are some new challenges. First, the regret of interest here in (3), by definition, is different from the regret considered in their work; thus, the non-private algorithm's theoretical guarantee on regrets cannot be directly obtained from their results, let along that of a private algorithm. Second, the newsvendor cost function is nonsmooth. The nonsmoothness will lead to an unstable estimator, prevent gradient-based optimization methods from being efficient, and invalidate uniform stability, which is a crucial property for theoretical analysis. Third and most importantly, the newsvendor function has a knot at the origin at which the curvature is infinite, implying an abrupt change in derivatives. The abrupt change is an informative signal and can expose much information (recall Example 1). Therefore, we need tailored algorithms for the DP-NV problem. Our tailored algorithms, in a nutshell, will follow a "smoothing-then-privatize" principle, where we first approximate the newsvendor cost function with a smooth surrogate function, and then apply private algorithms to the surrogate function.

3.1. Convolution Smoothing

Convolution smoothing is a tool to approximate a nonsmooth function with a smoothed surrogate generated from convolution between the original nonsmooth function and a properly chosen kernel function. To warrant high-quality approximations, we impose some conditions on kernel functions.

Condition 1 (Kernel Functions). *Let $K : \mathbb{R} \to \mathbb{R}_+$ be a nonnegative function satisfying*

- ***Integrate to 1****: $\int K = 1$;*
- ***Symmetry****: $K(u) = K(-u)$ for all $u \in \mathbb{R}$;*
- ***Monotonicity****: $K(u) \le K(v),\ \forall\, |u| \ge |v| \in \mathbb{R}$;*
- ***Finite first two moments****: $\kappa_i := \int_{-\infty}^{\infty} |u|^i K(u)\, du < \infty$ for $i = 1, 2$, and $\overline{K} := \sup_u K(u) < \infty$.*

Any functions that satisfy Condition 1 are suitable for our problem. Commonly used kernels are Gaussian, Logistic, Uniform, and Epanechnikov summarized in Table 3 under column "$K(u)$". Given a bandwidth $h > 0$, and a kernel function K, we follow notations in the literature and denote, for any $u \in \mathbb{R}$, $K_h(u) := K(u/h)/h$, $\mathcal{K}(u) := \int_{-\infty}^{u} K(v)\, dv$, and $\mathcal{K}_h(u) := \mathcal{K}(u/h)$. Then, against the newsvendor cost functon $c(u) = ru^+ + (1-r)(-u)^+ = |u|/2 + (r - 1/2)u$, the smoothed surrogate function is

$$c_h(u) := (c * K_h)(u) = \int_{-\infty}^{\infty} c(v) \cdot K_h(u - v)\, dv \tag{4}$$

$$= \frac{h}{2} \int_{-\infty}^{\infty} \left| \frac{u}{h} + v \right| \cdot K(v)\, dv + \left(r - \frac{1}{2} \right) u. \tag{5}$$

Intuitively, the value $c_h(u)$ is a weighted average over u's neighbors, and the weights are given by the adjusted kernel function $K_h(\cdot)$ so that a closer neighbor has a higher weight. With previously mentioned kernel functions, the integral term in (5) has closed forms as shown under the third column in Table 3. Mathematically, it can be shown that $c_h(\cdot)$ is convex, epi-graphically converges to $c(\cdot)$ as $h \to 0$, and

Table 3 Kernel Functions.

Kernels	$K(u)$	$\int_{-\infty}^{\infty} \lvert u/h+v\rvert K(v)\,dv$	$\mathcal{K}_h(u)$
Gaussian	$\frac{1}{\sqrt{2\pi}}e^{\frac{-u^2}{2}}$	$\sqrt{\frac{2}{\pi}}e^{-\frac{(u/h)^2}{2}}+u/h(1-2\Phi(-u/h))$	$\Phi(u/h)$
Logistic	$\frac{e^{-u}}{(1+e^{-u})^2}$	$u/h+2\ln\left(1+e^{-u/h}\right)$	$(1+e^{-u/h})^{-1}$
Uniform	$\frac{\mathbb{1}\{\lvert u\rvert\le 1\}}{2}$	$\begin{cases}\frac{u^2}{2h^2}+\frac{1}{2}, \text{ if } \lvert u/h\rvert\le 1\\ \lvert u/h\rvert, \text{ otherwise}\end{cases}$	$\min\{(u/h+1)/2,1\}\mathbb{1}\{u/h\ge -1\}$
Epanechnikov	$\frac{3}{4}(1-u^2)\mathbb{1}\{\lvert u\rvert\le 1\}$	$\begin{cases}-\frac{u^4}{8h^4}+\frac{3u^2}{4h^2}+\frac{3}{8}, \text{ if } \lvert u/h\rvert\le 1\\ \lvert u/h\rvert, \text{ otherwise}\end{cases}$	$\begin{cases}-\frac{u^3}{4h^3}+\frac{3u}{4h}+\frac{1}{2}, \text{ if } \lvert u/h\rvert\le 1\\ \mathbb{1}\{u/h\ge 1\}, \text{ otherwise}\end{cases}$

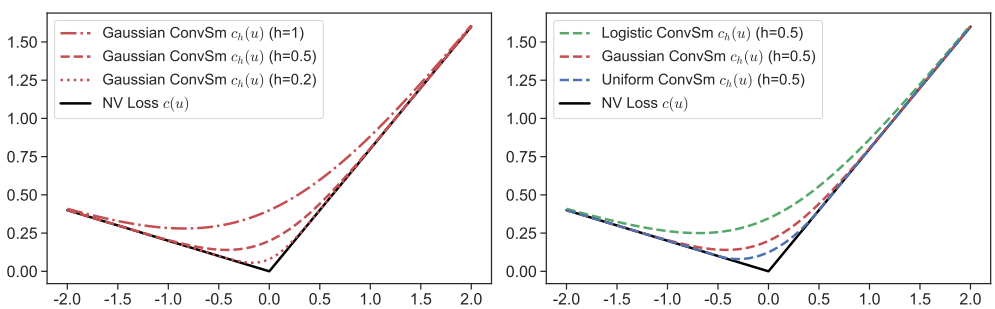
second-order differentiable (Fernandes et al. 2021):

$$c_h'(u)=\mathcal{K}_h(u)+r-1; \quad c_h''(u)=K_h(u)\ \ge 0.$$

These properties are graphically confirmed by Figure 3. Moreover, we observe from Figure 3a that a smaller

Figure 3 Smoothed Newsvendor Cost Function by Convolution Smoothing, $r=0.8$.

(a) Smoothed functions under various bandwidths. (b) Smoothed functions with various kernels.



bandwidth leads to a better approximation, and from Figure 3b that a kernel with lighter tails leads to a better approximation.

Similarly, the smoothed surrogate loss function $\ell_h(\boldsymbol{\theta};\boldsymbol{x},y):=c_h(y-\langle\boldsymbol{\theta},\boldsymbol{x}\rangle)$ possesses the same properties, and we provide a summary below for later reference.

Lemma 5 (Properties of ℓ_h). *Suppose kernel function K satisfies Condition 1. And define $\ell_h(\boldsymbol{\theta};\boldsymbol{x},y):=c_h(y-\langle\boldsymbol{\theta},\boldsymbol{x}\rangle)$, where c_h is a smoothed newsvendor cost function defined in (5). Suppose the feature space $\mathcal{X}$ is a subset of B_x-radius d-dimensional Euclidean ball, i.e. $\mathcal{X}\subseteq\mathcal{B}(B_x)\subseteq\mathbb{R}^d$. Let $L:=\bar{r}B_x$ where $\bar{r}:=\max\{r,1-r\}$, and let $\beta:=\overline{K}B_x^2/h$. Then surrogate function $\ell_h:\mathbb{R}^d\times\mathcal{X}\times\mathcal{Y}\to\mathbb{R}$ has following properties:*

1. *function ℓ_h is second-order differentiable in $\boldsymbol{\theta}$:*

$$\nabla\ell_h(\boldsymbol{\theta};\boldsymbol{x},y)=[\mathcal{K}_h(\langle\boldsymbol{\theta},x\rangle-y)-r]\cdot\boldsymbol{x}, \quad \nabla^2\ell_h(\boldsymbol{\theta};\boldsymbol{x},y)=K_h(y-\langle\boldsymbol{\theta},\boldsymbol{x}\rangle)\cdot\boldsymbol{x}\boldsymbol{x}^\top\succcurlyeq\mathbf{0}, \quad \forall\boldsymbol{\theta},\boldsymbol{x},y;$$

2. *function ℓ_h is L-Lipschitz continuous, and β-smooth in $\boldsymbol{\theta}$;*

3. *function ℓ_h is upper and lower bounded by affine functions of ℓ,*

$$\ell(\boldsymbol{\theta};\boldsymbol{x},y) \le \ell_h(\boldsymbol{\theta};\boldsymbol{x},y) \le \ell(\boldsymbol{\theta};\boldsymbol{x},y) + \frac{1}{2}h\kappa_1, \quad \forall \boldsymbol{\theta},\boldsymbol{x},y.$$

For notational brevity, we further use the subscript $_h$ to denote any functions and estimators involving convolution. Specifically, the empirical smoothed function is $\widehat{\mathcal{L}}_h(\boldsymbol{\theta}) := \frac{1}{n}\sum_{i=1}^n \ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)$, and population-level smoothed function is $\mathcal{L}_h(\boldsymbol{\theta}) := \mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}\left[\ell_h(\boldsymbol{\theta};\boldsymbol{x},y)\right]$. With smoothed surrogate loss functions, we are ready to design differentially private algorithms.

3.2. Objective Perturbation Policy

The first algorithm is Objective Perturbation (OP), which is formally given in Algorithm 1. The algorithm boils down to two steps: (i) solve a noisy regularized Empirical Risk Minimization (ERM) problem

$$\text{(Noisy Regularized ERM for OP)} \qquad \min_{\boldsymbol{\theta}\in\mathcal{C}} \widehat{\mathcal{L}}_h^{\mathsf{OP}}(\boldsymbol{\theta}) := \widehat{\mathcal{L}}_h(\boldsymbol{\theta}) + \lambda\left\|\boldsymbol{\theta}\right\|_2^2 + \frac{\langle \boldsymbol{b},\boldsymbol{\theta}\rangle}{n}, \tag{6}$$

whose objective function is perturbed by a Gaussian noise $\boldsymbol{b}\sim\mathcal{N}(\boldsymbol{0},\sigma^2\boldsymbol{I}_{d\times d})$ to obtain a private minimizer $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$; then (ii) output an order quantity $\langle\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}},\boldsymbol{x}_{n+1}\rangle$. The DP guarantee of $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$ can be obtained from a similar

Algorithm 1 Objective Perturbation (OP)

Parameters: DP parameters $\varepsilon>0,\delta>0$, kernel function K with bandwidth $h>0$, Lipschitz parameter $L=\bar{r}B_x$, smoothness parameter $\beta=\overline{K}B_x^2/h$, variance $\sigma^2=L^2\cdot(8\ln(2/\delta)+4\varepsilon)/\varepsilon^2$, feasible region $\mathcal{C}:=\mathcal{B}(B_\theta)$, projection operator $\mathsf{Proj}_{\mathcal{C}}(\boldsymbol{x}) := \arg\min_{\boldsymbol{y}\in\mathcal{C}}\left\|\boldsymbol{y}-\boldsymbol{x}\right\|_2$

Input: historical database $\mathcal{D}=\{(\boldsymbol{x}_i,y_i)\}_{i=1}^n\in\mathcal{X}^n\times\mathcal{Y}^n$; an upcoming feature vector $\boldsymbol{x}_{n+1}$

Output: order quantity

1: Set any $\lambda\ge\frac{2\beta}{n\varepsilon}$
2: Draw a noise vector $\boldsymbol{b}\sim\mathcal{N}(\boldsymbol{0},\sigma^2\mathbf{I}_{d\times d})$
3: Obtain private estimator $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\leftarrow\mathsf{Proj}_{\mathcal{C}}\left(\arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d}\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\boldsymbol{\theta}) := \widehat{\mathcal{L}}_h(\boldsymbol{\theta})+\lambda\left\|\boldsymbol{\theta}\right\|_2^2+\frac{\langle\boldsymbol{b},\boldsymbol{\theta}\rangle}{n}\right)$
4: Output order quantity $\langle\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}},\boldsymbol{x}_{n+1}\rangle$

analysis of Kifer et al. (2012, Theorem 2), and then Post-Processing Lemma 3 ensures the order quantity $\langle\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}},\boldsymbol{x}_{n+1}\rangle$ is also DP. We highlight here that it is the convolution smoothing that makes our analysis possible, without which the security is not guaranteed. Formally, the data security guarantee of Algorithm OP is stated below.

Theorem 1 (Data Security Guarantee of Algorithm OP). *Algorithm 1* OP *is (ε,δ)-DP.*

Next, we start to analyze the regret. We first notice that the regret $\mathcal{R}(\mathsf{OP};\mathbb{P}) := \mathbb{E}_{\mathcal{D},\mathsf{OP}}[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})]-\mathcal{L}(\boldsymbol{\theta}^*)$ can be decomposed into three parts:

$$\begin{aligned}
\mathcal{R}(\mathsf{OP};\mathbb{P}) &= \mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})\right]+\mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\mathcal{L}(\boldsymbol{\theta}^*)\right]\\
&\le \mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})\right]+\mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\mathcal{L}(\boldsymbol{\theta}^*)\right] && (7)\\
&= \mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})\right]+\mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right]+\left[\mathcal{L}_h(\boldsymbol{\theta}^*)-\mathcal{L}(\boldsymbol{\theta}^*)\right], && (8)
\end{aligned}$$

where the first line is by plugging in $\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}})$, the second line follows part 3 of Lemma 5 that $\ell_h \geq \ell$, and the third line is by first plugging in $\widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)$ and then using $\mathbb{E}_{\mathcal{D}}\left[\widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] = \mathcal{L}_h(\boldsymbol{\theta}^*)$. Therefore, it suffices to separately bound the three components in (8). Specifically, the first component can be controlled by uniform stability analysis, the second can be controlled by empirical risk analysis for regularized convex optimization problems, and the third can be upper bounded through analyzing the convolution smoothing approximation error. Combining the three bounds, we obtain the performance guarantee as follows.

Theorem 2 (Performance Guarantee of Algorithm OP). *Assume $\boldsymbol{\theta}^* \in \mathcal{C} := \mathcal{B}(B_\theta)$ and feature space $\mathcal{X} \subseteq \mathcal{B}(B_x)$. Let constant $C := B_x B_\theta \bar{r}$. Suppose $\delta \leq e^{-\varepsilon/2}$. Let K be a kernel function satisfying Condition 1. Further assume that the error term $\epsilon(\boldsymbol{x})$ in reformulated linear model (2) has a s-Lipschitz continuous CDF $F_{\epsilon|\boldsymbol{x}}$ for any $\boldsymbol{x} \in \mathcal{X}$. Then, for any distribution $\mathbb{P}$ satisfying stated assumptions, if we set $\lambda = \frac{1}{B_\theta}\sqrt{\frac{2L^2}{n} + \frac{d\sigma^2}{n^2}}$, $h = \frac{2\overline{K}B_x^2}{\lambda n \varepsilon}$, running Algorithm 1* OP *yields*

$$\sup_{\mathbb{P}} \mathcal{R}(\mathsf{OP};\mathbb{P}) \leq \frac{4\sqrt{2}C}{\sqrt{n}} + \frac{16C\sqrt{d\ln(1/\delta)}}{n\varepsilon} = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right),$$

when $\varepsilon \geq \frac{s\overline{K}\kappa_2}{\sqrt{\bar{r}^3}}\sqrt{\frac{B_\theta}{B_x}} \cdot \frac{1}{\sqrt[4]{n}}$, where $\mathcal{R}(\mathsf{OP};\mathbb{P}) = \mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n,\boldsymbol{b}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}})\right] - \mathcal{L}(\boldsymbol{\theta}^*)$ is the regret of Algorithm OP *under distribution $\mathbb{P}$.*

Theorem 2 shows the provable performance guarantee of Objective Perturbation. The regret upper bound is comprised of two terms; namely, learning regret $\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$ due to data sampling, and security-induced regret $\mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$. Both terms tend to 0 as n tends to infinity, implying that the proposed algorithm entails no (asymptotic) optimality compromise while guaranteeing data security at the specified level, even under the worst-case distribution. The λ chosen in Theorem 2 satisfies the requirement in Step 1 of the algorithm since we set $h = \frac{2\overline{K}B_x^2}{\lambda n\varepsilon}$, which implies $\lambda = \frac{2\overline{K}B_x^2}{hn\varepsilon} = \frac{2\beta}{n\varepsilon}$ by the definition of β in Lemma 5. To complement the theorem, we will show a minimax regret in Section 3.4 that nearly matches our regret upper bound (in terms of convergence rate) up to some logarithmic factors. This indicates that algorithm OP is statistically efficient and has near-optimal convergence rate. Other remarks are summarized below, and these remarks are applicable to DP algorithms in Section 3.3 as well.

Remark 1 (Degenerate to Sampling Case). *If we assume dimension $d = 1$ and $x_1 \equiv 1$, then our model covers data-driven sampling newsvendor as a special case.*

Remark 2 (Degenerate to Non-DP Case). *If we set noise $\boldsymbol{b} \equiv \mathbf{0}$ and do not use convolution smoothing, then our private ERM (6) degenerates to a standard non-private ERM with a $\|\cdot\|_2^2$-regularization term. If we further set $\lambda \asymp 1/\sqrt{n}$, then the regret of the non-private ERM will match the oracle complexity of learning $\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$.*

Remark 3 (High-dimensional Case). *Algorithm OP and the derived regret bound are valid in high-dimensional cases where $d \asymp n^k, \forall k \geq 0$. For example, a widely adopted demand forecast model in Bullwhip Effect literature is the Martingale Model of Forecast Evolution (MMFE) $D_n = D_0 + \sum_{t=0}^{n-1} \epsilon_{t,t+1}$ with normally*

distributed forecast update term $\epsilon_{t,t+1}$ that captures forecast update at time t for demand in time $t+1$. The forecast updates $\{\epsilon_{t,t+1}\}_{t=0}^{n-1}$ can be treated as features with increasing dimension $d=n$.

3.3. Gradient-based Policies

We may also leverage gradient-based optimization methods to develop private algorithms. The design philosophy of gradient-based policies is fundamentally distinct from that of OP. The gradient-based policies will introduce randomness during the solving process, while OP introduces randomness when establishing an optimization model. Specifically, we can perturb gradients with a zero-mean Gaussian noise in each iteration of gradient descent. The zero-mean Gaussian noise ensures (ε,δ)-DP while keeps gradients unbiased, thus guaranteeing good performance. Next, we will formally propose two gradient-based policies.

3.3.1. Gradient Perturbation

The Gradient Perturbation (GP) algorithm, formally described in Algorithm 2, boils down to solving an ERM problem

$$\text{(ERM for GP)} \qquad \min_{\boldsymbol{\theta}\in\mathcal{C}} \widehat{\mathcal{L}}_h(\boldsymbol{\theta})$$

with noisy gradient descent. Specifically, the algorithm follows standard Gradient Descent procedures, but

Algorithm 2 Gradient Perturbation (GP)

Parameters: DP parameters $\varepsilon>0,\delta>0$, kernel function K with bandwidth $h>0$, number of iterations T, noise variance $\sigma^2=\frac{8L^2T\ln(1/\delta)}{n^2\varepsilon^2}$, step size η, feasible region $\mathcal{C}:=\mathcal{B}(B_\theta)$, projection operator $\mathsf{Proj}_{\mathcal{C}}(\boldsymbol{x}):=\arg\min_{\boldsymbol{y}\in\mathcal{C}}\|\boldsymbol{y}-\boldsymbol{x}\|_2$

Input: dataset $\mathcal{D}=\{(\boldsymbol{x}_i,y_i)\}_{i=1}^n\in\mathcal{X}^n\times\mathcal{Y}^n$; an upcoming feature vector $\boldsymbol{x}_{n+1}$

Output: order quantity

1: Set initial point $\widehat{\boldsymbol{\theta}}_{h,1}^{\mathsf{GP}}=\mathbf{0}$
2: **for** $t=1$ to $T-1$ **do**
3: Draw a noise vector $\boldsymbol{w}_t\sim\mathcal{N}(\mathbf{0},\sigma^2\mathbf{I}_{d\times d})$
4: Update $\widehat{\boldsymbol{\theta}}_{h,t+1}^{\mathsf{GP}}\leftarrow\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}-\eta\cdot\left(\nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}};\mathcal{D})+\boldsymbol{w}_t\right)$
5: $\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}}\leftarrow\mathsf{Proj}_{\mathcal{C}}\left(\frac{1}{T}\sum_{t=1}^T\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}\right)$
6: Output order quantity $\langle\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}},\boldsymbol{x}_{n+1}\rangle$

gradients in each iteration are slightly perturbed by Gaussian noise, as shown in step 4 of Algorithm 2. The empirical gradient $\nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{OP}};\mathcal{D}):=\frac{1}{n}\sum_{i=1}^n\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{OP}};\boldsymbol{x}_i,y_i)$ exists everywhere because of convolution smoothing, and therefore this algorithm is well designed. It is noteworthy that, in step 5, we calculate the averaged noisy estimators over all iterations, instead of simply taking the value in the last iteration, in order to reduce fluctuations introduced by noises $\{\boldsymbol{w}_t\}_t$. Since gradient $\nabla\widehat{\mathcal{L}}_h$ has closed-forms (see Table 3 and part 1 in Lemma 5), our algorithm GP does not cause much additional computational burden over standard gradient descent.

Now, we start to analyze security guarantee. From Gaussian Mechanism (Lemma 1), we know that estimator $\widehat{\boldsymbol{\theta}}_{h,t}^{\text{GP}}$ in each iteration is $(\varepsilon_0, \delta_0)$-DP with proper ε_0 and δ_0. The Advanced Composition (Lemma 2) facilitates a T-fold adaptive composition, and thus estimator $\widehat{\boldsymbol{\theta}}_h^{\text{GP}}$ in step 5 is (ε, δ)-DP. Lastly, applying Post-processing Lemma 3 concludes the DP guarantee of the final order quantity.

Theorem 3 (Data Security Guarantee of Algorithm GP). *For any number of iterations $T \geq 1$, Algorithm 2 GP is (ε, δ)-DP.*

Following a similar regret decomposition in (7) and an analytical framework in learning theory that uniform stability plus shrinking ERM implies learnability, we can derive the performance guarantee of Algorithm GP below.

Theorem 4 (Performance Guarantee of Algorithm GP). *Assume $\boldsymbol{\theta}^* \in \mathcal{C} := \mathcal{B}(B_\theta)$ and feature space $\mathcal{X} \subseteq \mathcal{B}(B_x)$. Let constant $C := B_x B_\theta \bar{r}$. Let K be a given kernel function satisfying Condition 1. If we set step size $\eta = \frac{B_\theta}{\sqrt{2L^2n^3 + (d\sigma^2 + L^2 + \kappa_1 \overline{K} B_x^2)n^2 + 2L^2 n}}$, bandwidth $h = \eta \overline{K} B_x^2$, and number of iteration $T = n^2$, then the regret of* GP *is upper bounded as*

$$\sup_{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})} \mathcal{R}(\text{GP}; \mathbb{P}) \leq \frac{2\sqrt{2}C}{\sqrt{n}} + \frac{2\sqrt{2}C\sqrt{d \ln(1/\delta)}}{n\varepsilon} = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon}\right),$$

if $n \geq (1 + \kappa_1 \overline{K}/\bar{r}^2)/4$, where $\mathcal{R}(\text{GP}; \mathbb{P}) := \mathbb{E}_{\mathcal{D} \sim \mathbb{P}^n, \text{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\text{GP}})\right] - \mathcal{L}(\boldsymbol{\theta}^*)$ is the regret of Algorithm GP *under distribution $\mathbb{P}$.*

The regret upper bound in Theorem 4 also consists of two terms: learning regret $\mathcal{O}(\frac{1}{\sqrt{n}})$ and security-induced regret $\mathcal{O}(\frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon})$, demonstrating the asymptotic optimality of GP. Moreover, our convergence rate nearly matches the minimax regret in Theorem 7, indicating that GP is statistically efficient. The requirement $n \geq (1 + \kappa_1 \overline{K}/\bar{r}^2)/4$ is mild. If we choose kernels from Table 3, we have $\kappa_1 \leq 1$ and $\overline{K} \leq 1$. Therefore, the requirement becomes $n \geq 2$, which should hold naturally. One may notice that the regret upper bound of GP is similar to that of OP in terms of convergence rate. But which performs better in practice is unclear since derived results are merely regret upper bounds. We will compare their performances empirically in numerical study in Section 3.5.

3.3.2. Stochastic Gradient Perturbation

Stochastic Gradient Perturbation (SGP) is a variant of Gradient Perturbation. SGP uses only one data point to evaluate the gradient in each iteration, instead of using the entire dataset as Algorithm GP does. The Algorithm SGP is formally presented in Algorithm 3. The only difference appears in step 3, where we randomly draw a data point from $\mathcal{D}$ with replacement for gradient evaluation. Since the stochastic gradient is unbiased, it is not surprising to obtain similar theoretical guarantees.

Theorem 5 (Data Security Guarantee of Algorithm SGP). *For any number of iterations $T \geq 1$, Algorithm 3 SGP is (ε, δ)-DP.*

Algorithm 3 Stochastic Gradient Perturbation (SGP)

Parameters: DP parameters $\varepsilon > 0, \delta > 0$, kernel function K with bandwidth $h > 0$, number of itertaions T, noise variance $\sigma^2 = \frac{8TL^2 \ln(1/\delta)}{n^2\varepsilon^2}$, step size η, feasible region $\mathcal{C} := \mathcal{B}(B_\theta)$, projection operator $\mathsf{Proj}_{\mathcal{C}}(\boldsymbol{x}) := \arg\min_{\boldsymbol{y}\in\mathcal{C}} \|\boldsymbol{y} - \boldsymbol{x}\|_2$

Input: dataset $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n \in \mathcal{X}^n \times \mathcal{Y}^n$; an upcoming feature vector $\boldsymbol{x}_{n+1}$

Output: order quantity

Set initial point $\widehat{\boldsymbol{\theta}}_{h,1}^{\mathsf{SGP}} = \mathbf{0}$
for $t = 1$ to $T - 1$ **do**
    Uniformly sample a record $(\boldsymbol{x}_{(t)}, y_{(t)})$ from $\mathcal{D}$ with replacement
    Draw a noise vector $\boldsymbol{w}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_{d\times d})$
    Update $\widehat{\boldsymbol{\theta}}_{h,t+1}^{\mathsf{SGP}} \leftarrow \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}} - \eta \cdot \left(\nabla \ell_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}; \boldsymbol{x}_{(t)}, y_{(t)}) + \boldsymbol{w}_t\right)$
$\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}} \leftarrow \mathsf{Proj}_{\mathcal{C}}\left(\frac{1}{T}\sum_{t=1}^T \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}\right)$
Output order quantity $\langle \widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}, \boldsymbol{x}_{n+1}\rangle$

Theorem 6 (Performance Guarantee of Algorithm SGP). *Assume $\boldsymbol{\theta}^* \in \mathcal{C} := \mathcal{B}(B_\theta)$ and feature space $\mathcal{X} \subseteq \mathcal{B}(B_x)$. Let constant $C := B_x B_\theta \bar{r}$. Let K be a kernel function satisfying Condition 1. If we set step size $\eta = \frac{B_\theta}{\sqrt{2L^2n^3 + (d\sigma^2 + L^2 + \kappa_1 \overline{K} B_x^2/2)n^2 + 2L^2 n}}$, bandwidth $h = \frac{\eta \overline{K} B_x^2}{2}$, and number of iterations $T = n^2$, then the regret of* SGP *is upper bounded as*

$$\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathsf{SGP}; \mathbb{P}) \le \frac{2\sqrt{2}C}{\sqrt{n}} + \frac{4C\sqrt{d\ln(1/\delta)}}{n\varepsilon} = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right),$$

if $n \ge (1 + \kappa_1\overline{K}/\bar{r}^2)/4$, where $\mathcal{R}(\mathsf{SGP}; \mathbb{P}) := \mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n, \mathsf{SGP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}})\right] - \mathcal{L}(\boldsymbol{\theta}^*)$ is the regret of Algorithm SGP *under distribution $\mathbb{P}$.*

The proofs follow the same proof ideas as Theorems 3 and 4. Apparently, the regret upper bound is the same as that of Algorithm GP stated in Theorem 4, a (near) optimal convergence rate. Despite the same theoretical guarantees on data security and performance, SGP runs much faster than GP. We will empirically demonstrate the computational advantage later in the numerical studies.

3.4. Minimax Regret and Cost of Data Security

In this subsection, we will show that our proposed three DP algorithms have (near) optimal performance guarantees up to logarithmic factors $\ln d$ and $\ln(1/\delta)$. This will be done by characterizing the unavoidable regret entailed by DP requirements. Intuitively, because DP requires algorithms' output to be insensitive to one data point change, the algorithms should not effectively exploit every single data point to improve estimation, causing informational loss and thus an unavoidable regret. Formally, we quantify the unavoidable regret in the following Theorem.

Theorem 7 (Minimax Regret). *The minimax regret of DP-NV problem is given as*

$$\widetilde{\Omega}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d}}{n\varepsilon}\right) \le \inf_{\mathcal{A}\in\mathcal{F}_{\varepsilon,\delta}} \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathcal{A}; \mathbb{P}),$$

where $\widetilde{\Omega}$ hides the term $\ln d$ in the denominator of its second term; $\mathcal{F}_{\varepsilon,\delta}$ is the set of all (ε,δ)-DP algorithms from dataset space $\mathcal{X}^n \times \mathcal{Y}^n$ to estimator space $\mathcal{C}$; and $\mathcal{P}(\mathcal{X}\times\mathcal{Y})$ is the set of all distributions on $\mathcal{X}\times\mathcal{Y}$.

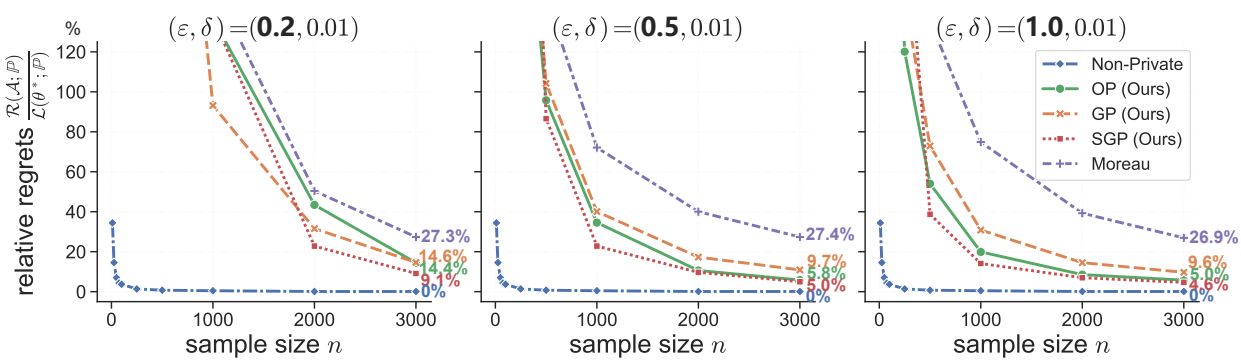
Essentially, the theorem states that among all (ε,δ)-DP algorithms, the best one that minimizes the worst-case regret still needs to suffer a regret that is lower bounded by $\widetilde{\Omega}\left(\frac{1}{\sqrt{n}}+\frac{\sqrt{d}}{n\varepsilon}\right)$. It is surprising that the regret upper bound $\mathcal{O}\left(\frac{1}{\sqrt{n}}+\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$ of our algorithms matches the lower bound up to some logarithmic factors, indicating that our proposed algorithms are (near) optimal. Comparing the minimax regret with regret upper bounds, we immediately conclude that the cost of data security that one must pay in the DP-NV problem is at least $\widetilde{\Omega}\left(\frac{\sqrt{d}}{n\varepsilon}\right)$, but no more than $\mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$. It remains an interesting open question to close the logarithmic gap.

3.5. Numerical Study

We run simulations on synthetic datasets to demonstrate our theoretical findings. We implement 5 algorithms, including a non-private model $\min_{\boldsymbol{\theta}\in\mathbb{R}^d}\widehat{\mathcal{L}}(\boldsymbol{\theta})$, our proposed private algorithms OP, GP, and SGP, and iterative Moreau Envelope (a widely used approach in DP literature for tackling nonsmooth loss functions, see Bassily et al. 2019). The iterative Moreau Envelope includes two steps: (i) first, approximate newsvendor cost function $c(\cdot)$ with an approximation function $c_\beta(\cdot)$ obtained from infimal convolution $c_\beta(u) := \inf_x\{c(x)+\frac{\beta}{2}(u-x)^2\}$ with a properly chosen $\beta>0$; (ii) then, run mini-batch SGP with well-calibrated noise. Specifically, we implement Algorithm 1 in Bassily et al. (2019). In our numerical studies, we take as the performance measure the excess generalization risk relative to the true minimal risk, i.e. relative regret $\frac{\mathcal{R}(\mathcal{A};\mathbb{P})}{\mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})} := \frac{\mathbb{E}_{\mathcal{D},\mathcal{A}}[\mathcal{L}(\widehat{\boldsymbol{\theta}}^{\mathcal{A}};\mathbb{P})]-\mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})}{\mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})}$, a suitable proxy of regrets.

In Figure 4, we conduct experiments under a benchmark setting as follows: Gaussian kernel is used for

Figure 4 **Relative Regrets of Various Policies.**



Note. (Benchmark setting: Gaussian kernel, endogenous error, $d=3$.) Demand follows $y=10x_1+5x_2-2x_3+\epsilon(\boldsymbol{x})$, where $x_1\equiv 1$, $x_2\sim\mathcal{N}(0,2^2)$, $x_3\sim\mathcal{N}(0,3^2)$, and $\epsilon(\boldsymbol{x})\sim\mathcal{N}(x_2-x_3+|x_2x_3|,3^2)$. We let $(x_2,x_3)\in\mathcal{B}(B_x)$ with $B_x=10$. We set $r=0.7$ and $B_\theta=2\|\boldsymbol{\theta}^*\|_2$. DP parameters (ε,δ) are set accordingly. And Gaussian kernel is used for OP, GP, and SGP. Simulations are repeatedly run 50 times with gradually increasing sample size n.

smoothing, an endogenous error term is set as the groundtruth, and the feature is low-dimensional with

three dimensions. It is clear from Figure 4 that our private algorithms outperform existing private methods, though they engender non-trivial but vanishing performance gaps compared to the non-private model. The performance gaps can be regarded as the practical cost of security. Moreover, as security requirement becomes less stringent (i.e., ε increases), the relative regret decreases, indicating a lower cost of security. We also report computational time in Table 4 for completeness. Remarkably, the running time of Algorithm

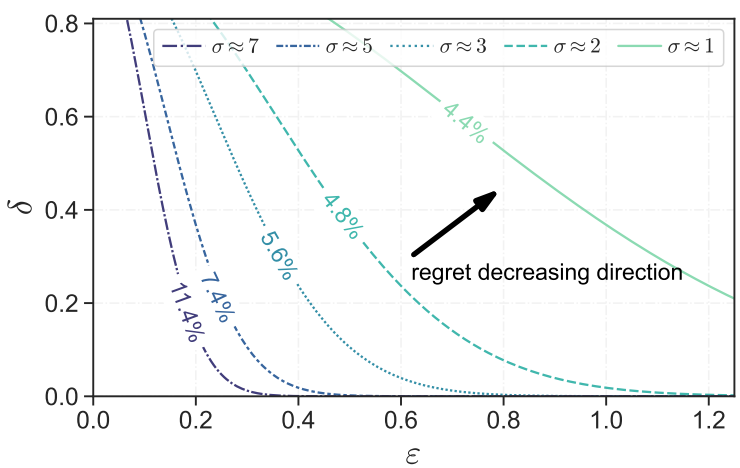
Table 4 **Running Time (mean±sd, in seconds), $d = 3$.**

Model	n=100	n=1000	n=3000
Non-Private	0.0±0.0	0.0±0.0	0.2±0.0
OP (Ours)	0.0±0.0	0.0±0.0	0.2±0.0
GP (Ours)	7.5±0.2	247.2±31.3	3656.0±387.5
SGP (Ours)	1.6±0.0	157.8±1.5	1391.3±85.6
Moreau	1.8±0.0	29.4±9.1	144.0±43.3

OP does not increase massively compared to its non-private counterpart under all sample sizes, while other methods' running time increases significantly with more data points being available.

Next, we examine the impact of DP parameters in a more direct way by running OP. As shown in Figure 5, regrets are smaller if (ε, δ) become larger, which matches our observation in Figure 4. Interestingly,

Figure 5 **Security-Optimality Tradeoff**



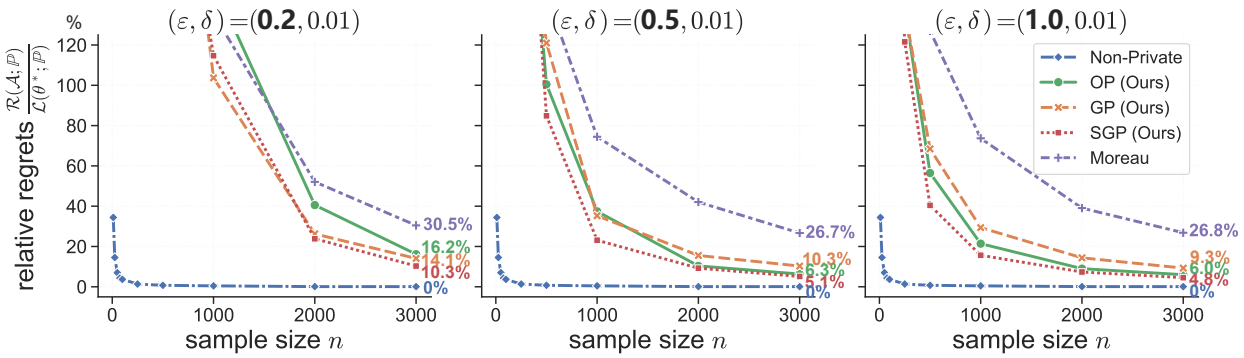
Note. (Benchmark setting: Gaussian kernel, endogenous error, $d = 3$.) Experiment settings are the same as in Figure 4. Each curve represents the set of all (ε, δ) combinations that the variances of injected noise, as specified in the legend, are the same. Values on curves are relative regrets by running OP with $n = 3000$. The σ in the legend actually means a pseudo standard deviation $\sigma := \sqrt{\ln(1/\delta)}/\varepsilon$.

when ε is extremely small, for example $\varepsilon = 0.2$, a mild increase in its value can significantly boost the performance. But as ε increases, the marginal benefit decreases. This observation suggests a rule of thumb

on the choice of ε that a moderate ε could be a good candidate, as it provides reasonable data security while not sacrificing much optimality.

Impact of Kernel. In Figure 6, we conduct experiments with Logistic kernel, i.e. we use Logistic kernel instead of Gaussian kernel for convolution smoothing. All trends in Figure 4 remain unchanged. However,

Figure 6 **Relative Regrets when Using Logistic Kernel.**

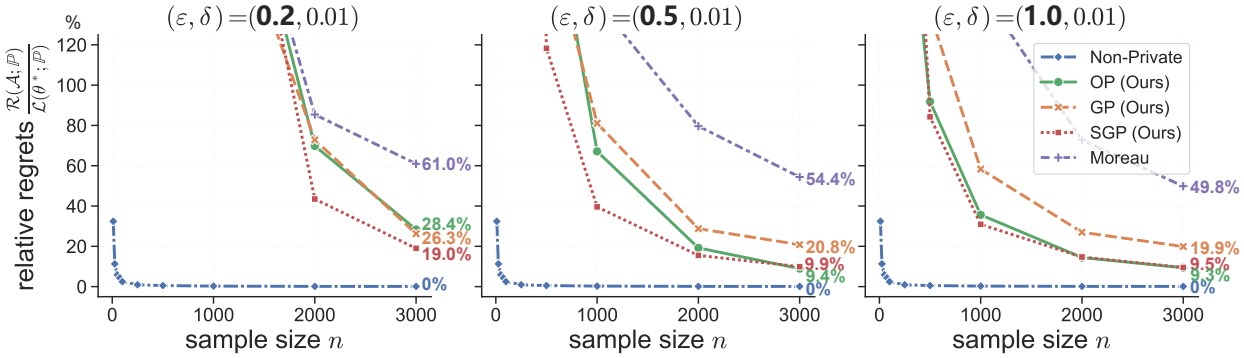


Note. (**Logistic kernel**, endogenous error, $d = 3$.) Experiment settings are the same as in Figure 4, except that Logistic kernel is used for smoothing.

the performances from using Logistic kernel are slightly worse than those from using Gaussian kernel by roughly 0.5%-1% when $n = 3000$. This is reasonable as Gaussian kernel has a lighter tail. Thus, the smoothed surrogate function can better approximate the original newsvendor cost function (recall Figure 3b).

Impact of Exogenous Error Term. In Figure 7, we conduct experiments with an exogenous error term. Compared to results under the benchmark setting shown in Figure 4, Figure 7 witnesses larger rela-

Figure 7 **Relative Regrets when Demand Error Term is Exogenous.**



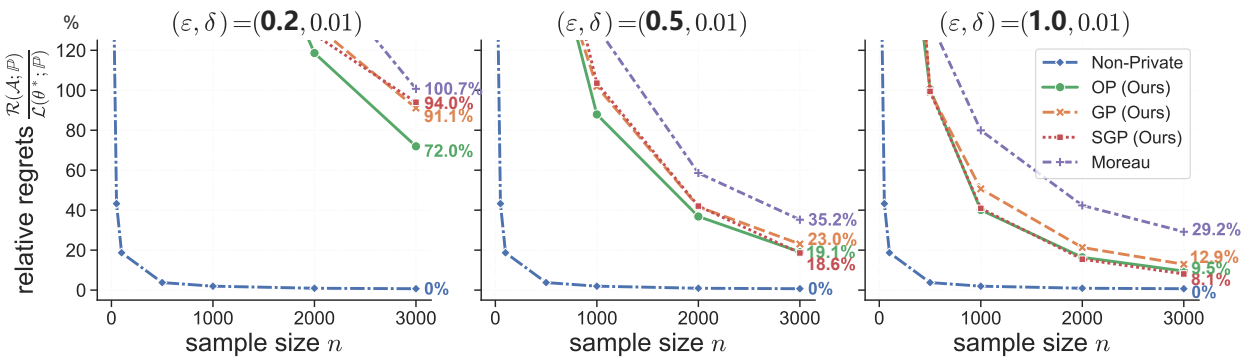
Note. (Gaussian kernel, **exogenous error**, $d = 3$.) Experiment settings are the same as in Figure 4, except exogenous error $\epsilon \sim \mathcal{N}(0, 3^2)$ is independent of $\boldsymbol{x}$.

tive regrets under all (ε, δ) levels. Though unexpected, it is reasonable as the oracle performs much better

in an exogenous case, making the performance measure's denominator $\mathcal{L}(\boldsymbol{\theta}^*)$ smaller, thus resulting in a higher relative regret. Hence, when the error term is endogenous, the proposed algorithms can actually achieve lower relative regrets. This observation practically favors our proposed algorithms, since endogenous errors are more likely to be the case in reality.

Impact of Feature Dimension. In Figure 8, we conduct experiments with a higher dimension $d = 21$. While experiment settings are slightly different from the benchmark setting, we generally observe higher

Figure 8 **Relative Regrets in a Higher-dimensional Case.**



Note. (Gaussian kernel, endogenous error, $\boldsymbol{d} = \mathbf{21}$.) Experiment setting are the same as in Figure 4, except $x_i \sim \mathcal{N}(0, (1/\sqrt{d})^2), \forall i = 2, \ldots, 21$, $\theta_1^\star = 10$, $[\theta_2^\star, \ldots, \theta_{21}^\star]$ are evenly spaced over $[-2, 5]$. The dimension $d = 21$ means 1 intercept term plus 20 features.

relative regrets compared to regrets in Figure 4. Intuitively, this is because we have more features to protect.

We conclude this subsection by giving practical guidelines based on observations in our simulations.

Remark 4 (Practical Guidelines). *A moderate value of ε is recommended. Moreover, given a security level (ε, δ), to reduce relative regrets, we should (i) choose a kernel with light tails, such as Gaussian; (ii) be brave to accept endogeneity issues, which helps reduce relative regrets; and (iii) properly implement dimensionality reduction techniques.*

3.6. Discussion on Proposed Algorithms

Comparison among OP, GP and SGP. Though the three proposed DP algorithms have similar theoretical guarantees and comparable empirical performance, one approach might be preferred over the others in certain circumstances. For example, OP is recommended if low coupling of the decision-making system is favored, because OP separates the modeling and solving processes. Moreover, because of the smoothness of the surrogate loss function, we have the opportunity to accelerate the solving process via second-order methods. However, OP requires to fully load datasets into memory, thus when the dataset is too large, OP might not be suitable. If such is the case, SGP is recommended, as it evaluates gradients on one data point only.

Where to inject noise. In the "Data-Model-Decision" pipeline, noise can be introduced in any step. For the DP-NV problem, we choose to inject noise in the Model step. The choice is not made arbitrarily but on purpose. First of all, injecting noise in the Data step is impractical. The features $\boldsymbol{x}$ and demand y of each data point are drawn from a distribution supported on $\mathcal{X} \times \mathcal{Y}$. To ensure DP, we should inject a Gaussian noise into data point $(\boldsymbol{x}, y)$. And the variance of that Gaussian noise must be of the same magnitude as the radius of $\mathcal{X} \times \mathcal{Y}$, which is usually very large. Moreover, the impact of noise injected in the Data step might be amplified by subsequent procedures, engendering additional optimality loss. Therefore, injecting noise into Data is not recommended. On the other hand, injecting noise in the Decision step, i.e. inject noise into a non-private order quantity directly, makes the final ordering decision inconsistent (see formal statement in Proposition 5 in Appendix). The inconsistency further renders a regret that is strictly positive and cannot be closed even when sample size goes to infinity. Considering both reasons, we opt to introduce noise in the Model step. Specifically, OP introduces noise when building models, while GP and SGP introduce noise when solving models.

4. Data Security: the Curse of Being Small

In the previous section, we proposed three DP algorithms and proved that they all achieve (near) optimal rates. However, our numerical studies also suggest the high expense of pursuing data security in practice. In this section, we examine the drivers of the cost of data security from various notions of "being small".

4.1. Driver 1: Small-sized Dataset

Recall the regret upper bound $\mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$ where the first term is the learning regret and the second is the security-induced regret. It is straightforward to identify a "security for free" region where learning regret $O(\frac{1}{\sqrt{n}})$ dominates security-induced regret $\mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$. Specifically, $n \geq \Omega\left(\frac{d\ln(1/\delta)}{\varepsilon^2}\right)$ characterizes this region, which implies that data security is almost free when the dataset contains sufficiently many data points. However, to warrant a reasonable security level, (ε, δ) are usually set to be small values, which in turn requires a large n to make security free. This partially explains the performance of our private algorithms in numerical studies. From another perspective, when a dataset is small, each of its data points has a greater impact on the final decision. Hiding any individual data point's contribution thus needs a high-variance noise. In light of these observations, we attribute the cost of data security to small-sized datasets.

4.2. Driver 2: Context-free Dataset

As guaranteeing security of dataset $\mathcal{D} := \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$ is costly, one may wonder what if decisions partially depend on $\mathcal{D}$; for example, only $\{y_i\}_{i=1}^n$ is used. The rationale is that if protecting $\boldsymbol{x}$ is costly, then we simply do not use it for decision-making. Naturally, the decisions will never disclose any information about $\boldsymbol{x}$. We carry out this idea with a PartillyDD (Partially Data-Driven) policy:

$$(\text{PartiallyDD}) \quad \widehat{q} := \arg\min_{q \in \mathbb{R}} \frac{1}{n} \sum_{i=1}^{n} c(y_i - q), \tag{9}$$

where $c(u)$ is the newsvendor cost function with critical ratio r. The PartiallyDD policy uses context-free dataset $\{y_i\}_{i=1}^n$ only and discards features $\{\boldsymbol{x}_i\}_{i=1}^n$. It is easy to show that the order quantity given by (9) admits an analytic expression $\widehat{q} = \inf\left\{t \in \mathbb{R} : \widehat{F}_y(t) \geq r\right\}$, where $\widehat{F}_y(t) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}\{y_i \leq t\}$ is the empirical CDF of y constructed from the dataset $\{y_i\}_{i=1}^n$. However, we find an unsatisfactory performance of PartiallyDD, indicating that protecting context through dropping context is financially impractical.

First, theoretically, PartiallyDD is asymptotically suboptimal as shown below.

Theorem 8 (Performance Guarantee of PartiallyDD). *Suppose the distribution $\mathbb{P}$ satisfies (i) feature $\boldsymbol{x}$ is Gaussian distributed, i.e., $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} := \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^\top] \succ \mathbf{0}$ being positive definite; (ii) $y = \langle \boldsymbol{\theta}^\star, \boldsymbol{x}\rangle + \epsilon^\star$ with an exogenous error term $\epsilon^\star$ having a PDF $f_{\epsilon^\star}$. Define $\mathcal{R}(\mathsf{PartiallyDD}; \mathbb{P}) := \mathbb{E}_{\boldsymbol{x},y}[c(y - \widehat{q})] - \mathbb{E}_{\boldsymbol{x},y}[c(y - \langle \boldsymbol{\theta}^*, \boldsymbol{x}\rangle)]$ as the regret of* PartiallyDD *policy under distribution $\mathbb{P}$, where $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta})$ Then, for any distribution $\mathbb{P}$ satisfying two stated assumptions, the regret of* PartiallyDD *is upper bounded as:*

$$\mathcal{R}(\mathsf{PartiallyDD}; \mathbb{P}) \leq \sqrt{\ln(1/\gamma)} \cdot \mathcal{O}\left(\frac{1}{\sqrt{n}}\right) + w,$$

with probability at least $1 - \gamma$, where $w := \mathbb{E}_{\boldsymbol{x}}[|q^* - \langle \boldsymbol{\theta}^*, \boldsymbol{x}\rangle|] \cdot \bar{r} > 0$ is a strictly positive constant with $q^* := \arg\min_{q\in\mathbb{R}} \mathbb{E}_{\mathbb{P}}[c(y - q)]$.

It is evident from Theorem 8 that PartiallyDD is an asymptotically suboptimal policy with a strictly positive performance gap $w > 0$ that cannot be closed. The existence of such a gap is reasonable as information in $\boldsymbol{x}$ that can improve decision quality is completely ignored. Comparing PartiallyDD with the fully data-driven algorithms in the previous section, we immediately obtain Table 5, which summarizes regrets of a non-private model, PartiallyDD, and OP, GP, SGP. Apparently, when the sample size is large,

Table 5 **Regrets $\mathcal{R}(\mathcal{A}; \mathbb{P})$ of Fully and Partially Data-Driven Algorithms.**

Fully Data-Driven		Partially Data-Driven
$\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$	$\mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$	$\mathcal{O}\left(\frac{1}{\sqrt{n}}\right) + w$
(Remark 2)	(by OP, GP, SGP)	(by PartiallyDD, constant $w > 0$)

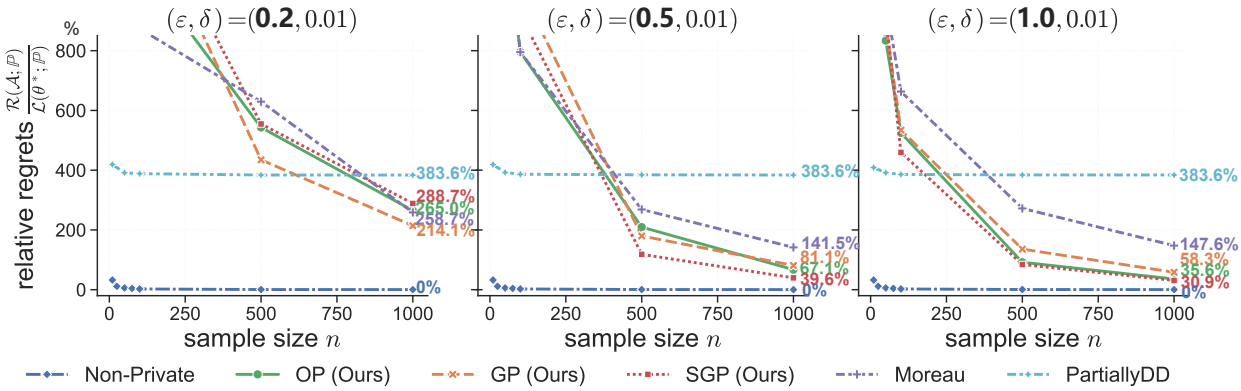
Note. "Fully Data-Driven" means the algorithms use full dataset $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$, including both features and demands, to train models, while "Partially Data-Driven" means only demands $\{y_i\}_{i=1}^n$ are used.

fully private data-driven algorithms are preferred over PartiallyDD. Roughly speaking, this is because the benefit from incorporating contextual information outweighs the cost from injecting noise to protect the additional information.

Second, practically, PartiallyDD also incurs a dramatic compromise in optimality. As shown in Figure 9, the relative regret of PartiallyDD is around 380% when $n = 1000$, and cannot be reduced further. In contrast, relative regrets of DP algorithms diminish gracefully with more samples.

Considering both theoretical and practical perspectives, we attribute the cost of data security to context-free datasets, and highlight that protecting context by ignoring context is financially unsound.

Figure 9 **Relative Regrets of Various Policies (incl. PartiallyDD).**



Note. Experiment settings are exactly same as in Figure 7, except that we additionally implement PartiallyDD.

4.3. Driver 3: Limited Number of Correlated Products

Another interesting finding is that "economies of scale" exists in the DP-NV problem. This phenomenon occurs when multiple products are serving the same finite market and their demands depend on the same features. Under this scenario, the total cost of data security across S products will roughly be $\sqrt{S}$ times that of the worst-performing product, instead of being linear in S.

To demonstrate this phenomenon, we consider a multi-product DP newsvendor selling $S \geq 1$ products indexed by a set $\mathcal{S} := \{1, \ldots, S\}$. The critical ratios are $\{r_s\}_{s\in\mathcal{S}} \in (0,1)^S$. For each product $s \in \mathcal{S}$, we assume the demand follows a linear form $y_s = \langle \boldsymbol{\theta}^{\star s}, \boldsymbol{x}\rangle + \epsilon^{\star s}$, where $\boldsymbol{x} \in \mathcal{X} \subseteq \mathbb{R}^d$ is the common feature shared by all products. We further assume there is an upper bound on total demand $\sum_{s\in\mathcal{S}} y_s \leq M$ that is independent of the number of products, which is reasonable since the market is finite. The dataset that the newsvendor has is $\mathcal{D} := \{(\boldsymbol{x}_i, y_{i1}, \ldots, y_{iS})\}_{i=1}^n$, where $\boldsymbol{x}_i$ is the common feature for data point i, and $(y_{i1}, \ldots, y_{iS})$ is the demand vector of S products. The aim is to find an (ε, δ)-DP algorithm that outputs order quantities $\widehat{\boldsymbol{q}} := (\widehat{q}_1, \ldots, \widehat{q}_S)$. A straightforward idea is to independently run previously proposed algorithms OP, GP, or SGP on each dataset $\mathcal{D}^s := \{(\boldsymbol{x}_i, y_{is})\}_{i=1}^n, \forall s \in \mathcal{S}$ with properly modified algorithm parameters. This idea is formally introduced below, along with its security and performance results.

Proposition 1 (Economies of Scale in a Multi-product DP-NV Problem). *Consider a multi-product DP-NV problem stated above. Let algorithm* $\mathsf{IndepRun}_s(\mathcal{D}^s, \varepsilon_0, \delta_0)$ *be: "run base algorithm (one of* OP, GP, SGP*) on* $\mathcal{D}^s := \{(\boldsymbol{x}_i, y_{is})\}_{i=1}^n$ *with* $(\varepsilon_0, \delta_0)$*". Let algorithm* $\mathsf{IndepRun}(\mathcal{D}, \varepsilon_0, \delta_0) := (\mathsf{IndepRun}_1(\mathcal{D}^1, \varepsilon_0, \delta_0), \ldots, \mathsf{IndepRun}_S(\mathcal{D}^S, \varepsilon_0, \delta_0))$*. For given* ε *and* δ*, if we set* $\varepsilon_0 = \frac{\varepsilon}{2\sqrt{2S\ln(2/\delta)}}$ *and* $\delta_0 = \frac{\delta}{2S}$*, then* $\mathsf{IndepRun}(\mathcal{D}, \varepsilon_0, \delta_0)$ *is an* (ε, δ)*-DP algorithm. Moreover, the total regret of* IndepRun *across* S *products is no more than* $\sqrt{S}$ *times that of the worst-performing product, i.e.*

$$\mathcal{R}(\mathsf{IndepRun}; \mathbb{P}) \leq \sqrt{S} \cdot \sup_{s\in\mathcal{S}} \mathcal{R}(\mathsf{IndepRun}_s; \mathbb{P}_s), \tag{10}$$

which holds for all distributions $\mathbb{P}$ whose marginal distribution $\mathbb{P}_s$ satisfies corresponding assumptions in base algorithms. Both total regret $\mathcal{R}(\mathsf{IndepRun};\mathbb{P})$ and individual regret $\mathcal{R}(\mathsf{IndepRun}_s;\mathbb{P}_s)$ are defined in the form of (3), but the former is associated with loss function $\ell(\boldsymbol{\theta}^{\mathcal{S}};\boldsymbol{x},\boldsymbol{y}) := \sum_{s\in\mathcal{S}}\ell(\boldsymbol{\theta}_s;\boldsymbol{x},y_s)$ with $\boldsymbol{\theta}^{\mathcal{S}} := (\boldsymbol{\theta}_1^\top,\ldots,\boldsymbol{\theta}_S^\top)^\top$, while the latter is associated with loss function $\ell(\boldsymbol{\theta}_s;\boldsymbol{x},y_s)$.

The Proposition confirms that the total regret is only $\sqrt{S}$ times that of the worst-performing product, as indicated by the multiplicative factor $\sqrt{S}$ on the r.h.s. of (10). Intuitively, this is because the market is finite, and when the newsvendor attains the worst performance on one product, she may not perform as poorly on other products due to the coupling constraint. The inequality (10) also indicates that simply scaling up one individual product's regret by S may overestimate the total regret.

We now turn to a more general case with correlated products. Suppose each product has k specific features that are unique to it, in addition to d common features. We assume the correlation exists so that other products' unique features can influence the demand for the product of interest. Specifically, demand functions now follow $y_s = \langle\boldsymbol{\theta}^{\star s},\boldsymbol{x}\rangle + \epsilon^{\star s}$, where $\boldsymbol{x}\in\mathcal{X}\subseteq\mathbb{R}^{d+kS}$ is the feature vector including d-dimensional common features and kS product-specific features. The assumption is sufficiently general, despite a potential dimensionality inflation in features. However, the inflation does not invalidate our findings on economies of scale, since (10) is independent of feature dimension. Thus, our result continues to apply to correlated products.

We end this subsection by emphasizing that the algorithm IndepRun is too naive to minimize the multiplicative factor $\sqrt{S}$ in (10), since the d-dimensional common features are overly protected, once in each of $\mathsf{IndepRun}_s$. We can expect to further reduce the factor $\sqrt{S}$ with a properly designed algorithm, and we leave this interesting question to future study.

5. Negative External Effects on Upstream Suppliers

Up to now, we have designed and examined private algorithms taking into account only the newsvendor's security and optimality performance. Nevertheless, it is natural to expect that any noise that the newsvendor introduces into her decision making process will be passed on to upstream suppliers, hampering the latter's operations. In this section, we will investigate and quantify how DP algorithms adopted by a newsvendor affect upstream suppliers. As security is costly for small-sized newsvendors, we restrict our attention to newsvendors with large-sized and full-context datasets. Single product is assumed for discussion convenience. Before we proceed, we make some mild assumptions on the underlying distribution $\mathbb{P}$ in the form of the reformulated linear demand model (2).

Assumption 1. *The unknown underlying joint distribution $\mathbb{P}$ of $(\boldsymbol{x},\epsilon(\boldsymbol{x}))$ satisfies*

- ***(Fixed Dimension)**: the dimension d is finite and fixed.*
- ***(Standardized $\boldsymbol{x}$)**: the first element $x_1\equiv 1$, and other elements satisfy $\mathbb{E}_{\mathbb{P}}[x_i]=0,\forall i=2,\ldots,d$.*

- ***(Locally Differentiable CDF)**: For any $\boldsymbol{x} \in \mathcal{X}$, the CDF $F_{\epsilon|\boldsymbol{x}}$ of $\epsilon(\boldsymbol{x})$ is twice-differentiable at 0 so that $f_{\epsilon|\boldsymbol{x}}(0) > 0$ and $f'_{\epsilon|\boldsymbol{x}}(0)$ exists.*
- ***(Widely Spread $\boldsymbol{x}$)**: The moment matrix $\boldsymbol{\Sigma} := \mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{x}\boldsymbol{x}^\top\right] \succ \mathbf{0}$ is positive definite.*

The first assumption is common for asymptotic analysis. The second assumption incorporates an intercept term and does not lose generality. The third assumption ensures the uniqueness of optimal $\boldsymbol{\theta}^*$. The last assumption is also common in the literature to improve estimation accuracy. We will focus on OP only since OP always returns a minimizer to an ERM.

5.1. Upstream Supplier Will Face Ambiguous Bias

When the newsvendor does not share data with the supplier and protects collected data using DP algorithms, it becomes impossible for the supplier to understand the true demand accurately. Intuitively, from the supplier's point of view, there are two reasons: (1) the order quantities depend on some statistics of the underlying demand distribution, estimating which is impossible without any data; (2) the only observable order quantities are contaminated by injected noises. Formally, we quantify the bias below.

Proposition 2 (Ambiguous Bias from Supplier's View). *For any distribution $\mathbb{P}$ that satisfies Assumption 1, the order quantity through running algorithm* OP *is*

$$\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x}) = q^*(\boldsymbol{x}) - 2\lambda \boldsymbol{u}^\top \boldsymbol{x} - \frac{1}{2}h^2\kappa_2 \boldsymbol{v}^\top \boldsymbol{x} + \mathcal{O}_p\left(\frac{1}{\sqrt{n}} + \frac{\sigma}{n}\right), \quad \forall \boldsymbol{x} \in \mathcal{X},$$

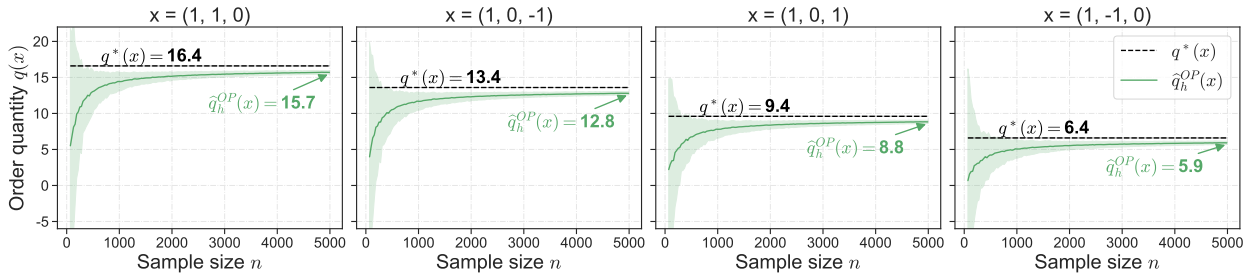
where $\sigma := \bar{r}\sqrt{\ln(1/\delta)}/\varepsilon$ is the standard deviation of injected noise, $\lambda = o(1)$ and $h = o(1)$ are algorithm's parameters; $\boldsymbol{u} := \boldsymbol{J}^{-1}\boldsymbol{\theta}^*$, $\boldsymbol{J} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0)\boldsymbol{x}\boldsymbol{x}^\top\right] \succ \mathbf{0}$, and $\boldsymbol{v} := \boldsymbol{J}^{-1}\mathbb{E}_{\boldsymbol{x}}\left[f'_{\epsilon|\boldsymbol{x}}(0)\boldsymbol{x}\right]$ are statistics of distribution $\mathbb{P}$. Higher-order infinitesimals of λ and h^2 are omitted.

Since statistics $\boldsymbol{u}$, $\boldsymbol{J}$, and $\boldsymbol{v}$ are unobservable from supplier's view, the supplier is thus facing ambiguous bias. In other words, the supplier cannot affirm whether the observed order quantity $\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x})$ is overstated or understated compared to the optimal quantity $q^*(\boldsymbol{x})$ no matter what auxiliary knowledge the supplier may have. As shown in Figure 10, the shadowed green region overlaps with the black line in all sub-figures when sample size is small, indicating the ambiguous bias in $\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x})$. As a consequence, the ambiguity may affect the supplier's operational decisions and reduce supply chain efficiency.

5.2. Increase in Variability

Because of the artificially injected noise, the variability of the order quantity is highly likely to increase as well. Let $\widehat{q}(\boldsymbol{x}) := \langle \widehat{\boldsymbol{\theta}}, \boldsymbol{x}\rangle$ be an order decision based on solving a non-private ERM $\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \widehat{\mathcal{L}}(\boldsymbol{\theta})$. Then, the variability measure of interest here is the increase in conditional variance defined as $\Delta V(\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x}), \widehat{q}(\boldsymbol{x})) := \mathsf{Var}_{\mathcal{D},\boldsymbol{b}}[\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x})|\boldsymbol{x}] - \mathsf{Var}_{\mathcal{D}}[\widehat{q}(\boldsymbol{x})|\boldsymbol{x}]$ for any given feature $\boldsymbol{x}$. Specifically, we only focus on the extra variance brought about by DP algorithms. We are not concerned with the variance from estimation, i.e., $\mathsf{Var}_{\mathcal{D}}[\widehat{q}(\boldsymbol{x})|\boldsymbol{x}]$, since it is relatively well studied in the literature. The increase in variability is found to depend on the variance of the injected noise and OP's parameters.

Figure 10 Order Quantities of OP and True Optimal Order Quantities under Various Feature Vectors.



Note. Black dashed lines are true optimal order quantities $q^*(\boldsymbol{x})$ for given $\boldsymbol{x}$, while green curves are private order quantities $\widehat{q}_h^{\mathrm{OP}}(\boldsymbol{x})$. Shadowed green region is the "3-sigma" region. $(\varepsilon, \delta) = (1, 0.01)$. Other experiment settings are same as in Figure 7, expect demand error term is assumed to be exogenous $\epsilon^\star \sim \mathcal{N}(0, 3^2)$; thus, by Lemma 4, $\boldsymbol{\theta}^* = (11.4, 5, -2)$.

Proposition 3 (Increase in Variability of Order Quantity). *Suppose the underlying distribution satisfies Assumption 1. Let* $\mathsf{Var}_{\mathcal{D},\mathcal{A}}[\widehat{q}^{\mathcal{A}}(\boldsymbol{x})|\boldsymbol{x}]$ *denote the conditional variance of data-driven ordering algorithm* $\mathcal{A}$ *given an upcoming feature vector* $\boldsymbol{x}$*, where the variance is taken over dataset sampling and algorithm's internal randomness. Then, the increase in conditional variance of order quantity* $\Delta V(\widehat{q}_h^{\mathrm{OP}}(\boldsymbol{x}), \widehat{q}(\boldsymbol{x})) := \mathsf{Var}_{\mathcal{D},\mathrm{OP}}[\widehat{q}_h^{\mathrm{OP}}(\boldsymbol{x})|\boldsymbol{x}] - \mathsf{Var}_{\mathcal{D}}[\widehat{q}(\boldsymbol{x})|\boldsymbol{x}]$ *due to DP algorithms (specifically,* OP*) is bounded as*

$$\Omega\left(\frac{\sigma^2}{n^2} - \frac{\lambda + h}{n}\right) \leq \Delta V(\widehat{q}_h^{\mathrm{OP}}(\boldsymbol{x}), \widehat{q}(\boldsymbol{x})) \leq \mathcal{O}\left(\frac{\sigma^2}{n^2}\right), \quad \forall \boldsymbol{x} \in \mathcal{X},$$

where $\sigma := \bar{r}\sqrt{\ln(1/\delta)}/\varepsilon$ *is the standard deviation of injected noise,* $\lambda = o(1)$*, and* $h = o(1)$ *are parameters for* OP*. Higher-order infinitesimals of* λ *and* h *are omitted.*

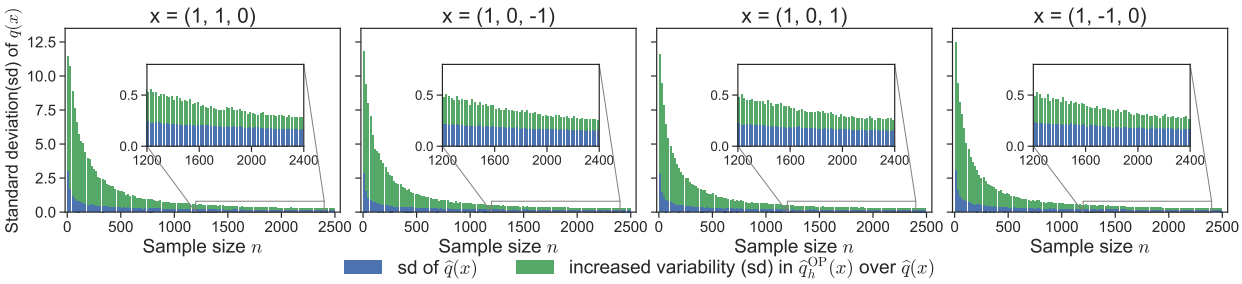
As the proposition states, the increase in variability is upper bounded by a vanishing value $\mathcal{O}(\sigma^2/n^2)$, which implies a weakening influence when more samples become available. Notably, when σ becomes 0, i.e. the model degenerates from a private to a non-private model, the upper bound becomes 0, indicating no additional variability. Similar interpretations can be obtained from the lower bound. Specially, the lower bound also hints that regularization terms and convolution smoothing are both helpful in stabilizing order quantities. But whether the lower bound takes positive or negative values is undetermined without specifying σ, n, λ, and h. In short, the upstream supplier may observe more volatile order quantities from a newsvendor adopting DP algorithms.

We run simulations to empirically verify our findings, and plot the standard deviations of order quantities in Figure 11 under four given feature vectors spanning the feature space $\mathcal{X}$. Green bars in the figures represent the increase in variability due to DP algorithms. It is evident that as sample size increases, the increase in variability keeps decreasing (shorter green bars) and is eventually exceeded by the variability due to data sampling, i.e. blue bars.

5.3. Decrease in Profit Share

We revisit a newsvendor-supplier supply chain channel under wholesale price contract that has been investigated in Lariviere and Porteus (2001), but from a DP perspective. In this channel, the newsvendor

Figure 11 Standard Deviations of Order Quantities under Various Feature Vectors.



Note. Bottom blue bars show the standard deviation (sd) of non-private order quantities; upper green bars represent increased sd due to DP algorithm. Experiment settings are exactly the same as in Figure 4. $(\varepsilon, \delta) = (1, 0.01)$.

needs to decide the number of items $q(\boldsymbol{x})$ bought at unit wholesale price w from the supplier after observing contextual information $\boldsymbol{x}$ but before observing demand y to maximize her expected profit $\Pi^N(\boldsymbol{x}) = \max_{q(\boldsymbol{x})} \mathbb{E}_{y|\boldsymbol{x}} [p \min\{y, q(\boldsymbol{x})\}] - wq(\boldsymbol{x})$ from selling products at retail price $p > w$. The supplier is assumed to have unlimited production capacity and produces one unit at cost $v < w$; thus, his profit is $\Pi^S(\boldsymbol{x}) = (w - v)q(\boldsymbol{x})$.

When demand y is a univariate random variable as discussed in Lariviere and Porteus (2001, Section 5), the supplier could almost fully encroach the newsvendor's profit by raising up the wholesale price w. Interestingly, when the newsvendor adopts DP algorithms, she can counter supplier encroachment, increasing her own profit share and bringing down that of the supplier.

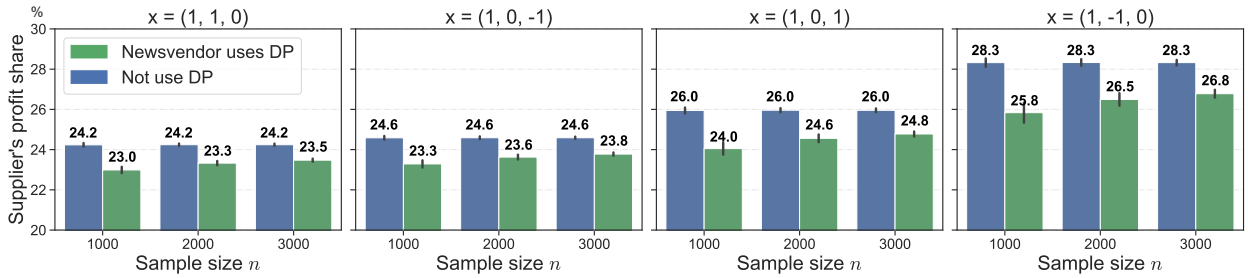
Proposition 4 (Supplier Gains a Lower Profit Share; informal). *Consider a newsvendor-supplier channel under the wholesale price contract with production cost $v > 0$, wholesale price $w > v$, and retail price $p > w$. Let the expected profit of the newsvendor under algorithm $\mathcal{A}$ be $\widehat{\Pi}^N(\boldsymbol{x}) = \mathbb{E}_{y|\boldsymbol{x}} [p \min\{y, \widehat{q}^{\mathcal{A}}(\boldsymbol{x})\} - w\widehat{q}^{\mathcal{A}}(\boldsymbol{x})], \forall \boldsymbol{x} \in \mathcal{X}$, and let the profit of the supplier be $\widehat{\Pi}^S(\boldsymbol{x}) = (w - v)\widehat{q}^{\mathcal{A}}(\boldsymbol{x})$. Then, under some regularity conditions and with high probability, adopting the private algorithm* OP *would raise up newsvendor's profit share (correspondingly, pull down supplier's share):*

$$\frac{\widehat{\Pi}^N(\boldsymbol{x})}{\widehat{\Pi}^N(\boldsymbol{x}) + \widehat{\Pi}^S(\boldsymbol{x})} = \frac{\mathsf{Rev}^N(\boldsymbol{x}) - C - U}{\mathsf{Rev}(\boldsymbol{x}) - C - U - \widehat{P}(\boldsymbol{x}, \mathcal{D})}, \quad \forall \boldsymbol{x} \in \mathcal{X}, \tag{11}$$

by setting a positive value of $\widehat{P}(\boldsymbol{x}, \mathcal{D}) := (w - v)(q^*(\boldsymbol{x}) - \widehat{q}_h^{\mathrm{OP}}(\boldsymbol{x}))$ in (11). The value $\widehat{P}(\boldsymbol{x}, \mathcal{D})$ is the supplier's revenue loss due to newsvendor's private learning process. Infinitesimals of $\widehat{P}(\boldsymbol{x}, \mathcal{D})$ are omitted. Other values on the right-hand-side of (11) are constants.

We leave the formal statement of this proposition to Appendix. The intuition behind the above proposition is that adopting DP algorithms will make the whole chain worse off, but the supplier will be worse off more; thus, the supplier will receive a lower profit share. The lower profit share is demonstrated in Figure 12. In all situations considered in Figure 12, the supplier will obtain a lower profit share after the newsvendor adopts DP algorithms. But the decrease in profit share narrows with more available data, which is

Figure 12 **Comparison on Supplier's Profit Shares before and after Newsvendor Uses DP Algorithms.**



Note. Blue (green) bars are supplier's profit shares before (after) newsvendor uses DP algorithms. Short thin black bars on top of each thick bar represent standard deviation. Experiment settings and demand generating process are the same as in Figure 7. $(\varepsilon, \delta) = (1, 0.01)$, $(p, w, v) = (10, 3, 1)$.

reasonable since estimations become more accurate, leading to a smaller positive value of $\widehat{P}(\boldsymbol{x}, \mathcal{D})$. In fact, the positive value of $\widehat{P}(\boldsymbol{x}, \mathcal{D})$ can be detected in earlier experiments, see Figure 10 for instance.

Our findings in this section have important implications for policy-makers and practitioners because implementing DP algorithms by a newsvendor may cause severe negative external effects on supply chain partners. First, newsvendor use of DP algorithms may push supply chain coordination to the brink of collapse. Second, it may also serve as a credible threat for the newsvendor in her negotiation with the supplier. She may seek commitment from the supplier not to conduct inference on her ordering decisions and to strengthen his storage security for these ordering data. Otherwise, the newsvendor may have no other recourse but to adopt DP algorithms. Hence, our analysis suggests a good opportunity for supply chain partners to come together to build a trustworthy environment where DP algorithms can be a credible last resort for data security.

We conclude this section with a remark summarizing DP algorithms' negative external effects.

Remark 5 (DP Algorithms' Negative External Effects). *A newsvendor's adoption of DP algorithms will negatively affect her supplier by sharing ambiguous bias, passing over higher variability, and encroaching profit share, which all reduce chain-level efficiency and profit.*

6. Conclusion and Future Research

In this paper, we discussed how to properly manage data security risks in supply chains, with a particular focus on preventing data from being inferred from operational decisions in the data-driven contextual newsvendor problem. To quantify data security, we borrowed the idea of Differential Privacy (DP). We first proposed several DP algorithms that guarantee data security and achieve (near) optimal performances simultaneously. The cost of data security was found to be at least $\widetilde{\Omega}\left(\frac{\sqrt{d}}{n\varepsilon}\right)$, but no more than $\mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$. We then identified three drivers of security cost; namely, small-sized datasets, context-free datasets, and datasets with a limited number of correlated products. We refer to the phenomenon that data security is financially costly for "small" firms as *the curse of being small*. Lastly, we examined

the negative external effects of DP algorithms and found that it may harm upstream suppliers in multiple ways.

There are many interesting extensions to explore in the future; for example, discrete demand or multi-period newsvendor problems. Besides the newsvendor, a network of many participants would also be a more realistic setting for supply chain analytics. Identifying data security issues therein and designing private algorithms would be fruitful endeavors.

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Appendix. Proofs of Statements

We want to highlight an important remark on notations before showing proofs. For proofs in Appendix, we will frequently adopt $\widehat{\mathcal{L}}(\boldsymbol{\theta})$ as a shorthand for $\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{D})$ *when no manipulations on dataset* $\mathcal{D}$. Always keep in mind that the hat symbol $\widehat{}$ reflects implicitly the dependence on the underlying dataset $\mathcal{D}$. We also adopt the same notation for the derivatives $\nabla\widehat{\mathcal{L}}(\boldsymbol{\theta})$ and minimizers $\widehat{\boldsymbol{\theta}}$. But when analysis involves manipulating dataset $\mathcal{D}$, we will explicitly express such dependence. Additionally, $\mathcal{L}(\boldsymbol{\theta})$ is the shorthand for population-level function $\mathcal{L}(\boldsymbol{\theta};\mathbb{P})$, and same rules are applied to population-level derivatives $\nabla\mathcal{L}(\boldsymbol{\theta})$ and minimizers $\boldsymbol{\theta}^*$.

A. Proofs for Section 3

A.1. Proof of Lemma 4

Proof. It is well known that the expected newsvendor cost $\mathcal{L}(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x},y}\left[c(y-\boldsymbol{\theta}^\top\boldsymbol{x})\right] = \mathbb{E}_{\boldsymbol{x},\epsilon^\star}\left[c(\boldsymbol{\theta}^{\star\top}\boldsymbol{x}+\epsilon^\star-\boldsymbol{\theta}^\top\boldsymbol{x})\right]$ is differentiable at any $\boldsymbol{\theta}\in\mathbb{R}^d$:

$$\begin{aligned}\nabla\mathcal{L}(\boldsymbol{\theta}) &= \mathbb{E}_{\boldsymbol{x},\epsilon^\star}\left[r\cdot(-\boldsymbol{x})\cdot\mathbb{1}\left\{\boldsymbol{\theta}^{\star\top}\boldsymbol{x}+\epsilon^\star-\boldsymbol{\theta}^\top\boldsymbol{x}\geq 0\right\}+(1-r)\cdot\boldsymbol{x}\cdot\mathbb{1}\left\{\boldsymbol{\theta}^\top\boldsymbol{x}-\boldsymbol{\theta}^{\star\top}\boldsymbol{x}-\epsilon^\star>0\right\}\right]\\ &= \mathbb{E}_{\boldsymbol{x},\epsilon^\star}\left[\boldsymbol{x}\cdot\mathbb{1}\left\{\boldsymbol{\theta}^\top\boldsymbol{x}-\boldsymbol{\theta}^{\star\top}\boldsymbol{x}-\epsilon^\star\geq 0\right\}\right]-r\mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{x}\right]\end{aligned}$$

The gradient at $\boldsymbol{\theta}^* := \boldsymbol{\theta}^\star + [F_{\epsilon^\star}^{-1}(r),0,\ldots,0]$ is

$$\begin{aligned}\nabla\mathcal{L}(\boldsymbol{\theta}^*) &= \mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{x}\cdot\mathbb{E}_{\epsilon^\star}\left[\mathbb{1}\left\{F_{\epsilon^\star}^{-1}(r)x_1-\epsilon^\star\geq 0\right\}\right]\right]-r\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{x}]\\ &= \mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{x}\cdot\mathrm{Pr}_{\epsilon^\star}\left[\epsilon^\star\leq F_{\epsilon^\star}^{-1}(r)\right]\right]-r\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{x}]\\ &= \mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{x}\cdot r\right]-r\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{x}]\\ &= \mathbf{0},\end{aligned}$$

The second equality is from our assumption that $x_1\equiv 1$, and the third equality is due to another assumption that $F_{\epsilon^\star}$ is continuous at r-th quantile, which implies $F_{\epsilon^\star}(F_{\epsilon^\star}^{-1}(r))=r$. Therefore, $\boldsymbol{\theta}^*$ is an optimal solution to $\min_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta})$. □

A.2. Proof of Lemma 5

Proof. Gradients in part 1 of this Lemma can be easily calculated from function $c_h(\cdot)$. As for part 2, L-Lipschitz continuity is by noticing $\sup_{\boldsymbol{\theta},\boldsymbol{x},y}\|\nabla\ell_h(\boldsymbol{\theta};\boldsymbol{x},y)\|_2=\bar{r}B_x$; β-smooth is by noticing the supremum of maximum eigenvalue of Hessian matrix satisfies $\sup_{\boldsymbol{\theta};\boldsymbol{x},y}\lambda_{\max}\left(\nabla^2\ell_h(\boldsymbol{\theta};\boldsymbol{x},y)\right)=\sup_{\boldsymbol{\theta};\boldsymbol{x},y}\lambda_{\max}\left(K_h\left(y-\boldsymbol{\theta}^\top\boldsymbol{x}\right)\cdot\boldsymbol{x}\boldsymbol{x}^\top\right)\leq\overline{K}B_x^2/h$. To prove part 3, it is equivalent to show $c(u)\leq c_h(u)\leq c(u)+\frac{1}{2}h\kappa_1,\forall u\in\mathbb{R}$. Recall that $c(u)=|u|/2+(r-1/2)u$. Let $g_h(u):=c_h(u)-c(u)$, we have $g_h'(u)=\mathcal{K}_h(u)\geq 0,\forall u<0$ and $g_h'(u)=\mathcal{K}_h(u)-1\leq 0,\forall u>0$. Since $g_h(\cdot)$ is continuous on $\mathbb{R}$, it takes maximum value at $u=0$ with $g_h(0)=c_h(0)-c(0)=\frac{h}{2}\int_{-\infty}^{\infty}|v|\,K(v)\,dv$, which proves the right-hand-side in part 3. Moreover, this upper bound is tight by our argument. It remains to show, for any $h>0$, $\lim_{u\to\infty}g_h(u)\geq 0$ and $\lim_{u\to-\infty}g_h(u)\geq 0$. To prove this result, we can first calculate $g_h(u)$ explicitly:

$$2g_h(u)=\int_{-\infty}^{\infty}|u+vh|\,K(v)\,dv-|u|=2h\int_{|u|/h}^{\infty}vK(v)\,dv+u\cdot\int_{-u/h}^{u/h}K(v)\,dv-|u|\,,\quad\forall u\in\mathbb{R}. \tag{12}$$

We notice that, when u tends to positive or negative infinity, both limits of $g_h(u)$ exist and equal 0, which completes the proof.

□

A.3. Lemma 6 and Proof

Lemma 6 (Objective Perturbation). *(Kifer et al. 2012, Theorem 2, rephrased) Let $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$ be a dataset. Let $\widehat{\mathcal{L}}(\boldsymbol{\theta}; \mathcal{D}) = \frac{1}{n}\sum_{i=1}^{n} \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)$ be a convex loss function with continuous Hessian. Let L be the upper bound on $\|\nabla \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)\|_2$ and let β be an upper bound on the eigenvalue of $\nabla^2 \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)$. Assume that for any $\boldsymbol{\theta} \in \mathcal{B}(B_\theta)$ and for all data points $(\boldsymbol{x}_i, y_i)$, the rank of $\nabla^2 \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)$ is at most one. Then $\widehat{\boldsymbol{\theta}} = \arg\min \widehat{\mathcal{L}}(\boldsymbol{\theta}; \mathcal{D}) + \lambda \|\boldsymbol{\theta}\|_2 + \boldsymbol{b}^\top \boldsymbol{\theta}/n$ is (ε, δ)-DP if*

1. *the noise vector $\boldsymbol{b}$ is sampled from a multivariate Gaussian $\mathcal{N}(0, \sigma^2 \boldsymbol{I}_{d\times d})$ with variance $\sigma^2 \geq L^2 \cdot (8\ln(2/\delta) + 4\varepsilon)/\varepsilon^2$;*
2. *the regularization coefficient $\lambda \geq 2\beta/(n\varepsilon)$.*

Proof. For completeness and for later analysis in next section, we provide a brief proof for Lemma 6. The proof follows the same proof idea for Theorem 2 in Kifer et al. (2012), with a slight modification. A similar proof can be found in Iyengar et al. (2019). Let $\mathcal{A}(\mathcal{D}) := \arg\min_{\boldsymbol{\theta}} \widehat{\mathcal{L}}_h^{\text{OP}}(\boldsymbol{\theta}; \mathcal{D})$, and we are going to show, for any $\boldsymbol{v}$ and $\mathcal{D} \sim \mathcal{D}'$,

$$\frac{\Pr_{\mathcal{A}}[\mathcal{A}(\mathcal{D}) = \boldsymbol{v}]}{\Pr_{\mathcal{A}}[\mathcal{A}(\mathcal{D}') = \boldsymbol{v}]} \leq e^{\varepsilon}, \quad \text{w.p. at least } 1-\delta.$$

Because of first-order-condition, we know that $\boldsymbol{b}(\mathcal{A}(\mathcal{D}); \mathcal{D}) = -n\nabla \widehat{\mathcal{L}}_h(\mathcal{A}(\mathcal{D}); \mathcal{D}) - 2n\lambda \mathcal{A}(\mathcal{D})$ for any $\mathcal{D}$. Changing variables according to the function inverse theorem A35 in Billingsley (2017), we get

$$\frac{\Pr_{\mathcal{A}}[\mathcal{A}(\mathcal{D}) = \boldsymbol{v}]}{\Pr_{\mathcal{A}}[\mathcal{A}(\mathcal{D}') = \boldsymbol{v}]} = \frac{pdf(\boldsymbol{b}(\boldsymbol{v}; \mathcal{D}))}{pdf(\boldsymbol{b}(\boldsymbol{v}; \mathcal{D}'))} \cdot \frac{|\det(\nabla \boldsymbol{b}(\boldsymbol{v}; \mathcal{D}'))|}{|\det(\nabla \boldsymbol{b}(\boldsymbol{v}; \mathcal{D}))|}, \quad \forall \boldsymbol{v}, \tag{13}$$

where $\det(\cdot)$ is the determinant of a given matrix. Suppose the differing data point that $\mathcal{D}$ has is $(\boldsymbol{x}_k, y_k)$, while $\mathcal{D}'$ has $(\boldsymbol{x}'_k, y'_k)$. We note that the Global Sensitivity (GS) of $\boldsymbol{b}(\boldsymbol{v}; \cdot)$ is

$$\|\boldsymbol{b}(\boldsymbol{v}; \mathcal{D}) - \boldsymbol{b}(\boldsymbol{v}; \mathcal{D}')\|_2 = \|\nabla \ell_h(\boldsymbol{v}; \boldsymbol{x}'_k, y'_k) - \nabla \ell_h(\boldsymbol{v}; \boldsymbol{x}_k, y_k)\|_2 \leq 2L.$$

Therefore, according to Gaussian Mechanism Lemma 1, setting $\sigma^2 \geq (2L)^2 \cdot 2\ln(1.25/\delta)/(\varepsilon/2)^2$ can ensure

$$\frac{pdf(\boldsymbol{b}(\boldsymbol{v}; \mathcal{D}))}{pdf(\boldsymbol{b}(\boldsymbol{v}; \mathcal{D}'))} \leq e^{\frac{\varepsilon}{2}}, \quad \text{with prob. at least } 1-\delta. \tag{14}$$

However, using Gaussian Mechanism Lemma here is too naive to provide a tight constant. A more involved analysis in Kifer et al. (2012, Lemma 14) shows that a tighter $\sigma^2 = L^2 \cdot (8\ln(2/\delta) + 4\varepsilon)/\varepsilon^2$ is enough to guarantee (14).

We then come to control the ratio between two determinants. The matrix $\nabla \boldsymbol{b}(\boldsymbol{v}; \mathcal{D})$ and $\nabla \boldsymbol{b}(\boldsymbol{v}; \mathcal{D}')$ have following relationship:

$$\begin{aligned}
\nabla \boldsymbol{b}(\boldsymbol{v}; \mathcal{D}) &= -n\nabla^2 \widehat{\mathcal{L}}_h(\boldsymbol{v}; \mathcal{D}) - 2n\lambda \boldsymbol{I} \\
&= -n\nabla^2 \widehat{\mathcal{L}}_h(\boldsymbol{v}; \mathcal{D}') + \nabla^2 \ell_h(\boldsymbol{v}; \boldsymbol{x}'_k, y'_k) - \nabla^2 \ell_h(\boldsymbol{v}; \boldsymbol{x}_k, y_k) - 2n\lambda \boldsymbol{I} \\
&= \nabla \boldsymbol{b}(\boldsymbol{v}; \mathcal{D}') + \nabla^2 \ell_h(\boldsymbol{v}; \boldsymbol{x}'_k, y'_k) - \nabla^2 \ell_h(\boldsymbol{v}; \boldsymbol{x}_k, y_k),
\end{aligned}$$

where $E := \nabla^2 \ell_h(\boldsymbol{v}; \boldsymbol{x}'_k, y'_k) - \nabla^2 \ell_h(\boldsymbol{v}; \boldsymbol{x}_k, y_k)$ is the difference. For a given d-by-d symmetric matrix X, let $\lambda_1(X) \geq \lambda_2(X) \geq \cdots \geq \lambda_d(X)$ be eigenvalues of X. Specifically, let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$ be the eigenvalues of

$n\nabla^2\widehat{\mathcal{L}}_h(\boldsymbol{v};\mathcal{D})$, and let $\lambda_1' \geq \lambda_2'$ be two eigenvalues of E since E is at most rank 2. Then, the ratio between determinants can be evaluated as:

$$\begin{aligned}
\frac{|\det(\nabla \boldsymbol{b}(\boldsymbol{v};\mathcal{D}'))|}{|\det(\nabla \boldsymbol{b}(\boldsymbol{v};\mathcal{D}))|} &= \frac{\left|\det(-n\nabla^2\widehat{\mathcal{L}}_h(\boldsymbol{v};\mathcal{D}) - 2n\lambda\boldsymbol{I} - E)\right|}{\left|\det(-n\nabla^2\widehat{\mathcal{L}}_h(\boldsymbol{v};\mathcal{D}) - 2n\lambda\boldsymbol{I})\right|} \\
&= \frac{\left|\Pi_{i=1}^d \lambda_i(n\nabla^2\widehat{\mathcal{L}}_h(\boldsymbol{v};\mathcal{D}) + 2n\lambda\boldsymbol{I} + E)\right|}{\left|\Pi_{i=1}^d \lambda_i(n\nabla^2\widehat{\mathcal{L}}_h(\boldsymbol{v};\mathcal{D}) + 2n\lambda\boldsymbol{I})\right|} \\
&\leq \frac{(\lambda_1 + 2n\lambda + |\lambda_1'|)\cdot(\lambda_2 + 2n\lambda + |\lambda_2'|)\,\Pi_{i=3}^d\,(\lambda_i + 2n\lambda)}{\Pi_{i=1}^d\,(\lambda_i + 2n\lambda)} \\
&\leq \left(1 + \frac{|\lambda_1'|}{2n\lambda}\right)\left(1 + \frac{|\lambda_2'|}{2n\lambda}\right).
\end{aligned}$$

The first inequality is due to Weyl's theorems on eigenvalues for Hermitian matrices, which can be applied here because all involved matrices are positive semi-definite. Since ℓ_h is β-smooth, the two eigenvalues of $\nabla^2\ell_h$ should satisfy $\max\{|\lambda_1'|, |\lambda_2'|\} \leq \beta$. To require

$$\left(1 + \frac{\beta}{2n\lambda}\right)^2 \leq e^{\frac{\varepsilon}{2}}, \tag{15}$$

it suffices to have $\lambda \geq \frac{2\beta}{n\varepsilon}$. Therefore, we have

$$\frac{|\det(\nabla \boldsymbol{b}(\boldsymbol{v};\mathcal{D}'))|}{|\det(\nabla \boldsymbol{b}(\boldsymbol{v};\mathcal{D}))|} \leq e^{\frac{\varepsilon}{2}}. \tag{16}$$

Plugging (14) and (16) into (13), we finally obtain,

$$\frac{\Pr_{\mathcal{A}}[\mathcal{A}(\mathcal{D}) = \boldsymbol{v}]}{\Pr_{\mathcal{A}}[\mathcal{A}(\mathcal{D}') = \boldsymbol{v}]} \leq e^{\varepsilon}, \quad \text{with prob. at least } 1-\delta,$$

if $\sigma^2 \geq L^2 \cdot (8\ln(2/\delta) + 4\varepsilon)/\varepsilon^2$ and $\lambda \geq \frac{2\beta}{n\varepsilon}$.

□

A.4. Proof of Theorem 1

Proof of Theorem 1 According to Lemma 6, the minimizer $\widehat{\boldsymbol{\theta}}_h^{\text{OP}}$ to (6) satisfies (ε,δ)-DP, because (1) noise vector $\boldsymbol{b}$ is sampled from a multivariate Gaussian $\mathcal{N}(0, \sigma^2\mathbf{I}_{d\times d})$ with variance $\sigma^2 \geq (8\ln(2/\delta) + 4\varepsilon)/\varepsilon^2$; and (2) the regularization coefficient $\lambda \geq 2\beta/(n\varepsilon)$. By Post-Processing Lemma 3, the final output order quantity $\left\langle \widehat{\boldsymbol{\theta}}_h^{\text{OP}}, \boldsymbol{x}_{n+1}\right\rangle$ is therefore (ε,δ)-DP.

□

A.5. Proof of Theorem 2

We first introduce the concept of Uniform Stability, and a related Lemma, which will help with our regret analysis.

Definition 4 (τ-Uniform Stability). *A randomized algorithm $\mathcal{A}: \mathcal{X}^n \times \mathcal{Y}^n \to \mathbb{R}^d$ is τ-uniform stable with respect to function $\ell: \mathbb{R}^d \times \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ if for any pair of neighboring datasets $\mathcal{D} \sim \mathcal{D}'$ that differ in one data point only, we have*

$$\sup_{\boldsymbol{x},y} \mathbb{E}_{\mathcal{A}}\left[\ell(\mathcal{A}(\mathcal{D}); \boldsymbol{x}, y) - \ell(\mathcal{A}(\mathcal{D}'); \boldsymbol{x}, y)\right] \leq \tau.$$

Lemma 7 (Uniform Stability Lemma). *(Bousquet and Elisseeff 2002) Let $\mathcal{A}: \mathcal{X}^n \times \mathcal{Y}^n \to \mathbb{R}^d$ be a τ-uniform stable algorithm w.r.t. loss function $\ell: \mathbb{R}^d \times \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. Let $\mathbb{P}$ be a distribution over $\mathcal{X} \times \mathcal{Y}$, and $\mathcal{D} \sim \mathbb{P}^n$ be samples i.i.d. drawn from $\mathbb{P}$. Then, we have*

$$\mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n,\mathcal{A}}\left[\mathcal{L}(\mathcal{A}(\mathcal{D})) - \widehat{\mathcal{L}}(\mathcal{A}(\mathcal{D}))\right] \leq \tau,$$

Table 6 Notations for Objective Perturbation

Type	Description	Abbr.	Functions	Minimizers
Empirical	ERM	$\widehat{\mathcal{L}}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^{n}\ell(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)$	$\widehat{\boldsymbol{\theta}}$
	Regularized ERM	$\widehat{\mathcal{L}}^{\#}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^{n}\ell(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)+\lambda\left\|\boldsymbol{\theta}\right\|_2^2$	$\widehat{\boldsymbol{\theta}}^{\#}$
	Private regularized ERM	$\widehat{\mathcal{L}}^{\mathsf{OP}}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^{n}\ell(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)+\lambda\left\|\boldsymbol{\theta}\right\|_2^2+\frac{\boldsymbol{b}^\top\boldsymbol{\theta}}{n}$	$\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}$
Smoothed	ERM	$\widehat{\mathcal{L}}_h(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^{n}\ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)$	$\widehat{\boldsymbol{\theta}}_h$
	Regularized ERM	$\widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^{n}\ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)+\lambda\left\|\boldsymbol{\theta}\right\|_2^2$	$\widehat{\boldsymbol{\theta}}_h^{\#}$
	Private regularized ERM	$\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^{n}\ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)+\lambda\left\|\boldsymbol{\theta}\right\|_2^2+\frac{\boldsymbol{b}^\top\boldsymbol{\theta}}{n}$	$\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$
Stochastic	Risk minimization	$\mathcal{L}(\boldsymbol{\theta})$	$\mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}\left[\ell(\boldsymbol{\theta};\boldsymbol{x},y)\right]$	$\boldsymbol{\theta}^*$
	Smoothed RM	$\mathcal{L}_h(\boldsymbol{\theta})$	$\mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}\left[\ell_h(\boldsymbol{\theta};\boldsymbol{x},y)\right]$	$\boldsymbol{\theta}_h^*$

The following Table 6 summarizes notations we are going to use for this proof.

Proof. We analyze the algorithm's performance without projection, which gives a looser upper bound. In following analysis, we notationally suppress the dependences on $\mathcal{D}$ and $\boldsymbol{b}$ unless explicitly manipulating them. Recall the risk decomposition

$$\mathcal{R}(\mathsf{OP};\mathbb{P})=\mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})\right]+\mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right]+\mathcal{L}_h(\boldsymbol{\theta}^*)-\mathcal{L}(\boldsymbol{\theta}^*). \tag{17}$$

Therefore, we can separately analyze above three terms.

1) **(Uniform Stability)** We follow the standard stability analysis for strongly convex loss functions. For any $\mathcal{D}$ and $\boldsymbol{b}$, suppose that there is a dataset $\mathcal{D}'$ that differs from $\mathcal{D}$ in only one data point $(\boldsymbol{x}',y')$, then

$$\begin{aligned}\lambda\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D})-\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')\right\|_2^2 &\le \widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}');\mathcal{D})-\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}),\mathcal{D})\\
&=\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}');\mathcal{D}')-\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D});\mathcal{D}')\\
&\quad+\frac{\ell_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}');\boldsymbol{x},y)-\ell_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D});\boldsymbol{x},y)}{n}\\
&\quad+\frac{\ell_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D});\boldsymbol{x}',y')-\ell_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}');\boldsymbol{x}',y')}{n}\\
&\le\frac{2L\cdot\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D})-\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')\right\|_2}{n},\end{aligned} \tag{18}$$

The first inequality is due to strong convexity of $\widehat{\mathcal{L}}_h^{\mathsf{OP}}$. The last inequality is due to the fact that $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')$ is the minimizer to $\min_{\boldsymbol{\theta}}\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\boldsymbol{\theta};\mathcal{D}')$, and the fact that ℓ_h is L-Lipschitz continuous. The inequality (18) implies that $\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D})-\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')\right\|_2\le\frac{2L}{n\lambda}$, $\forall\mathcal{D}\sim\mathcal{D}',\boldsymbol{b}$. Recall that function $\ell(\cdot)$ is L-Lipschitz continuous in $\boldsymbol{\theta}$, thus

$$\ell(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}))-\ell(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}'))\le L\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D})-\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')\right\|_2\le\frac{2L^2}{n\lambda},\ \forall\mathcal{D}\sim\mathcal{D}',\boldsymbol{b}. \tag{19}$$

Applying uniform stability Lemma 7, we get

$$\mathbb{E}_{\boldsymbol{b},\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})\right]\le\frac{2L^2}{n\lambda}. \tag{20}$$

2) **(Shrinking ERM)** We now fix a dataset $\mathcal{D}$ and a noise vector $\boldsymbol{b}$, and come to bound $\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})-\widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)$. By strong convexity of $\widehat{\mathcal{L}}_h^{\mathsf{OP}}$ and Cauchy's inequality, we have

$$\lambda\left\|\widehat{\boldsymbol{\theta}}_h^{\#}-\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right\|_2^2\le\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\#})-\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})=\widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\#})-\widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})+\frac{\boldsymbol{b}^\top\widehat{\boldsymbol{\theta}}_h^{\#}}{n}-\frac{\boldsymbol{b}^\top\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}}{n}\le\frac{\left\|\boldsymbol{b}\right\|_2\left\|\widehat{\boldsymbol{\theta}}_h^{\#}-\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right\|_2}{n},$$

which gives $\left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}\right\|_2 \le \frac{\|\boldsymbol{b}\|_2}{n\lambda}$. We further notice that

$$\begin{aligned}
\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*) &= \left[\widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}) - \lambda\left\|\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}\right\|_2^2\right] - \left[\widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta}^*) - \lambda\|\boldsymbol{\theta}^*\|_2^2\right] \\
&\le \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}) - \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\#}) + \lambda\left(\|\boldsymbol{\theta}^*\|_2^2 - \left\|\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}\right\|_2^2\right) \\
&\le \left[\widehat{\mathcal{L}}_h^{\mathrm{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}) - \frac{\boldsymbol{b}^\top\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}}{n}\right] - \left[\widehat{\mathcal{L}}_h^{\mathrm{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}) - \frac{\boldsymbol{b}^\top\widehat{\boldsymbol{\theta}}_h^{\#}}{n}\right] + \lambda\|\boldsymbol{\theta}^*\|_2^2 \\
&\le \frac{\|\boldsymbol{b}\|_2 \cdot \left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}\right\|_2}{n} + \lambda\|\boldsymbol{\theta}^*\|_2^2 \\
&\le \frac{\|\boldsymbol{b}\|_2^2}{n^2\lambda} + \lambda\|\boldsymbol{\theta}^*\|_2^2, \qquad (21)
\end{aligned}$$

where the first inequality is from $\widehat{\boldsymbol{\theta}}_h^{\#}$ is the minimizer to $\widehat{\mathcal{L}}_h^{\#}$; the second inequality is by plugging minimizer $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}$ into the second bracket, and dropping a negative term $-\lambda\left\|\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}\right\|_2^2$. The reasoning here holds for any $\mathcal{D}$ and $\boldsymbol{b}$, and therefore we have

$$\mathbb{E}_{\mathcal{D},\mathrm{OP}}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] \le \frac{\mathbb{E}_{\boldsymbol{b}}\left[\|\boldsymbol{b}\|_2^2\right]}{n^2\lambda} + \lambda\|\boldsymbol{\theta}^*\|_2^2 \le \frac{d\sigma^2}{n^2\lambda} + \lambda B_\theta^2. \qquad (22)$$

3) **(Convolution approximation error)** We first notice that the population-level approxiamtion error is

$$\mathcal{L}_h(\boldsymbol{\theta}^*) - \mathcal{L}(\boldsymbol{\theta}^*) = \mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}\left[c_h(y - \boldsymbol{\theta}^{*\top}\boldsymbol{x}) - c(y - \boldsymbol{\theta}^{*\top}\boldsymbol{x})\right] =: \mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{\epsilon(\boldsymbol{x})}\left[g_h(\epsilon(\boldsymbol{x}))\right]\right].$$

From the proof of part 3 of Lemma 5, we know that function $g_h(u)$ has a closed-form (12) and can be upper bounded as:

$$g_h(u) = h\int_{|u|/h}^{\infty} vK(v)\,dv + \frac{1}{2}\left(u\cdot\int_{-u/h}^{u/h} K(v)\,dv - |u|\right) \le h\int_{|u|/h}^{\infty} vK(v)\,dv. \quad \forall u\in\mathbb{R}$$

since the term in the parenthesis is negative. Therefore, the population-level convolution approximation error can be upper bounded as, $\forall \boldsymbol{x}\in\mathcal{X}$,

$$\begin{aligned}
\mathbb{E}_{\epsilon(\boldsymbol{x})}\left[g_h(\epsilon(\boldsymbol{x}))\right] &\le h\int_{-\infty}^{\infty}\int_{|u|/h}^{\infty} vK(v)\,dv\,dF_{\epsilon|\boldsymbol{x}}(u) && (F_{\epsilon|\boldsymbol{x}}(u) \text{ is the conditional CDF of } \epsilon(\boldsymbol{x})) \\
&= h\int_0^{\infty}\int_{-vh}^{vh} vK(v)\,dF_{\epsilon|\boldsymbol{x}}(u)\,dv && (\text{interchangeable, by Fubini's Theorem}) \\
&= h\int_0^{\infty} vK(v)\cdot\left(F_{\epsilon|\boldsymbol{x}}(vh) - F_{\epsilon|\boldsymbol{x}}(-vh)\right)dv.
\end{aligned}$$

Since we assume $\epsilon(\boldsymbol{x})$'s CDF $F_{\epsilon|\boldsymbol{x}}$ is Lipschitz continuous, i.e., there exists a constant $s\ge 0$ such that

$$F_{\epsilon|\boldsymbol{x}}(vh) - F_{\epsilon|\boldsymbol{x}}(-vh) \le 2svh.$$

Under this assumption, we have

$$\begin{aligned}
\mathbb{E}_{\epsilon(\boldsymbol{x})}\left[g_h(\epsilon(\boldsymbol{x}))\right] &\le h^2\cdot 2s\int_0^{\infty} v^2K(v)\,dv \\
&= h^2\cdot s\int_{-\infty}^{\infty}|v|^2K(v)\,dv =: h^2\cdot M, \quad \forall h>0,
\end{aligned}$$

where $M := s\kappa_2$ is a constant that depends on the Lipschitz parameter of $F_{\epsilon|\boldsymbol{x}}$ and kernel function $K(\cdot)$. Since the upper bound $h^2\cdot M$ holds for any $\boldsymbol{x}\in\mathcal{X}$, the overall convolution approximation error $\mathcal{L}_h(\boldsymbol{\theta}^*) - \mathcal{L}(\boldsymbol{\theta}^*)$ is therefore upper bounded as

$$\mathcal{L}_h(\boldsymbol{\theta}^*) - \mathcal{L}(\boldsymbol{\theta}^*) = \mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{\epsilon(\boldsymbol{x})}\left[g_h(\epsilon(\boldsymbol{x}))\right]\right] \le \mathbb{E}_{\boldsymbol{x}}\left[h^2M\right] = h^2M. \qquad (23)$$

Lastly, combining three parts together and plugging (20), (22), and (23) into (17), we get

$$\mathcal{R}(\mathsf{OP};\mathbb{P}) \le \frac{2L^2}{n\lambda} + \frac{d\sigma^2}{n^2\lambda} + \lambda B_\theta^2 + h^2 M. \tag{24}$$

Set $\lambda = \frac{1}{B_\theta}\sqrt{\frac{2L^2}{n} + \frac{d\sigma^2}{n^2}}$ and $h = \frac{2\overline{K}B_x^2}{\lambda n\varepsilon}$, we get

$$\mathcal{R}(\mathsf{OP};\mathbb{P}) \le 2B_\theta\sqrt{\frac{2L^2}{n} + \frac{d\sigma^2}{n^2}} + \frac{4\overline{K}^2 B_x^4 M B_\theta^2}{2L^2 n\varepsilon^2 + d\sigma^2\varepsilon^2}.$$

To ensure the first term on the r.h.s. dominates the second term, it suffices to have $2B_\theta\sqrt{\frac{2L^2}{n} + \frac{d\sigma^2}{n^2}} \ge \frac{4\overline{K}^2 B_x^4 M B_\theta^2}{2L^2 n\varepsilon^2 + d\sigma^2\varepsilon^2}$. A sufficient condition to make it true is $\varepsilon \ge \frac{\overline{K}s\kappa_2}{\sqrt{\bar{r}^3}} \cdot \sqrt{\frac{B_\theta}{B_x}} \cdot \frac{1}{\sqrt[4]{n}}$. Consequently,

$$\begin{aligned}
\mathcal{R}(\mathsf{OP};\mathbb{P}) &\le 4B_\theta\sqrt{\frac{2L^2}{n} + \frac{d\sigma^2}{n^2}} \\
&= 4B_\theta B_x \bar{r}\sqrt{\frac{2}{n} + \frac{d(8\ln(1/\delta) + 4\varepsilon)}{n^2\varepsilon^2}} \\
&\le 4B_\theta B_x \bar{r}\sqrt{\frac{2}{n} + \frac{16d\ln(1/\delta)}{n^2\varepsilon^2}} && \text{(since } \delta \le e^{-\varepsilon/2} \Leftrightarrow 4\varepsilon \le 8\ln(1/\delta)\text{)} \\
&= \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right),
\end{aligned}$$

if $\varepsilon \ge \frac{\overline{K}s\kappa_2}{\sqrt{\bar{r}^3}} \cdot \sqrt{\frac{B_\theta}{B_x}} \cdot \frac{1}{\sqrt[4]{n}}$. The desired Theorem then follows by noticing that the above analysis can be applied to any distribution $\mathbb{P}$ that satisfies stated assumptions.

□

A.6. Proof of Theorem 3

Proof. We use Gaussian Mechanism (Lemma 1) and Advanced Composition Lemma (Lemma 2) to derive a nearly same magnitude that the variance of a injected noise should be. We notice that, for any $\boldsymbol{\theta} \in \mathbb{R}^d$, the Global Sensitivity of the empirical gradient function $\nabla\widehat{\mathcal{L}}_h$ is

$$\mathrm{GS} = \sup_{\mathcal{D}\sim\mathcal{D}'} \left\|\nabla\widehat{\mathcal{L}}_h(\boldsymbol{\theta},\mathcal{D}) - \nabla\widehat{\mathcal{L}}_h(\boldsymbol{\theta},\mathcal{D}')\right\|_2 \le \frac{2}{n}\sup_{\boldsymbol{x},y}\|\nabla\ell_h(\boldsymbol{\theta};\boldsymbol{x},y)\|_2 \le \frac{2L}{n}. \tag{25}$$

Therefore, by Gaussian Mechanism, setting the variance of injected noise as $\sigma^2 = \frac{8L^2\ln(1.25/\delta_0)}{n^2\varepsilon_0^2}$ ensures (ε_0,δ_0)-DP for each iteration. By Advanced Composition Lemma (Lemma 2), to ensure an adaptive algorithm with T iterations to be (ε,δ)-DP, it suffices to set $\varepsilon_0 = \frac{\varepsilon}{2\sqrt{2T\ln(2/\delta)}}$ and $\delta_0 = \frac{\delta}{2T}$. Substituting these values into the expression of σ^2, we obtain $\sigma^2 = \frac{64TL^2\ln(2.5T/\delta)\ln(2/\delta)}{n^2\varepsilon^2}$. The derived variance σ^2 nearly matches the magnitude of that of injected noise in Algorithm GP.

To get rid of dependency on $\ln T$ and further reduce the constant from 64 to 8, we can utilize Moments Accountant method (Abadi et al. 2016) or Rényi differential privacy (Mironov 2017) to track security loss more carefully, see for example Smith and Ullman (2021) Lemma 1.1, where they show that $\sigma^2 = \frac{8L^2T\ln(1/\delta)}{n^2\varepsilon^2}$ is enough to ensure (ε,δ)-DP.

□

A.7. Proof of Theorem 4

Proof. We analyze the algorithm's performance without projection, which gives a looser upper bound. We follow *uniform stability* and *shrinking ERM* framework to complete our proof. The regret can be decomposed into two parts,

$$\mathcal{R}(\mathsf{GP};\mathbb{P}) = \mathbb{E}_{\mathsf{GP},\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}};\mathbb{P}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}};\mathcal{D})\right] + \mathbb{E}_{\mathsf{GP},\mathcal{D}}\left[\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*)\right], \tag{26}$$

where the expectation $\mathbb{E}_{\mathsf{GP}}[\cdot]$ means taking expectation over algorithm's randomness, i.e., sampling noise vectors $\{\boldsymbol{w}_t\}_{t=1}^T$.

During we derive an upper bound for the first term in (26), for notation brevity, the superscribe ^{GP} is omitted, and we use $\widehat{\boldsymbol{\theta}}_{h,t}$ and $\widehat{\boldsymbol{\theta}}'_{h,t}$ to represent vectors in t-th iteration trained on neighboring datasets $\mathcal{D}$ and $\mathcal{D}'$, respectively. Their different data point is indexed by k, and let $\mathcal{D}^{-k}$, $\mathcal{D}'^{-k}$ denote datasets with $n-1$ data points by removing k-th data point. So $\mathcal{D}^{-k}$ and $\mathcal{D}'^{-k}$ are identical. We first conjecture that, if step size $\eta \le 2/\beta$, then

$$\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2 \le \frac{2L\eta t}{n}, \quad \forall t = 1,\dots,T. \tag{27}$$

This conjecture can be proved by induction. When $t=1$, by the setting of initial points, obviously it is true. Then suppose it is true for t-th iteration, it remains to check $(t+1)$-th iteration,

$$\begin{aligned}
\left\|\widehat{\boldsymbol{\theta}}_{h,t+1} - \widehat{\boldsymbol{\theta}}'_{h,t+1}\right\|_2 &= \left\|\left(\widehat{\boldsymbol{\theta}}_{h,t} - \eta(\nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t};\mathcal{D}) + \boldsymbol{w}_t)\right) - \left(\widehat{\boldsymbol{\theta}}'_{h,t} - \eta(\nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}'_{h,t};\mathcal{D}') + \boldsymbol{w}_t)\right)\right\|_2 \\
&\le \left\|\left(\widehat{\boldsymbol{\theta}}_{h,t} - \eta\frac{(n-1)}{n}\nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t};\mathcal{D}^{-k})\right) - \left(\widehat{\boldsymbol{\theta}}'_{h,t} - \eta\frac{n-1}{n}\nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}'_{h,t};\mathcal{D}'^{-k})\right)\right\|_2 \\
&\quad + \frac{\eta}{n}\left\|\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t};\boldsymbol{x}_k,y_k) - \nabla\ell_h(\widehat{\boldsymbol{\theta}}'_{h,t};\boldsymbol{x}'_k,y'_k)\right\|_2 \\
&\le \left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2 + \frac{2L\eta}{n} \\
&\le \frac{2L\eta(t+1)}{n},
\end{aligned}$$

where the second inequality follows the non-expansiveness property of gradient update rule from Lemma 3.6 in Hardt et al. (2016). Therefore, our conjecture (27) is correct. With this conjecture, we are able to establish uniform stability:

$$\ell(\widehat{\boldsymbol{\theta}}_h;\boldsymbol{x},y) - \ell(\widehat{\boldsymbol{\theta}}'_h,\boldsymbol{x},y) \le L\cdot\left\|\frac{1}{T}\sum_{t=1}^T\left(\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right)\right\|_2 \le \frac{L}{T}\sum_{t=1}^T\frac{2L\eta t}{n} = \frac{L^2\eta(1+T)}{n}, \quad \forall \boldsymbol{x},y.$$

Hence, by uniform stability Lemma 7, we have

$$\mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n,\mathsf{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}})\right] \le \frac{L^2\eta(1+T)}{n} \tag{28}$$

It remains to control the second term in (26). Let us fix a dataset $\mathcal{D}$. By the relationship in Lemma 5 part 3 and convexity of $\widehat{\mathcal{L}}_h(\cdot)$, we can show

$$\begin{aligned}
\mathbb{E}_{\mathsf{GP}}\left[\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*)\right] &\le \mathbb{E}_{\mathsf{GP}}\left[\widehat{\mathcal{L}}_h\left(\frac{1}{T}\cdot\sum_{t=1}^T\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}\right) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] + \frac{1}{2}h\kappa_1 \\
&\le \frac{1}{T}\cdot\mathbb{E}_{\mathsf{GP}}\left[\sum_{t=1}^T\left(\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right)\right] + \frac{1}{2}h\kappa_1 \\
&\le \frac{1}{T}\cdot\mathbb{E}_{\mathsf{GP}}\left[\sum_{t=1}^T\left\langle\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}} - \boldsymbol{\theta}^*, \nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}})\right\rangle\right] + \frac{1}{2}h\kappa_1.
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{T} \cdot \mathbb{E}_{\mathsf{GP}} \left[\sum_{t=1}^{T} \left\langle \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}} - \boldsymbol{\theta}^*, \nabla \widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}) + \boldsymbol{w}_t \right\rangle \right] + \frac{1}{2} h \kappa_1 \\
&\leq \frac{1}{T} \cdot \mathbb{E}_{\mathsf{GP}} \left[\frac{\|\boldsymbol{\theta}^*\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \left\| \nabla \widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}) + \boldsymbol{w}_t \right\|_2^2 \right] + \frac{1}{2} h \kappa_1, \\
&\leq \frac{1}{T} \cdot \mathbb{E}_{\mathsf{GP}} \left[\frac{\|\boldsymbol{\theta}^*\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \left\| \nabla \widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}) \right\|_2^2 \right] + \frac{\eta}{2T} \sum_{t=1}^{T} \mathbb{E}_{\mathsf{GP}} \left[\|\boldsymbol{w}_t\|_2^2 \right] + \frac{1}{2} h \kappa_1, \\
&\leq \frac{\|\boldsymbol{\theta}^*\|_2^2}{2T\eta} + \frac{\eta L^2}{2} + \frac{\eta d \sigma^2}{2} + \frac{1}{2} h \kappa_1, \qquad (29)
\end{aligned}
$$

where the second and third lines are due to convexity of $\widehat{\mathcal{L}}_h$; the fourth line comes from the fact that $\boldsymbol{w}_t$ is drawn from a zero-mean Gaussian distribution and is independent of $\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}$; the fifth line follows from a classic gradient descent analysis (Shalev-Shwartz and Ben-David 2014, Lemma 14.1) and from the gradient descent update rule in our algorithm; the sixth line comes again from the zero-mean $\boldsymbol{w}_t$ being independent; the last line is due to L-Lipschitz continuity of $\widehat{\mathcal{L}}_h$ and the fact that $\left\| \frac{\boldsymbol{w}_t}{\sigma} \right\|_2^2$ is a χ^2 random variable with d-degree.

Thus, plugging (28) and (29) into (26), and then letting $h = \eta \overline{K} B_x^2$ (which ensures $\eta\beta = 1$, making our conjecture (27) to be true), we obtain

$$
\mathcal{R}(\mathsf{GP}; \mathbb{P}) \leq \frac{L^2(1+T)\eta}{n} + \frac{\|\boldsymbol{\theta}^*\|_2^2}{2T\eta} + \frac{\eta L^2}{2} + \frac{\eta d \sigma^2}{2} + \frac{\eta \overline{K} B_x^2 \kappa_1}{2} \qquad (30)
$$

It is safe to replace $\|\boldsymbol{\theta}^*\|_2^2$ with its upper bound B_θ^2. Then optimize the r.h.s over η. Specifically, take $\eta = \sqrt{\frac{B_\theta^2}{2T} / \left(\frac{L^2(1+T)}{n} + \frac{L^2}{2} + \frac{d\sigma^2}{2} + \frac{\kappa_1 \overline{K} B_x^2}{2} \right)}$, and replace T with n^2. Plugging these values and stated σ into (30), we obtain

$$
\begin{aligned}
\sup_{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})} \mathcal{R}(\mathsf{GP}; \mathbb{P}) &\leq B_\theta \sqrt{\frac{2L^2}{n} + \frac{8dL^2 \ln(1/\delta)}{n^2 \varepsilon^2} + \frac{L^2 + \kappa_1 \overline{K} B_x^2}{n^2} + \frac{2L^2}{n^3}} \\
&= B_\theta B_x \bar{r} \sqrt{\frac{4}{n} + \frac{8d \ln(1/\delta)}{n^2 \varepsilon^2} + \frac{1 + \kappa_1 \overline{K} / \bar{r}^2}{n^2}} && (\text{by } L := B_x \bar{r}) \\
&\leq B_\theta B_x \bar{r} \sqrt{\frac{8}{n} + \frac{8d \ln(1/\delta)}{n^2 \varepsilon^2}} && (\text{since } n \geq \frac{1 + \kappa_1 \overline{K} / \bar{r}^2}{4}) \\
&= \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon} \right),
\end{aligned}
$$

which completes the proof (The sup is by noticing that the above analysis applies to any distribution $\mathbb{P}$).

□

A.8. Proof of Theorem 5

Proof. our stochastic gradient perturbation is a variant of Algorithm 1 in Bassily et al. (2014) by removing a multiplicative factor n in their update rule. Therefore, the variance of injected noise in our algorithm is $1/n^2$-times smaller, and can still guarantee (ε, δ)-DP.

□

A.9. Proof of Theorem 6

We analyze the algorithm's performance without projection, which gives a looser upper bound.

Proof. The proof follows exactly the same idea for the proof of Theorem 4. Thus, we omit most details here, and only point out key steps. We first decomposed the regret into two parts,

$$\mathcal{R}(\text{SGP};\mathbb{P}) = \mathbb{E}_{\text{SGP},\mathcal{D}\sim\mathbb{P}^n}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\text{SGP}};\mathbb{P}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\text{SGP}};\mathcal{D})\right] + \mathbb{E}_{\text{SGP},\mathcal{D}\sim\mathbb{P}^n}\left[\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\text{SGP}}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*)\right], \tag{31}$$

where the expectation $\mathbb{E}_{\text{SGP}}[\cdot]$ means taking expectation over algorithm's randomness. Note that there are two types of randomness, one randomness comes from sampling record $(\boldsymbol{x}_{(t)}, y_{(t)})$ for evaluating gradients, and another randomness is from sampling noise vector $\boldsymbol{w}_t$.

We first control the first term in (31) by using uniform stability Lemma 7. Because of L-Lipschitz continuity of ℓ, we have

$$\mathbb{E}_{\text{SGP}}\left[\ell(\widehat{\boldsymbol{\theta}}^{\text{SGP}}(\mathcal{D});\boldsymbol{x},y) - \ell(\widehat{\boldsymbol{\theta}}_h^{\text{SGP}}(\mathcal{D}');\boldsymbol{x},y)\right] \le L\cdot\mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_h^{\text{SGP}}(\mathcal{D}) - \widehat{\boldsymbol{\theta}}_h^{\text{SGP}}(\mathcal{D}')\right\|_2\right], \quad \forall \boldsymbol{x}, y, \mathcal{D}\sim\mathcal{D}';$$

therefore, it suffices to control the expected deviation between returned vectors trained on two neighboring datasets. For notation brevity, the superscribe ^{SGP} and dependences on $\mathcal{D}$ are omitted throughout this proof, and we use $\widehat{\boldsymbol{\theta}}_h := \widehat{\boldsymbol{\theta}}_h^{\text{SGP}}(\mathcal{D})$ and $\widehat{\boldsymbol{\theta}}'_h := \widehat{\boldsymbol{\theta}}_h^{\text{SGP}}(\mathcal{D}')$. And we use $\widehat{\boldsymbol{\theta}}_{h,t}$ and $\widehat{\boldsymbol{\theta}}'_{h,t}$ to represent vectors in t-th iteration trained on $\mathcal{D}$ and $\mathcal{D}'$, respectively. Firstly, we conjecture that, when step size $\eta \le 2/\beta$, we have

$$\mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2\right] \le \frac{2L\eta t}{n}, \quad \forall t = 1,\ldots,n^2. \tag{32}$$

This can be proved by induction. When $t=1$, by the setting of initial points $\widehat{\boldsymbol{\theta}}_{h,1} = \widehat{\boldsymbol{\theta}}'_{h,1} = \mathbf{0}$, obviously it is true. Then suppose that (32) is true for t-th iteration, it remains to check $(t+1)$-th iteration. Let us fix a sequence of noise vector $\{\boldsymbol{w}_t\}_{t=1}^{n^2}$, then,

$$\begin{aligned}\left\|\widehat{\boldsymbol{\theta}}_{h,t+1} - \widehat{\boldsymbol{\theta}}'_{h,t+1}\right\|_2 &= \left\|\left(\widehat{\boldsymbol{\theta}}_{h,t} - \eta(\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t};\boldsymbol{x}_{(t)},y_{(t)}) + \boldsymbol{w}_t)\right) - \left(\widehat{\boldsymbol{\theta}}'_{h,t} - \eta(\nabla\ell_h(\widehat{\boldsymbol{\theta}}'_{h,t};\boldsymbol{x}'_{(t)},y'_{(t)}) + \boldsymbol{w}_t)\right)\right\|_2 \\ &= \left\|\left(\widehat{\boldsymbol{\theta}}_{h,t} - \eta\cdot\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t};\boldsymbol{x}_{(t)},y_{(t)})\right) - \left(\widehat{\boldsymbol{\theta}}'_{h,t} - \eta\cdot\nabla\ell_h(\widehat{\boldsymbol{\theta}}'_{h,t};\boldsymbol{x}'_{(t)},y'_{(t)})\right)\right\|_2\end{aligned} \tag{33}$$

Notice that $(\boldsymbol{x}_{(t)}, y_{(t)})$ and $(\boldsymbol{x}'_{(t)}, y'_{(t)})$ are uniformly drawn from $\mathcal{D}$ and $\mathcal{D}'$, it implies that, with probability $1/n$, we have $(\boldsymbol{x}_{(t)}, y_{(t)}) \ne (\boldsymbol{x}'_{(t)}, y'_{(t)})$; and with probability $1-1/n$, we have $(\boldsymbol{x}_{(t)}, y_{(t)}) = (\boldsymbol{x}'_{(t)}, y'_{(t)})$. When not equal, (33) $\le \left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2 + 2\eta L$ because of triangular inequality. When equal, gradient update rule is 1-expansive, i.e., (33) $\le \left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2$ (Hardt et al. 2016, Lemma 3.6). Consequently, taking expectation over the randomness of algorithm, we have

$$\begin{aligned}\mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t+1} - \widehat{\boldsymbol{\theta}}'_{h,t+1}\right\|_2\right] &\le \left(1-\frac{1}{n}\right)\mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2\right] + \frac{1}{n}\mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2 + 2\eta L\right] \\ &\le \mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2\right] + \frac{2\eta L}{n} \\ &\le \frac{2\eta L(t+1)}{n}.\end{aligned}$$

Therefore, by induction, our conjecture (32) is true. Recall that $\widehat{\boldsymbol{\theta}}_h = \frac{1}{T}\sum_{t=1}^T \widehat{\boldsymbol{\theta}}_{h,t}$ is the averaged vector over all iterations, thus,

$$\mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_h - \widehat{\boldsymbol{\theta}}'_h\right\|_2\right] \le \frac{1}{T}\sum_{t=1}^T \mathbb{E}_{\text{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2\right] \le \frac{1}{T}\sum_{t=1}^T \frac{2\eta L t}{n} = \frac{(1+T)\eta L}{n}$$

Thus, by uniform stability Lemma 7, the above inequality implies:

$$\mathbb{E}_{\mathsf{SGP},\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}})\right] \leq \frac{(1+T)\eta L^2}{n}. \tag{34}$$

Then, we come to bound the second term in (31). By convexity of $\widehat{\mathcal{L}}_h(\cdot)$, when we fix algorithm's sampling noise randomness SGP^{sn} and sampling record randomness SGP^{sr}, following first three lines for (29), we can show

$$\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*) \leq \frac{1}{T}\sum_{t=1}^{T}\left[\left\langle \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}} - \boldsymbol{\theta}^*, \nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}})\right\rangle\right] + \frac{1}{2}h\kappa_1. \tag{35}$$

Since record $(\boldsymbol{x}_{(t)}, y_{(t)})$ is uniformly sampled over $\mathcal{D}$ and independent across iterations, the expected gradient over sampling record (sr) randomness is unbiased, i.e., the relationship $\mathbb{E}_{\mathsf{SGP}^{sr}}\left[\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}; \boldsymbol{x}_{(t)}, y_{(t)})\right] = \nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}), \forall t$ is true. Thus, substituting the relationship into (35) gives

$$\begin{aligned}
\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*) &\leq \mathbb{E}_{\mathsf{SGP}^{sr}}\left[\frac{1}{T}\sum_{t=1}^{T}\left\langle \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}} - \boldsymbol{\theta}^*, \nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}; \boldsymbol{x}_{(t)}, y_{(t)})\right\rangle\right] + \frac{1}{2}h\kappa_1 \\
&\leq \frac{1}{T}\cdot\mathbb{E}_{\mathsf{SGP}^{sr}}\left[\frac{\|\boldsymbol{\theta}^*\|_2^2}{2\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\left\|\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}; \boldsymbol{x}_{(t)}, y_{(t)}) + \boldsymbol{w}_t\right\|_2^2\right] \\
&\quad - \mathbb{E}_{\mathsf{SGP}^{sr}}\left[\frac{1}{T}\sum_{t=1}^{T}\left\langle \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}} - \boldsymbol{\theta}^*, \boldsymbol{w}_t\right\rangle\right] + \frac{1}{2}h\kappa_1,
\end{aligned}$$

Since $\boldsymbol{w}_t$ are drawn from a zero-mean multivariate Gaussian, taking expectation over $\{\boldsymbol{w}_t\}_t$ on both sides removes the second term on the r.h.s. And following the same argument for (29), we obtain

$$\mathbb{E}_{\mathsf{SGP}}\left[\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*)\right] \leq \frac{\|\boldsymbol{\theta}^*\|_2^2}{2T\eta} + \frac{\eta L^2}{2} + \frac{\eta d\sigma^2}{2} + \frac{1}{2}h\kappa_1. \tag{36}$$

Therefore, substituting (34) and (36) into (31), and then plugging in $h = \frac{\eta\overline{K}B_x^2}{2}$ results in

$$\mathcal{R}(\mathsf{SGP};\mathbb{P}) \leq \frac{(1+T)\eta L^2}{n} + \frac{\|\boldsymbol{\theta}^*\|_2^2}{2T\eta} + \frac{\eta L^2}{2} + \frac{\eta d\sigma^2}{2} + \frac{\eta\kappa_1\overline{K}B_x^2}{4}. \tag{37}$$

Lastly, taking $\eta^* = \sqrt{\frac{\|B_\theta\|_2^2}{2T} / \left(\frac{(1+T)L^2}{n} + \frac{L^2}{2} + \frac{d\sigma^2}{2} + \frac{\kappa_1\overline{K}B_x^2}{4}\right)}$ gives the desired upper bound:

$$\begin{aligned}
\sup_{\mathbb{P}\in(\mathcal{X}\times\mathcal{Y})}\mathcal{R}(\mathsf{SGP}_h;\mathbb{P}) &\leq \sqrt{2}B_\theta\sqrt{\frac{L^2}{n} + \frac{8dL^2\ln(1/\delta)}{n^2\varepsilon^2} + \frac{L^2+\kappa_1\overline{K}B_x^2}{2n^2} + \frac{L}{n^3}} \\
&\leq \sqrt{2}B_\theta B_x\bar{r}\sqrt{\frac{2}{n} + \frac{8d\ln(1/\delta)}{n^2\varepsilon^2} + \frac{1+\kappa_1\overline{K}/\bar{r}^2}{2n^2}} \\
&\leq \sqrt{2}B_\theta B_x\bar{r}\sqrt{\frac{4}{n} + \frac{8d\ln(1/\delta)}{n^2\varepsilon^2}} \qquad \left(\text{since } n \geq \frac{1+\kappa_1\overline{K}/\bar{r}^2}{n}\right) \\
&= \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right),
\end{aligned}$$

where the sup operator is by noticing that the above analysis can be applied to any distributions.

□

A.10. Proof of Theorem 7

Proof. The proof includes four steps. In first three steps, we gradually find out better lower bounds for the minimax risk of an (ε, δ)-DP algorithm $\mathsf{A} : \mathcal{X}^n \times \mathcal{Y}^n \to \mathcal{C}$, where $\mathcal{C} := \mathcal{B}(B_\theta)$. These lower bounds finally lead to a minimax risk of an (ε', δ')-DP d-dimensional classification problem. The sample complexity of DP classification problems for achieving a certain accuracy is well studied in DP literature, from which we can derive the accuracy under a specific sample size. Consequently, the accuracy provides an overall lower bound.

1) **Step 1: relax the feasible region of the** inf **problem & restrict the feasible region of the** sup **problem**

We first lower bound the minimax risk by relaxing relax the feasible region of the inf problem. Note that, in the minimax risk $\inf_{\mathsf{A}\in\mathcal{F}_{\varepsilon,\delta}} \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathsf{A};\mathbb{P})$, the inf is taken over $\mathcal{F}_{\varepsilon,\delta}$, a set of mappings from $\mathcal{X}^n \times \mathcal{Y}^n$ to an Euclidean ball $\mathcal{C} = \mathcal{B}(B_\theta) := \{\boldsymbol{\theta} \in \mathbb{R}^d : \|\boldsymbol{\theta}\|_2 \le B_\theta\}$. If we relax the output space from an Euclidean ball $\mathcal{C}$ to a superset $\mathcal{C}_\infty := \{\boldsymbol{\theta} \in \mathbb{R}^d : \|\boldsymbol{\theta}\|_\infty \le B_\theta\}$ a box set with $\|\boldsymbol{\theta}\|_\infty := \max_{j=1,\dots,d} |\theta_j|$, the corresponding feasible set of mappings $\mathcal{F}^\infty_{\varepsilon,\delta}$, which includes all DP mappings from $\mathcal{X}^n \times \mathcal{Y}^n$ to $\mathcal{C}_\infty$, is therefore much larger. Thus, we get a lower bound by relaxing the feasible region of the inf problem:

$$\inf_{\mathsf{A}\in\mathcal{F}_{\varepsilon,\delta}} \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathsf{A};\mathbb{P}) \ge \inf_{\mathsf{A}\in\mathcal{F}^\infty_{\varepsilon,\delta}} \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathsf{A};\mathbb{P}) \tag{38}$$

We further discover a lower bound by restricting the feasible region of the sup problem. Specifically, we restrict $\mathbb{P}$ to a smaller set of probability measures $\mathbb{P}_\mathcal{S}$ where $\mathcal{S} \in \mathcal{X}^n \times \mathcal{Y}^n$ is a n-samples dataset with a specific structure, and $\mathbb{P}_\mathcal{S}$ is a distribution generated from the given dataset $\mathcal{S}$ by assigning each sample probability $1/n$:

$$(38) \ge \inf_{\mathsf{A}\in\mathcal{F}^\infty_{\varepsilon,\delta}} \sup_{\mathbb{P}_\mathcal{S}} \mathcal{R}(\mathsf{A};\mathbb{P}_\mathcal{S})$$

The inequality holds as $\mathcal{S}$ will be restricted to a specific structure to be elaborated later. As a result of that, $\mathbb{P}_\mathcal{S}$ is optimized over a more restrictive region, leading to a smaller objective value. Therefore, the r.h.s in above inequality is a lower bound.

The dataset $\mathcal{S}$ is constructed as follows: let contextual matrix $\boldsymbol{X} := \{\boldsymbol{x}_i\}_{i=1}^n$ be an $n \times d$ matrix, where $\boldsymbol{x}_i \in \mathcal{X} := \{\frac{-B_x}{\sqrt{d}}, \frac{B_x}{\sqrt{d}}\}^d$. Denote the average vector by $\overline{\boldsymbol{x}} = \frac{1}{n}\sum_{i=1}^n \boldsymbol{x}_i$, and its sign vector by $sign(\overline{\boldsymbol{x}}) := (sign(\bar{x}_1), \dots, sign(\bar{x}_d)) \in \{-1, 1\}^d$. Then, the dataset $\mathcal{S}$ is constructed as

$$\mathcal{S} = (B_\theta \cdot \boldsymbol{X} sign(\overline{\boldsymbol{x}}), \boldsymbol{X}).$$

By our construction, dataset $\mathcal{S}$ depends only on contextual matrix $\boldsymbol{X}$, therefore, the $\sup_{\mathbb{P}_\mathcal{S}}$ is actually optimizing over $\boldsymbol{X}$. We will use $\mathcal{S}(\boldsymbol{X})$ to indicate such dependence when necessary. We conclude this step by writing down the lower bound explicitly for later reference:

$$\begin{aligned}\inf_{\mathsf{A}\in\mathcal{F}_{\varepsilon,\delta}} \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathsf{A};\mathbb{P}) &\ge \inf_{\mathsf{A}\in\mathcal{F}^\infty_{\varepsilon,\delta}} \sup_{\mathbb{P}_\mathcal{S}} \mathcal{R}(\mathsf{A};\mathbb{P}_\mathcal{S}) \\ &= \inf_{\mathsf{A}\in\mathcal{F}^\infty_{\varepsilon,\delta}} \sup_{\mathbb{P}_\mathcal{S}} \mathbb{E}_{\mathsf{A},\mathcal{D}\sim\mathbb{P}^n_\mathcal{S}} [\mathcal{L}(\mathsf{A}(\mathcal{D});\mathbb{P}_\mathcal{S})] - \min_{\boldsymbol{\theta}\in\mathcal{C}_\infty} \mathcal{L}(\boldsymbol{\theta};\mathbb{P}_\mathcal{S}),\end{aligned} \tag{39}$$

where $\mathcal{R}(\mathsf{A};\mathbb{P}_\mathcal{S})$ is the excess generalization risk of A under distribution $\mathbb{P}_\mathcal{S}$.

2) **Step 2: reduce excess generalization risk to excess empirical risk**

We notice that, since $\mathbb{P}_{\mathcal{S}}$ is an n-valued distribution generated from $\mathcal{S}$, for any $\boldsymbol{\theta}$, the population risk $\mathcal{L}(\boldsymbol{\theta};\mathbb{P}_{\mathcal{S}})$ w.r.t. $\mathbb{P}_{\mathcal{S}}$ is equal to an empirical risk $\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S})$ w.r.t. $\mathcal{S}$ by definitions of $\mathcal{L}$ and $\widehat{\mathcal{L}}$. Therefore,

$$\begin{aligned}(39) &= \inf_{\mathsf{A}\in\mathcal{F}^{\infty}_{\varepsilon,\delta}} \sup_{\mathbb{P}_{\mathcal{S}}} \mathbb{E}_{\mathsf{A},\mathcal{D}\sim\mathbb{P}^n_{\mathcal{S}}}\left[\widehat{\mathcal{L}}(\mathsf{A}(\mathcal{D});\mathcal{S})\right] - \min_{\boldsymbol{\theta}\in\mathcal{C}_\infty}\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S})\\ &= \inf_{\mathsf{A}\in\mathcal{F}^{\infty}_{\varepsilon,\delta}} \sup_{\mathbb{P}_{\mathcal{S}}} \mathbb{E}_{\mathsf{A}}\left[\mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n_{\mathcal{S}}}\left[\widehat{\mathcal{L}}(\mathsf{A}(\mathcal{D});\mathcal{S})\right]\right] - \min_{\boldsymbol{\theta}\in\mathcal{C}_\infty}\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S}) \qquad (40)\end{aligned}$$

The inner expectation in the first term is taken over $\mathcal{D}\sim\mathbb{P}^n_{\mathcal{S}}$, that is, we i.i.d. draw n samples with replacement from a given $\mathcal{S}$. Thus, we can treat "subsampling $\mathcal{D}$ from $\mathcal{S}$, then run $\mathsf{A}(\mathcal{D})$" as a new algorithm B, which takes $\mathcal{S}$ as the input and outputs an estimator $\mathsf{A}(\mathcal{D})$. By a mild revision of notations only, we have

$$(40) = \inf_{\substack{\mathsf{B}\text{: subsampling, then run A,}\\ \text{where } \mathsf{A}\in\mathcal{F}^{\infty}_{\varepsilon,\delta}}} \sup_{\mathcal{S}} \mathbb{E}_{\mathsf{B}}\left[\widehat{\mathcal{L}}(\mathsf{B}(\mathcal{S});\mathcal{S})\right] - \min_{\boldsymbol{\theta}\in\mathcal{C}_\infty}\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S}). \qquad (41)$$

It would be helpful to check whether algorithm B is DP. Following the definition of DP, we consider two neighboring dataset $\mathcal{T} := ((\boldsymbol{x}_1,y_1),\ldots,(\boldsymbol{x}_k,y_k),\cdots,(\boldsymbol{x}_n,y_n))$ and $\mathcal{T}' := ((\boldsymbol{x}_1,y_1),\ldots,(\boldsymbol{x}'_k,y'_k),\cdots,(\boldsymbol{x}_n,y_n))$ that differ in k-th sample only. Datasets $\mathcal{T}$ and $\mathcal{T}'$ here not necessarily follow the specific structure in Step 1; instead, they are conceptual here for checking if B is DP only. Let the set $\mathcal{I}\in\{1,\ldots,n\}^n$ be an index set with indices from i.i.d. sampling with replacement, and let $\mathcal{T}(\mathcal{I})$ be the resulting dataset with index set $\mathcal{I}$. Denote the number of different samples between $\mathcal{T}(\mathcal{I})$ and $\mathcal{T}'(\mathcal{I})$ by $\Delta(\mathcal{I}) := |\mathcal{T}(\mathcal{I})\backslash\mathcal{T}'(\mathcal{I})|$. As $\mathcal{T}$ and $\mathcal{T}'$ are neighboring and differ in k-th sample only, the value $\Delta(\mathcal{I})$ follows an n-trial Binomial distribution with success probability $1/n$; thus, it should be small with high probability. Specifically, we should have

$$\Pr_{\mathcal{I}}\left[\Delta(\mathcal{I})\geq z+1\right] = \Pr\left[\mathsf{Binomial}(n,1/n)\geq z+1\right] \leq \exp\left(-z^2/3\right), \quad \forall z>0,$$

where the inequality follows from multiplicative Chernoff upper tail bound. Equivalently, the above implies that $\Delta(\mathcal{I})\geq 3\sqrt{\ln(1/\gamma)}+1 := u$ with probability at most γ. Now, we are ready to check if B is DP by definition: for any subset $\mathcal{U}$ of the output space of $\mathsf{B}(\mathcal{T})$, we have

$$\begin{aligned}\Pr_{\mathsf{B}}\left[\mathsf{B}(\mathcal{T})\in\mathcal{U}\right] &\leq \Pr_{\mathsf{B}|\mathcal{I}}\left[\mathsf{B}(\mathcal{T})\in\mathcal{U}|\Delta(\mathcal{I})\leq u\right]\cdot\Pr_{\mathcal{I}}\left[\Delta(\mathcal{I})\leq u\right]+\gamma\\ &= \Pr_{\mathsf{A}|\mathcal{I}}\left[\mathsf{A}(\mathcal{T}(\mathcal{I}))\in\mathcal{U}|\Delta(\mathcal{I})\leq u\right]\cdot\Pr_{\mathcal{I}}\left[\Delta(\mathcal{I})\leq u\right]+\gamma\\ &\leq \left(e^{\Delta(\mathcal{I})\cdot\varepsilon}\Pr_{\mathsf{A}|\mathcal{I}}\left[\mathsf{A}(\mathcal{T}'(\mathcal{I}))\in\mathcal{U}|\Delta(\mathcal{I})\leq u\right]+\Delta(\mathcal{I})\delta e^{\Delta(\mathcal{I})\varepsilon}\right)\cdot\Pr_{\mathcal{I}}\left[\Delta(\mathcal{I})\leq u\right]+\gamma\\ &\leq e^{u\varepsilon}\Pr_{\mathsf{B}|\mathcal{I}}\left[\mathsf{B}(\mathcal{T}')\in\mathcal{U}|\Delta(\mathcal{I})\leq u\right]\cdot\Pr_{\mathcal{I}}\left[\Delta(\mathcal{I})\leq u\right]+(u\delta e^{u\varepsilon}+\gamma)\\ &\leq e^{\varepsilon'}\Pr_{\mathsf{B}}\left[\mathsf{B}(\mathcal{T}')\in\mathcal{U}\right]+\delta',\end{aligned}$$

where the third line follows from the fact that A is (ε,δ)-DP and Group Privacy Lemma (Vadhan 2017, Lemma 2.2). Therefore, the algorithm B is (ε',δ')-DP with $\varepsilon' := u\varepsilon$ and $\delta' := u\delta e^{u\varepsilon}+\gamma$. However, algorithm B is very restrictive as it must follow a "subsampling, then run A" framework. If we remove the framework requirement and only require $\mathsf{B}\in\mathcal{F}^{\infty}_{\varepsilon',\delta'}$ a set of (ε',δ')-DP mappings from $\mathcal{X}^n\times\mathcal{Y}^n$ to $\mathcal{C}_\infty$, we will get a lower bound to (41):

$$\begin{aligned}(41) &= \inf_{\substack{\mathsf{B}\in\mathcal{F}^{\infty}_{\varepsilon',\delta'}\text{: subsampling, then run A,}\\ \text{where } \mathsf{A}\in\mathcal{F}^{\infty}_{\varepsilon,\delta}}} \sup_{\mathcal{S}} \mathbb{E}_{\mathsf{B}}\left[\widehat{\mathcal{L}}(\mathsf{B}(\mathcal{S});\mathcal{S})\right] - \min_{\boldsymbol{\theta}\in\mathcal{C}_\infty}\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S})\\ &\geq \inf_{\mathsf{B}\in\mathcal{F}^{\infty}_{\varepsilon',\delta'}} \sup_{\mathcal{S}} \mathbb{E}_{\mathsf{B}}\left[\widehat{\mathcal{L}}(\mathsf{B}(\mathcal{S});\mathcal{S})\right] - \min_{\boldsymbol{\theta}\in\mathcal{C}_\infty}\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S}). \qquad (42)\end{aligned}$$

3) **step 3: convert excess empirical risk to DP binary classification error**

Now, we start to analyze the excess empirical risk $\mathbb{E}_{\mathsf{B}}\left[\widehat{\mathcal{L}}(\mathsf{B}(\mathcal{S});\mathcal{S})\right] - \min_{\boldsymbol{\theta}\in\mathcal{C}_\infty}\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S})$ for any given dataset $\mathcal{S}$. Recall that the dataset $\mathcal{S}$ is constructed as $(\boldsymbol{X}\cdot sign(\overline{\boldsymbol{x}})\cdot B_\theta, \boldsymbol{X})$ in Step 1. Hence, the empirical minimizer is $\widehat{\boldsymbol{\theta}} := \arg\min_{\boldsymbol{\theta}\in\mathcal{C}_\infty}\widehat{\mathcal{L}}(\boldsymbol{\theta};\mathcal{S}) = sign(\overline{\boldsymbol{x}})\cdot B_\theta \in \mathcal{C}_\infty$, and the empirical risk $\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}};\mathcal{S}) = 0$. We therefore only need to focus our attention on $\mathbb{E}_{\mathsf{B}}\left[\widehat{\mathcal{L}}(\mathsf{B}(\mathcal{S});\mathcal{S})\right]$. It is straightforward to show

$$
\begin{aligned}
\mathbb{E}_{\mathsf{B}}\left[\widehat{\mathcal{L}}(\mathsf{B}(\mathcal{S});\mathcal{S})\right] &= \mathbb{E}_{\mathsf{B}}\left[\frac{1}{n}\sum_{i=1}^{n}\left(r\cdot(y_i - \mathsf{B}(\mathcal{S})^\top \boldsymbol{x}_i)^+ + (1-r)\cdot(\mathsf{B}(\mathcal{S})^\top\boldsymbol{x}_i - y_i)^+\right)\right] \\
&\geq \min\{r, 1-r\}\mathbb{E}_{\mathsf{B}}\left[\frac{1}{n}\sum_{i=1}^{n}\left|\left(sign(\overline{\boldsymbol{x}})\cdot B_\theta - \mathsf{B}(\mathcal{S})\right)^\top \boldsymbol{x}_i\right|\right] \\
&\geq \min\{r, 1-r\}\mathbb{E}_{\mathsf{B}}\left[\left|\left(sign(\overline{\boldsymbol{x}})\cdot B_\theta - \mathsf{B}(\mathcal{S})\right)^\top \overline{\boldsymbol{x}}\right|\right] \qquad \text{(by triangular inequality)}
\end{aligned}
$$

By simple algebra, the absolute value in the expectation operator is

$$
\begin{aligned}
\left|\left(sign(\overline{\boldsymbol{x}})\cdot B_\theta - \mathsf{B}(\mathcal{S})\right)^\top \overline{\boldsymbol{x}}\right| &= \left|\sum_{j=1}^{d}\left[sign(\overline{\boldsymbol{x}})_j\cdot B_\theta - \mathsf{B}(\mathcal{S})_j\right]\cdot sign(\overline{\boldsymbol{x}})_j\,|\overline{x}_j|\right| \\
&= \left|\sum_{j=1}^{d}|\overline{x}_j|\left(B_\theta - \mathsf{B}(\mathcal{S})_j\cdot sign(\overline{\boldsymbol{x}})_j\right)\right| \\
&= \sum_{j=1}^{d}|\overline{x}_j|\left(B_\theta - \mathsf{B}(\mathcal{S})_j\cdot sign(\overline{\boldsymbol{x}})_j\right) \qquad \text{(Since } |\mathsf{B}(\mathcal{S})_j| \leq B_\theta\text{)} \\
&\geq B_\theta\cdot\sum_{j=1}^{d}|\overline{x}_j|\,\mathbb{1}\left\{sign(\overline{\boldsymbol{x}})_j \neq sign(\mathsf{B}(\mathcal{S})_j)\right\}.
\end{aligned}
$$

Putting above analysis together, we further obtain a lower bound:

$$
\begin{aligned}
(42) &\geq \min\{r, 1-r\}B_\theta\cdot\inf_{\mathsf{B}\in\mathcal{F}^\infty_{\varepsilon',\delta'}}\sup_{\mathcal{S}}\mathbb{E}_{\mathsf{B}}\left[\sum_{j=1}^{d}|\overline{x}_j|\,\mathbb{1}\left\{sign(\overline{\boldsymbol{x}})_j \neq sign(\mathsf{B}(\mathcal{S})_j)\right\}\right] \\
&= \min\{r, 1-r\}B_\theta\cdot\inf_{\mathsf{B}\in\mathcal{F}^\infty_{\varepsilon',\delta'}}\sup_{\boldsymbol{X}\in\mathcal{X}^n}\mathbb{E}_{\mathsf{B}}\left[\sum_{j=1}^{d}|\overline{x}_j|\,\mathbb{1}\left\{sign(\overline{\boldsymbol{x}})_j \neq sign(\mathsf{B}(\mathcal{S}(\boldsymbol{X}))_j)\right\}\right] \\
&\geq \min\{r, 1-r\}B_\theta\cdot\inf_{\substack{\mathsf{C}:\mathcal{X}^n\to\{-1,1\}^d;\\ \mathsf{C}\text{ is }(\varepsilon',\delta')\text{-DP}}}\sup_{\boldsymbol{X}\in\mathcal{X}^n}\mathbb{E}_{\mathsf{C}}\left[\sum_{j=1}^{d}|\overline{x}_j|\,\mathbb{1}\left\{sign(\overline{\boldsymbol{x}})_j \neq \mathsf{C}(\boldsymbol{X})_j\right\}\right], \qquad (43)
\end{aligned}
$$

where the second line is due to the dependence between dataset $\mathcal{S}$ and feature matrix $\boldsymbol{X}$ that $\mathcal{S}$ does not provide more information than $\boldsymbol{X}$; the third line follows a similar idea to (42), and $\mathsf{C}:\mathcal{X}^n\to\{-1,1\}^d$ is a DP binary classifier. The expectation term in (43) can be thought of as the worst-case expected error of estimating the $sign$ vector of the (column-wise) average vector of $\boldsymbol{X}$ when using a DP binary classifier C.

4) **step 4: bounding the classification error with sample complexity**

To facilitate further analysis, we use $\mathsf{Error}(\mathsf{C},\boldsymbol{X}) := \mathbb{E}_{\mathsf{C}}\left[\sum_{j=1}^{d}|\overline{x}_j|\,\mathbb{1}\left\{sign(\overline{\boldsymbol{x}})_j \neq \mathsf{C}(\boldsymbol{X})_j\right\}\right]$ to denote the expected error of a DP binary classification algorithm C for a given $n\times d$ matrix $\boldsymbol{X}$. Denote the smallest number of samples to achieve a certain minimax risk $\alpha > 0$ as the sample complexity:

$$
\mathsf{S}(\alpha,\varepsilon,\delta) := \min\{n : \inf_{\mathsf{C}\in\mathcal{F}_{\varepsilon,\delta},\text{classifier }\mathsf{C}}\sup_{\boldsymbol{X}\in\mathcal{X}^n}\mathsf{Error}(\mathsf{C},\boldsymbol{X}) \leq \alpha\}
$$

By Proposition 1 and Lemma D.2 in Asi et al. (2021), for ε', δ' defined in step 3, the sample complexity satisfies

$$\mathsf{S}(\alpha, \varepsilon', \delta') \geq \Omega\left(\frac{\sqrt{d}}{\alpha\varepsilon' \ln d}\right).$$

The lower bound on sample complexity implies a lower bound on the minimax risk:

$$\inf_{\mathsf{C}\in\mathcal{F}_{\varepsilon',\delta'},\text{classifier C}} \sup_{\boldsymbol{X}\in\mathcal{X}^n} \mathsf{Error}(\mathsf{C}, X) \geq \Omega\left(\frac{\sqrt{d}}{n\varepsilon' \ln d}\right) = \Omega\left(\frac{\sqrt{d}}{n\varepsilon \ln d}\right).$$

Combing preceding four steps, we obtain

$$\inf_{\mathsf{A}\in\mathcal{F}_{\varepsilon,\delta}} \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathsf{A};\mathbb{P}) \geq \widetilde{\Omega}\left(\frac{\sqrt{d}}{n\varepsilon}\right), \tag{44}$$

where a logarithmic factor $\ln d$ in the denominator of r.h.s of (44) is hidden. Moreover, since the oracle complexity of stochastic convex optimization is $\Omega(\frac{1}{\sqrt{n}})$, the minimax risk is therefore further lower bounded by the maximum between oracle complexity and (44):

$$\begin{aligned}\inf_{\mathsf{A}\in\mathcal{F}_{\varepsilon,\delta}} \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathsf{A};\mathbb{P}) &\geq \widetilde{\Omega}\left(\max\left(\frac{1}{\sqrt{n}}, \frac{\sqrt{d}}{n\varepsilon}\right)\right)\\ &= \widetilde{\Omega}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d}}{n\varepsilon}\right),\end{aligned}$$

where the second line gives the desired bound.

□

A.11. Proposition 5 and Proof

Proposition 5 (Inconsistent Order Quantity by Output Perturbation). *Consider a sampling case DP-NV problem without features. Assume demand $y \sim \mathbb{P} \in \mathcal{P}([0,\bar{y}])$ where $\mathbb{P}$ has a CDF F and is supported on $[0,\bar{y}]$ with finite $\bar{y} > 0$. Let optimal order quantity $q^* := \inf_{y\in\mathbb{R}}\{y : F(y) \geq r\}$, and let non-private data-driven order quantity be $\widehat{q} := \inf_{y\in\mathbb{R}}\{y : \widehat{F}(y) \geq r\}$. For any given (ε,δ), let $\widehat{q}^{\mathcal{G}} := \widehat{q} + \mathcal{N}(0,\sigma^2)$ with proper σ by Gaussian Mechanism 1. Then, there exists a strictly positive constant $C > 0$ such that*

$$\Pr\left[\left|\widehat{q}^{\mathcal{G}} - q^*\right| \geq \epsilon\right] \geq C > 0, \quad \forall \epsilon \in (0, \max\{q^*, \bar{y} - q^*\}),$$

where the probability is taken over Gaussian noise. In other words, $\widehat{q}^{\mathcal{G}}$ is inconsistent.

Proof. For given demand observations $\boldsymbol{y} := \{y_i\}_{i=1}^n$ and critical ratio $r \in (0,1)$, we first notice that $\widehat{q} := \inf_{y\in\mathbb{R}}\left\{y : \widehat{F}(y) \geq r\right\} = \boldsymbol{y}_{(\lceil nr\rceil)}$, where $\boldsymbol{y}_{(\lceil nr\rceil)}$ is the $\lceil nr\rceil^{th}$-smallest value. As a result, the Global Sensitivity of $\widehat{q}$ is $\mathrm{GS}_{\widehat{q}} = \sup_{\boldsymbol{y}\sim\boldsymbol{y}'} |\widehat{q}(\boldsymbol{y}) - \widehat{q}(\boldsymbol{y}')| = \bar{y}$. This is achievable by a pair of neighboring demand samples with $\lceil nr\rceil - 1$ points at 0 and $n - \lceil nr\rceil$ points at $\bar{y}$, and differ only in $\lceil nr\rceil^{th}$-smallest point. Whether the $\lceil nr\rceil^{th}$-smallest point is at 0 or $\bar{y}$ determines whether the order quantity is 0 or $\bar{y}$. Consequently, by Gaussian Mechanism 1, there exists a valid $\sigma \geq \sqrt{2\ln(1.25/\delta)}\cdot\bar{y}/\varepsilon$ such that $\widehat{q}^{\mathcal{G}} := \widehat{q} + \mathcal{N}(0,\sigma^2)$ is (ε,δ)-DP. Then for any given $\boldsymbol{y}$ and $\epsilon \in (0, \max\{q^*, \bar{y} - q^*\})$, we have

$$\begin{aligned}\Pr\left[\left|\widehat{q}^{\mathcal{G}} - q^*\right| \geq \epsilon\right] &= \Pr\left[\left|\widehat{q} + \mathcal{N}(0,\sigma^2) - q^*\right| \geq \epsilon\right]\\ &\geq \Pr\left[0 + \mathcal{N}(0,\sigma^2) - q^* \geq \epsilon\right] + \Pr\left[\bar{y} + \mathcal{N}(0,\sigma^2) - q^* \leq -\epsilon\right]\\ &= \Pr\left[\mathcal{N}(0,\sigma^2) \geq q^* + \epsilon\right] + \Pr\left[\mathcal{N}(0,\sigma^2) \geq \bar{y} + \epsilon - q^*\right],\end{aligned} \tag{45}$$

where all probabilities are taken with respect to algorithm's randomness, i.e. the Gaussian noise. Since a standard Gaussian distribution $z \sim \mathcal{N}(0, 1^2)$ admits a tail bound that $\Pr[z \geq t] \geq \frac{1}{\sqrt{2\pi}} \frac{t}{t^2+1} \cdot e^{-t^2/2}, \forall t > 0$, it naturally leads to a lower bound for (45):

$$(45) \geq \frac{1}{\sqrt{2\pi}} \left(\frac{t_1}{t_1^2+1} \cdot e^{-t_1^2/2} + \frac{t_2}{t_2^2+1} \cdot e^{-t_2^2/2} \right) > 0,$$

with $t_1 := (q^* + \epsilon)/\sigma$ and $t_2 := (\bar{y} + \epsilon - q^*)/\sigma$.

□

B. Proofs for Section 4

B.1. Lemma 8 and Proof

Lemma 8. *Let $\mathbb{P}$ be a distribution that satisfies the assumptions in Theorem 8. Denote $\mathsf{z} := \left\| \boldsymbol{\Sigma}^{1/2} \boldsymbol{\theta}^\star \right\|_2$ Then,*

1. *The CDF $F_y(t) = \left(\Phi * f_{\epsilon^\star,1/\mathsf{z}}\right)(t/\mathsf{z})$ is the convolution between CDF Φ of a standard normal random variable and an adjusted PDF $f_{\epsilon^\star,h}(t) := f_{\epsilon^\star}(t/h)/h$.*
2. *The CDF F_y is differentiable and strictly increasing; therefore, $q^* := F_y^{-1}(r) = \arg\min\{t \in \mathbb{R} : F_y(t) \geq r\}$ is unique and well-defined.*

Proof. To show part 1, we need to check the CDF of demand y. Let $f_{\boldsymbol{x}}$ be the PDF of $\boldsymbol{x}$ and $f_{\boldsymbol{z}}$ be the PDF of a standard multivariate Gaussian $\boldsymbol{z} := (z_1, \ldots, z_d) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_{d\times d})$ with independent coordinates. Rescaling and rotating give $\boldsymbol{x} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{z}$. Denote $\mathsf{z} := \left\| \boldsymbol{\Sigma}^{1/2} \boldsymbol{\theta}^\star \right\|_2$, then we have:

$$\begin{aligned}
F_y(t) = \Pr_y[y \leq t] &= \Pr_{\boldsymbol{x},\epsilon}[\langle \boldsymbol{\theta}^\star, \boldsymbol{x} \rangle + \epsilon^\star \leq t] \\
&= \int_{-\infty}^{\infty} \Pr_{\boldsymbol{x}}[\langle \boldsymbol{\theta}^\star, \boldsymbol{x} \rangle \leq t - u] \cdot f_{\epsilon^\star}(u)\, du \\
&= \int_{-\infty}^{\infty} \Pr_{\boldsymbol{z}}\left[\left\langle \boldsymbol{\theta}^\star, \boldsymbol{\Sigma}^{1/2}\boldsymbol{z} \right\rangle \leq t - u\right] \cdot f_{\epsilon^\star}(u)\, du && \text{(rescale and rotate)} \\
&= \int_{-\infty}^{\infty} \Pr_{z_1}\left[z_1 \leq (t-u)/\left\| \boldsymbol{\Sigma}^{1/2}\boldsymbol{\theta}^\star \right\|_2\right] \cdot f_{\epsilon^\star}(u)\, du && \text{(by rotational invariance of Gaussian } \boldsymbol{z}) \\
&= \int_{-\infty}^{\infty} \Phi\left(\frac{t-u}{\mathsf{z}}\right) \cdot f_{\epsilon^\star}(u)\, du \\
&= \int_{-\infty}^{\infty} \Phi\left(\frac{t}{\mathsf{z}} - v\right) \cdot f_{\epsilon^\star}(v\mathsf{z})\mathsf{z}\, dv && \text{(change variable } v := u/\mathsf{z}) \\
&= \int_{-\infty}^{\infty} \Phi\left(\frac{t}{\mathsf{z}} - v\right) \cdot f_{\epsilon^\star,1/\mathsf{z}}(v)\, dv && \text{(by definition of } f_{\epsilon^\star,h}) \\
&= \left(\Phi * f_{\epsilon^\star,1/\mathsf{z}}\right)(t/\mathsf{z}), && \text{(by the definition of convolution (4))}
\end{aligned}$$

where $f_{\epsilon^\star,h}(t) := f_{\epsilon^\star}(t/h)/h$ is an adjusted function of the pdf $f_{\epsilon^\star}$.

As for part 2 of Lemma 8, it is a corollary of part 1 by applying differentiation property of convolution that the derivative of convolution can be obtained by taking derivative first on either functions and then doing convolution, i.e., $(f * g)' = (f' * g) = (f * g')$. Therefore, we have

$$F_y'(t) = \frac{d\left(\Phi * f_{\epsilon^\star,1/\mathsf{z}}\right)(t/\mathsf{z})}{dt} = \left(\Phi' * f_{\epsilon^\star,1/\mathsf{z}}\right)(t/\mathsf{z}) \cdot \frac{1}{\mathsf{z}} > 0, \quad \forall t,$$

which implies F_y is differentiable and strictly increasing, ensuring the uniqueness of $q^* := F_y^{-1}(r)$.

□

B.2. Proof of Theorem 8

Proof. Before proving the Theorem, we highlight that this proof heavily relies on the technical Lemma 8 and reasoning therein.

We come to prove Theorem 8 now. First of all, because of $\bar{r}$-Lipschitz continuity of newsvendor loss function $c(\cdot)$, the excess generalization risk can be upper bounded as

$$\begin{aligned}\mathbb{E}_{\boldsymbol{x},y}\left[c(y-\widehat{q})\right]-\mathbb{E}_{\boldsymbol{x},y}\left[c(y-\langle\boldsymbol{\theta}^*,\boldsymbol{x}\rangle)\right] &\leq \mathbb{E}_{\boldsymbol{x}}\left[|\widehat{q}-\langle\boldsymbol{\theta}^*,\boldsymbol{x}\rangle|\right]\cdot\bar{r}\\ &\leq |\widehat{q}-q^*|\cdot\bar{r}+\mathbb{E}_{\boldsymbol{x}}\left[|q^*-\langle\boldsymbol{\theta}^*,\boldsymbol{x}\rangle|\right]\cdot\bar{r},\end{aligned} \tag{46}$$

where $q^* := F_y^{-1}(r)$ is unique and well-defined by Lemma 8 part 2. Since the second term on the r.h.s of (46) is a constant and strictly greater than 0, it remains to control the distance $|\widehat{q}-q^*|$.

By Lemma 8 part 2, we know that the CDF F_y is continuous and strictly increasing; thus $F_y^{-1}(F_y(u))=u$ for any u. Let the bold symbol $\boldsymbol{y}$ denote a dataset $\{y_i\}_{i=1}^n$, then for any $\epsilon>0$, by standard analysis, we have:

$$\begin{aligned}\Pr_{\boldsymbol{y}}\left[\widehat{q}-q^*\geq\epsilon\right] &= \Pr_{\boldsymbol{y}}\left[\widehat{q}\geq F_y^{-1}(F_y(q^*+\epsilon))\right] && \text{(apply } F_y^{-1}(F_y(u))=u\text{)}\\ &= \Pr_{\boldsymbol{y}}\left[\sum_{i=1}^n \mathbb{1}\left\{y_i\leq F_y(q^*+\epsilon)\right\}\leq nr\right]\\ &= \Pr\left[\mathsf{Binomial}(n,F_y(q^*+\epsilon))\leq nr\right]\\ &\leq \exp\left(-nD_{KL}\left(r\,\|\,F_y(q^*+\epsilon)\right)\right),\end{aligned}$$

where the last line follows the classic Chernoff tail bound on Binomial random variable, and $D_{KL}\left(p_1\,\|\,p_2\right) := p_1\ln\left(\frac{p_1}{p_2}\right)+(1-p_1)\ln\left(\frac{1-p_1}{1-p_2}\right)$ is the Kullback–Leibler (KL) divergence between two Bernoulli random variables with success probabilities p_1 and p_2. Similarly we can obtain the bound for another side that

$$\Pr_{\boldsymbol{y}}\left[\widehat{q}-q^*\leq-\epsilon\right]\leq\exp\left(-nD_{KL}\left(r\,\|\,F_y(q^*-\epsilon)\right)\right).$$

Combining both sides results in

$$\Pr_{\boldsymbol{y}}\left[|\widehat{q}-q^*|\geq\epsilon\right]\leq\exp\left(-nD_{KL}\left(r\,\|\,F_y(q^*+\epsilon)\right)\right)+\exp\left(-nD_{KL}\left(r\,\|\,F_y(q^*-\epsilon)\right)\right). \tag{47}$$

Since F_y is strictly increasing, there must exist $\tau>0, s>0$ such that for any q in the punctured τ-neighborhood of q^* defined as $\{q\in\mathbb{R}:|q-q^*|\leq\tau, q\neq q^*\}$, the slope $(F_y(q)-F_y(q^*))/(q-q^*)\geq s$. Consequently, if we assume ϵ is small enough so that $\epsilon\in(0,\tau)$, then $q^*+\epsilon$ is in the puctured neighborhood; and

$$\begin{aligned}F_y(q^*+\epsilon)&\geq F_y(q^*)+s\epsilon=r+s\epsilon;\\ F_y(q^*-\epsilon)&\leq F_y(q^*)-s\epsilon=r-s\epsilon.\end{aligned}$$

Therefore,

$$\begin{aligned}(47) &\leq \exp\left(-nD_{KL}\left(r\,\|\,r+s\epsilon\right)\right)+\exp\left(-nD_{KL}\left(r\,\|\,r-s\epsilon\right)\right) && \text{(by KL-divergence)}\\ &\leq 2\exp\left(-2ns^2\epsilon^2\right),\end{aligned}$$

where the last inequality is by applying Pinsker's inequality that $D_{KL}\left(a\,\|\,b\right)\geq 2(a-b)^2$. Setting $2\exp\left(-2ns^2\epsilon^2\right)=\gamma$ and solving for ϵ, we get $\epsilon=\frac{1}{s}\sqrt{\frac{\ln(2/\gamma)}{2n}}$. That implies, with probability at least $1-\gamma$ over data sampling process, the order quantity $\widehat{q}$ by PartiallyDD satisfies

$$|\widehat{q}-q^*|\leq\frac{1}{s}\sqrt{\frac{\ln(2/\gamma)}{2n}},$$

for large enough n. Plugging the above inequality back into (46), we obtain the desired Theorem.

□

B.3. Proof of Proposition 1

Proof. The (ε, δ)-DP is a corollary of Advanced Composition Lemma 2, and the values $(\varepsilon_0, \delta_0)$ are set accordingly. By setting $(\varepsilon_0, \delta_0)$ as stated values, the individual regret is:

$$\sup_{s\in\mathcal{S}} \mathcal{R}(\mathsf{IndepRun_s}; \mathbb{P}_s) = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{Sd\ln(S/\delta)}}{n\varepsilon}\right), \tag{48}$$

which directly follows from the regret results of proposed algorithms, see Theorem 2, 4, 6.

To show the total regret $\mathcal{R}(\mathsf{IndepRun}; \mathbb{P})$, we look at Algorithm OP as an example; results on other algorithms can be obtained from a similar analysis with proper modifications. We first notice that, independently running S models is equivalent to the following problem,

$$\min_{\boldsymbol{\theta}^{\mathcal{S}}} \frac{1}{n}\sum_{i=1}^{n} \ell_h(\boldsymbol{\theta}^{\mathcal{S}}; \boldsymbol{x}_i; \boldsymbol{y}_i) + \lambda \left\|\boldsymbol{\theta}^{\mathcal{S}}\right\|_2 + \frac{\langle \boldsymbol{\theta}^{\mathcal{S}}, \boldsymbol{b}\rangle}{n}, \tag{49}$$

where $\ell_h(\boldsymbol{\theta}^{\mathcal{S}}; \boldsymbol{x}_i, \boldsymbol{y}_i) = \sum_{s\in\mathcal{S}} \ell_h(\boldsymbol{\theta}_s; \boldsymbol{x}_i, y_{is})$, which is the loss function associated with $\mathcal{R}(\mathsf{IndepRun}; \mathbb{P})$. Noise $\boldsymbol{b} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2)$ is a dS-dimensional random vector with $\sigma_0 = 2L^{\mathcal{S}}\sqrt{8\ln(2/\delta_0)}/\varepsilon_0$. Thus, upper bounding the total regret amounts to upper bounding the regret of (49). We can show that

1. $\ell_h(\boldsymbol{\theta}^{\mathcal{S}})$ is $\bar{r}^{\mathcal{S}} B_x \sqrt{S} =: L^{\mathcal{S}}$-Lipschitz continuous;
2. $\ell_h(\boldsymbol{\theta}^{\mathcal{S}})$ is $\overline{K} B_x^2 \sqrt{S}/h =: \beta^{\mathcal{S}}$-smooth.

This is because, for a pair of estimators $\boldsymbol{\theta}^{\mathcal{S}}, \boldsymbol{\theta}^{\mathcal{S}\prime}$ trained on a pair of neighboring datasets, we have

$$\begin{aligned}
\text{(Lipschitzness)}\quad \ell_h(\boldsymbol{\theta}^{\mathcal{S}}) - \ell_h(\boldsymbol{\theta}^{\mathcal{S}\prime}) &= \sum_{s\in\mathcal{S}} [\ell_h(\boldsymbol{\theta}_s) - \ell_h(\boldsymbol{\theta}'_s)] \\
&\le \bar{r}^{\mathcal{S}} B_x \sum_{s\in\mathcal{S}} \|\boldsymbol{\theta}_s - \boldsymbol{\theta}'_s\|_2 && \text{(because } \ell_h \text{ is } \bar{r}^{\mathcal{S}} B_x\text{-Lip cts)} \\
&\le \bar{r}^{\mathcal{S}} B_x \sqrt{S} \left\|\boldsymbol{\theta}^{\mathcal{S}} - \boldsymbol{\theta}^{\mathcal{S}\prime}\right\|_2; && \text{(by Cauchy's inequality)} \\
\text{(Smoothness)}\quad \nabla\ell_h(\boldsymbol{\theta}^{\mathcal{S}}) - \nabla\ell_h(\boldsymbol{\theta}^{\mathcal{S}\prime}) &= \sum_{s\in\mathcal{S}} [\nabla\ell(\boldsymbol{\theta}_s) - \nabla\ell_h(\boldsymbol{\theta}'_s)] \\
&\le \overline{K} B_x^2/h \sum_{s\in\mathcal{S}} \|\boldsymbol{\theta}_s - \boldsymbol{\theta}'_s\|_2 && \text{(because } \ell_h \text{ is } \overline{K} B_x^2/h\text{-smooth)} \\
&\le \overline{K} B_x^2 \sqrt{S}/h \left\|\boldsymbol{\theta}^{\mathcal{S}} - \boldsymbol{\theta}^{\mathcal{S}\prime}\right\|_2. && \text{(by Cauchy's inequality)}
\end{aligned}$$

Therefore, following the analysis in the proof of Theorem 2, we can upper bound the total regret as

$$\mathcal{R}(\mathsf{IndepRun}; \mathbb{P}) \le \mathcal{O}\left(\underbrace{\frac{(L^{\mathcal{S}})^2}{n\lambda}}_{\text{by uniform stability}} + \underbrace{\frac{dS\sigma_0^2}{n^2\lambda} + \lambda M/B_\theta^2}_{\text{by shrinking ERM}} + \underbrace{\frac{S^3}{\lambda^2 n^2 \varepsilon_0^2}}_{\text{approximation error}}\right). \tag{50}$$

The approximation error on the r.h.s of (50) is slightly different from (24). Differences and reasons are summarized below: (1) in the second part which is by shrinking ERM, the additional S in the numerator of first term is due to increased dimensionality in $\boldsymbol{b}$; and M/B_θ^2 comes from the upper bound of demand M. (2) in the third part which is approximation error, the numerator S^3 originates from two sources: regularizer λ now should satisfy $\lambda \ge \frac{2\beta S}{n\varepsilon}$ because of the increase in dimensionality which modifies the exponent on the l.h.s of (15) to $2S$, thus contributing a S^2; The increase in smoothness parameter contributes another S.

After getting (50), we can set $\lambda = (\frac{(L^S)^2}{n} + \frac{dS\sigma_0^2}{n^2})/(M/B_\theta^2)$. Following remaining analysis steps for Theorem 2, we get

$$\mathcal{R}(\mathsf{IndepRun}; \mathbb{P}) = \mathcal{O}\left(\sqrt{\frac{S}{n}} + \frac{\sqrt{S \cdot Sd\ln(S/\delta)}}{n\varepsilon}\right), \quad \forall \mathbb{P}, \tag{51}$$

if $\varepsilon \gtrsim S^{5/4}/n^{1/4}$. Comparing (48) and (51) gives the proposition immediately. □

C. Proofs for Section 5

C.1. Lemma 9 and Proof

Lemma 9 (Population-Level Minimizers' (Asymptotic) Bias). *Let $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$, $\boldsymbol{\theta}_h^* := \arg\min_{\boldsymbol{\theta}} \mathcal{L}_h(\boldsymbol{\theta})$, and $\boldsymbol{\theta}_h^{\#*} := \arg\min_{\boldsymbol{\theta}} \mathcal{L}_h^{\#}(\boldsymbol{\theta})$, and $\boldsymbol{\theta}_h^{\mathrm{OP}*} := \arg\min_{\boldsymbol{\theta}} \mathcal{L}_h^{\mathrm{OP}}(\boldsymbol{\theta})$. Denote $\boldsymbol{J} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0) \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \succ \mathbf{0}$. Then, under Assumption 1, if $\lambda = o(1)$ and $h = o(1)$, we have*

(i) $\boldsymbol{\theta}_h^* - \boldsymbol{\theta}^* = -\frac{1}{2}h^2\kappa_2\boldsymbol{v} + o(h^2)$, *where* $\boldsymbol{v} = \boldsymbol{J}^{-1}\mathbb{E}_{\boldsymbol{x}}\left[f'_{\epsilon|\boldsymbol{x}}(0)\boldsymbol{x}\right]$;

(ii) $\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^* = -2\lambda\boldsymbol{J}^{-1}\boldsymbol{\theta}_h^* + o(\lambda) + o(h^2)$;

(iii) $\boldsymbol{\theta}_h^{\mathrm{OP}*} = \boldsymbol{\theta}_h^{\#*}$.

Proof. We first express gradient $\nabla\mathcal{L}_h(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x},y}\left[\left[\mathcal{K}_h\left(\boldsymbol{\theta}^\top\boldsymbol{x} - y\right) - r\right] \cdot \boldsymbol{x}\right]$ and Hessian matrix $\nabla^2\mathcal{L}_h(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x},y}\left[K_h\left(y - \boldsymbol{\theta}^\top\boldsymbol{x}\right) \cdot \boldsymbol{x}\boldsymbol{x}^\top\right]$ for any given $\boldsymbol{\theta} \in \mathbb{R}^d$ as functions of an error $\boldsymbol{\delta} := \boldsymbol{\theta} - \boldsymbol{\theta}^*$. Though these expressions are well known in literature (Fernandes et al. 2021), we give detailed proofs for completeness and for usage in later analysis. We first note that, conditional on feature $\boldsymbol{x}$, we have

$$\begin{aligned}
\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[\mathcal{K}\left(\frac{\boldsymbol{\delta}^\top\boldsymbol{x} - \epsilon}{h}\right)\Big|\boldsymbol{x}\right] &= \int_{-\infty}^{\infty} \mathcal{K}\left(\frac{\boldsymbol{\delta}^\top\boldsymbol{x} - t}{h}\right) dF_{\epsilon|\boldsymbol{x}}(t) \\
&= \frac{1}{h}\int_{-\infty}^{\infty} F_{\epsilon|\boldsymbol{x}}(t) K\left(\frac{\boldsymbol{\delta}^\top\boldsymbol{x} - t}{h}\right) dt && \text{(integration by parts)} \\
&= \int_{-\infty}^{\infty} F_{\epsilon|\boldsymbol{x}}(\boldsymbol{\delta}^\top\boldsymbol{x} - uh) K(u)\, du && (\text{let } u = (\boldsymbol{\delta}^\top\boldsymbol{x} - t)/h) \\
&= r + \int_{-\infty}^{\infty} \left[F_{\epsilon|\boldsymbol{x}}(\boldsymbol{\delta}^\top\boldsymbol{x} - uh) - F_{\epsilon|\boldsymbol{x}}(0)\right] K(u)\, du && (F_{\epsilon|\boldsymbol{x}}(0) = r \text{ and } \int K = 1),
\end{aligned} \tag{52}$$

and

$$\begin{aligned}
\mathbb{E}_{y|\boldsymbol{x}}\left[K_h(y - \boldsymbol{\theta}^\top\boldsymbol{x})\big|\boldsymbol{x}\right] &= \int_{-\infty}^{\infty} \frac{1}{h} K\left(\frac{\boldsymbol{\theta}^{*\top}\boldsymbol{x} + t - \boldsymbol{\theta}^\top\boldsymbol{x}}{h}\right) dF_{\epsilon|\boldsymbol{x}}(t) \\
&= \int_{-\infty}^{\infty} K(u) f_{\epsilon|\boldsymbol{x}}(\boldsymbol{\delta}^\top\boldsymbol{x} + uh)\, du. && (\boldsymbol{\delta} := \boldsymbol{\theta} - \boldsymbol{\theta}^*; \text{ let } u = \frac{t - \boldsymbol{\delta}^\top\boldsymbol{x}}{h})
\end{aligned} \tag{53}$$

Following (52) and (53), the gradient and Hessian matrix can be expressed as

$$\nabla\mathcal{L}_h(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[\mathcal{K}_h\left(\boldsymbol{\delta}^\top\boldsymbol{x} - \epsilon\right) - r\right] \cdot \boldsymbol{x}\right] = \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot \int_0^{\boldsymbol{\delta}^\top\boldsymbol{x} - uh} f_{\epsilon|\boldsymbol{x}}(t)\, dt\, du \cdot \boldsymbol{x}\right]; \tag{54}$$

$$\nabla^2\mathcal{L}_h(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[K_h(\epsilon - \boldsymbol{\delta}^\top\boldsymbol{x})\right] \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] = \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot f_{\epsilon|\boldsymbol{x}}(\boldsymbol{\delta}^\top\boldsymbol{x} + uh)\, du \cdot \boldsymbol{x}\boldsymbol{x}^\top\right]. \tag{55}$$

With above expressions, we are ready to prove three statements in the Lemma.

(i) According to (54), the gradient at $\boldsymbol{\theta}^*$, i.e., $\boldsymbol{\delta} = \mathbf{0}$ is

$$\begin{aligned}
\nabla \mathcal{L}_h(\boldsymbol{\theta}^*) &= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot \int_0^{-uh} f_{\epsilon|\boldsymbol{x}}(t) - f_{\epsilon|\boldsymbol{x}}(0)\, dt\, du \cdot \boldsymbol{x}\right] \qquad \left(\int K(u)\cdot(-uh) f_{\epsilon|\boldsymbol{x}}(0)\, du = 0\right) \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot \int_0^{-uh} t f'_{\epsilon|\boldsymbol{x}}(0) + o(t)\, dt\, du \cdot \boldsymbol{x}\right] \qquad \text{(by Taylor expansion)} \\
&= \frac{1}{2} h^2 \kappa_2 \mathbb{E}_{\boldsymbol{x}}\left[f'_{\epsilon|\boldsymbol{x}}(0)\boldsymbol{x}\right] + o(h^2).
\end{aligned}$$

And according to (55), the Hessian matrix at $\boldsymbol{\theta}^*$, i.e., $\boldsymbol{\delta} = \mathbf{0}$, is

$$\begin{aligned}
\nabla^2 \mathcal{L}_h(\boldsymbol{\theta}^*) &= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) f_{\epsilon|\boldsymbol{x}}(uh)\, du \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot \left[f_{\epsilon|\boldsymbol{x}}(0) + f'_{\epsilon|\boldsymbol{x}}(0) uh + o(uh)\right] du \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \qquad \text{(by Taylor expansion)} \\
&= \boldsymbol{J} + o(h),
\end{aligned}$$

where $\boldsymbol{J} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0) \cdot \boldsymbol{x}\boldsymbol{x}^\top\right]$. The Taylor expansions can be applied because of our assumption that $f_{\epsilon|\boldsymbol{x}}$ is differentiable at 0. We further notice that, with $\boldsymbol{\delta}_h^* := \boldsymbol{\theta}_h^* - \boldsymbol{\theta}^*$,

$$-\nabla \mathcal{L}_h(\boldsymbol{\theta}^*) = \nabla \mathcal{L}_h(\boldsymbol{\theta}_h^*) - \nabla \mathcal{L}_h(\boldsymbol{\theta}^*) = \nabla^2 \mathcal{L}_h(\boldsymbol{\theta}^*)\boldsymbol{\delta}_h^* + o(\|\boldsymbol{\delta}_h^*\|_2),$$

where the first equality is due to First-Order-Condition that $\nabla \mathcal{L}_h(\boldsymbol{\theta}_h^*) = \mathbf{0}$, and the second equality follows from Taylor expansion of $\nabla \mathcal{L}_h$ at $\boldsymbol{\theta}^*$. By omitting the higher-order infinitesimal $o(\|\boldsymbol{\delta}_h^*\|_2)$, we have $\boldsymbol{\delta}_h^* = -\left(\nabla^2 \mathcal{L}_h(\boldsymbol{\theta}^*)\right)^{-1} \nabla \mathcal{L}_h(\boldsymbol{\theta}^*)$. Replacing the gradient and Hessian matrix with their expressions derived above gives the first statement.

(ii) As for the second part, from First-Order-Conditions, we know $\nabla \mathcal{L}_h(\boldsymbol{\theta}_h^*) = \mathbf{0}$ and $\nabla \mathcal{L}_h^{\#*}(\boldsymbol{\theta}_h^{\#*}) = \mathbf{0}$. According to (54), we can express

$$\begin{aligned}
\nabla \mathcal{L}_h^{\#}(\boldsymbol{\theta}_h^{\#*}) &= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \int_0^{(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}^*)^\top \boldsymbol{x} - uh} f_{\epsilon|\boldsymbol{x}}(t)\, dt\, du \cdot \boldsymbol{x}\right] + 2\lambda \boldsymbol{\theta}_h^{\#*} \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \int_0^{\boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh} f_{\epsilon|\boldsymbol{x}}(t)\, dt\, du \cdot \boldsymbol{x}\right] \\
&\quad + \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \int_{\boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh}^{\boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh + (\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x}} f_{\epsilon|\boldsymbol{x}}(t)\, dt\, du \cdot \boldsymbol{x}\right] + 2\lambda \boldsymbol{\theta}_h^{\#*} \\
&= \nabla \mathcal{L}_h(\boldsymbol{\theta}_h^*) + \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \int_{\boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh}^{\boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh + (\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x}} f_{\epsilon|\boldsymbol{x}}(t)\, dt\, du \cdot \boldsymbol{x}\right] + 2\lambda \boldsymbol{\theta}_h^{\#*}. \qquad (56)
\end{aligned}$$

Since $\|\boldsymbol{\delta}_h^*\|_2 = O(h^2)$ and $\boldsymbol{\theta}_h^{\#*} \nearrow \boldsymbol{\theta}_h^*$ as $\lambda \searrow 0$, the upper limit $u(\boldsymbol{\delta}_h^*) := \boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh + (\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x}$ and lower limit $l(\boldsymbol{\delta}_h^*) := \boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh$ of the most inner integration on the r.h.s. of (56) will tend to 0; thus we can approximate the integration by Taylor expansion at 0:

$$\begin{aligned}
\int_{l(\boldsymbol{\delta}_h^*)}^{u(\boldsymbol{\delta}_h^*)} f_{\epsilon|\boldsymbol{x}}(t)\, dt &= \left[F_{\epsilon|\boldsymbol{x}}(0) + f_{\epsilon|\boldsymbol{x}}(0) u(\boldsymbol{\delta}_h^*) + \frac{1}{2} f'_{\epsilon|\boldsymbol{x}}(0) u^2(\boldsymbol{\delta}_h^*) + o(u^2(\boldsymbol{\delta}_h^*))\right] \\
&\quad - \left[F_{\epsilon|\boldsymbol{x}}(0) + f_{\epsilon|\boldsymbol{x}}(0) l(\boldsymbol{\delta}_h^*) + \frac{1}{2} f'_{\epsilon|\boldsymbol{x}}(0) l^2(\boldsymbol{\delta}_h^*) + o(l^2(\boldsymbol{\delta}_h^*))\right] \\
&= f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x} + \frac{1}{2} f'_{\epsilon|\boldsymbol{x}}(0)\left[2(\boldsymbol{\delta}_h^{*\top}\boldsymbol{x} - uh)(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x} + \left[(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x}\right]^2\right]
\end{aligned}$$

$$
\begin{aligned}
&+ o(h^2) + o(\|\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*\|_2^2) \\
&= f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x} - uhf'_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x} + \frac{1}{2} f'_{\epsilon|\boldsymbol{x}}(0) \left[(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x}\right]^2 \\
&+ o(h^2) + o(\|\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*\|_2^2),
\end{aligned}
$$

where the last equality is due to $\boldsymbol{\delta}_h^{*\top}\boldsymbol{x}(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x} = O(h^2)o(1) = o(h^2)$. Plugging above value into (56), we can obtain

$$
\begin{aligned}
\nabla \mathcal{L}_h^{\#}(\boldsymbol{\theta}_h^{\#*}) &= \mathbb{E}_{\boldsymbol{x}} \left[\left[f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x} + \frac{1}{2} f'_{\epsilon|\boldsymbol{x}}(0) \left[(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x}\right]^2 \right] \cdot \boldsymbol{x} \right] + 2\lambda \boldsymbol{\theta}_h^{\#*} \\
&\quad + o(h^2) + o(\|(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)\|_2^2) \\
&= \boldsymbol{J} \cdot (\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*) + \frac{1}{2} \mathbb{E}_{\boldsymbol{x}} \left[f'_{\epsilon|\boldsymbol{x}}(0) \left[(\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*)^\top \boldsymbol{x}\right]^2 \cdot \boldsymbol{x} \right] + 2\lambda \boldsymbol{\theta}_h^{\#*} + o(h^2) + o(\|\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*\|_2^2) \\
&= \boldsymbol{J} \cdot (\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*) + 2\lambda \boldsymbol{\theta}_h^{\#*} + o(h^2) + o(\|\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*\|_2) = \mathbf{0}.
\end{aligned}
$$

Rearranging the above expression gives $\boldsymbol{\theta}_h^{\#*} = (\boldsymbol{I} + 2\lambda \boldsymbol{J}^{-1})^{-1} \boldsymbol{\theta}_h^* + o(h^2) + o(\|\boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}_h^*\|_2)$. Then the second part of this Lemma directly follows from inverse matrix approximation with second-order Neumann series that $(\boldsymbol{I} + A)^{-1} \cong I - A + A^2$ if $A \to \mathbf{0}$.

(iii) The proof for the third part of this Lemma is shown below:

$$
\begin{aligned}
\boldsymbol{\theta}_h^{\mathrm{OP}*} := \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}_h^{\mathrm{OP}}(\boldsymbol{\theta}) &= \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathbb{E}_{(\boldsymbol{x},y),\boldsymbol{b}} \left[\ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) + \lambda \|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top \boldsymbol{\theta}}{n} \right] \\
&= \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathbb{E}_{(\boldsymbol{x},y)} \left[\ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) \right] + \lambda \|\boldsymbol{\theta}\|_2^2 \\
&= \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}_h^{\#}(\boldsymbol{\theta}) = \boldsymbol{\theta}_h^{\#*}.
\end{aligned}
$$

□

Lemma 10 (Lemma 9.21 in Wainwright 2019, rephrased). *Denote $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$, $\widehat{\boldsymbol{\theta}} := \arg\min_{\boldsymbol{\theta}} \widehat{\mathcal{L}}(\boldsymbol{\theta})$. For any given radius r_0, if $\widehat{\mathcal{L}}(\boldsymbol{\theta}^* + \boldsymbol{\delta}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*) > 0$, $\forall \boldsymbol{\delta} \in \{\boldsymbol{\delta} \in \mathbb{R}^d : \|\boldsymbol{\delta}\|_2 = r_0\}$, then $\left\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\right\|_2 \le r_0$.*

C.2. Proof of Proposition 2

Proof. Recall that, for any upcoming feature vector $\boldsymbol{x}$, the private order quantity is $\widehat{q}_h^{\mathrm{OP}} = \left\langle \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}, \boldsymbol{x} \right\rangle$; thus, we only need to show $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} = \boldsymbol{\theta}^* - 2\lambda \boldsymbol{u} - \frac{1}{2} h^2 \kappa_2 \boldsymbol{v} + \mathcal{O}_p\left(\frac{1}{\sqrt{n}} + \frac{\sigma}{n}\right)$, with $\boldsymbol{u}, \boldsymbol{v}$, and $\boldsymbol{J}$ as stated. From Lemma 9, we know that $\boldsymbol{\theta}_h^{\mathrm{OP}*} - \boldsymbol{\theta}^* = -2\lambda \boldsymbol{u} - \frac{1}{2} h^2 \kappa_2 \boldsymbol{v} + o(\lambda) + o(h^2)$. In following analysis, we will ignore the infinitesimals of λ and h^2 for conciseness. Therefore, it remains to show $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} \xrightarrow{\mathbb{P}\times\mathbb{Q}} \boldsymbol{\theta}_h^{\mathrm{OP}*}$ at the given rate $\frac{1}{\sqrt{n}} + \frac{\sigma}{n}$, or equivalently to show $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} \xrightarrow{\mathbb{P}} \boldsymbol{\theta}_h^{\mathrm{OP}*}(\boldsymbol{b})$ at rate $\frac{1}{\sqrt{n}}$ and $\boldsymbol{\theta}_h^{\mathrm{OP}*}(\boldsymbol{b}) \xrightarrow{\mathbb{Q}} \boldsymbol{\theta}_h^{\mathrm{OP}*}$ at rate $\frac{\sigma}{n}$ simultaneously, where $\boldsymbol{\theta}_h^{\mathrm{OP}*}(\boldsymbol{b}) := \arg\min_{\boldsymbol{\theta} \in \mathcal{C}} \mathcal{L}_h^{\#}(\boldsymbol{\theta}) + \boldsymbol{b}^\top \boldsymbol{\theta}/n$ for a given $\boldsymbol{b}$. The symbol $\mathbb{P}$ on rightward arrow represents dataset sampling randomness, and $\mathbb{Q}$ represents algorithm's randomness.

We first fix a noise vector $\boldsymbol{b}$ and show the former convergence $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} \xrightarrow{\mathbb{P}} \boldsymbol{\theta}_h^{\mathrm{OP}*}(\boldsymbol{b})$. Let us treat $\ell_h(\boldsymbol{\theta}) + \boldsymbol{b}^\top \boldsymbol{\theta}/n$ as a new loss function of interest. Because we assume that feature space $\mathcal{X}$ and parameter space $\mathcal{C}$ are compact, and assume that the new loss function has a finite expectation. Therefore, by uniform laws of large numbers, we have

$$
\sup_{\boldsymbol{\theta} \in \mathcal{C}} \left| \left(\widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta}) + \boldsymbol{b}^\top \boldsymbol{\theta}/n \right) - \left(\mathcal{L}_h^{\#}(\boldsymbol{\theta}) + \boldsymbol{b}^\top \boldsymbol{\theta}/n \right) \right| \xrightarrow{\mathbb{P}} 0.
$$

Furthermore, we know that function $\mathcal{L}_h^{\#}(\boldsymbol{\theta}) + \boldsymbol{b}^\top \boldsymbol{\theta}/n$ is strongly convex, implying a unique and well-separated minimizer $\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b})$ for any given noise $\boldsymbol{b}$. Consequently, the sequence of empirical minimizers $\{\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\}_{n=1}^{\infty}$ will converge to $\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b})$, i.e., $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}} \xrightarrow{\mathbb{P}} \boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b})$. More importantly, the convergence rate can be shown to be $\mathcal{O}_p(\frac{1}{\sqrt{n}})$ by standard Bayes's risk analysis, which is tight.

Now, we come to show the latter convergence $\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b}) \xrightarrow{\mathbb{Q}} \boldsymbol{\theta}_h^{\mathsf{OP}*}$. We conjecture that $\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b}) = \boldsymbol{\theta}_h^{\mathsf{OP}*} + \mathcal{O}_p(\frac{\sigma}{n})$, and highlight that the stochastic convergence rate $\mathcal{O}_p\left(\frac{\sigma}{n}\right)$ is *NOT* tight; instead, infinitesimals of λ and h^2 should be included to get a tighter bound $\mathcal{O}_p\left(\frac{\sigma}{n} + o(\lambda) + o(h^2)\right)$. However, we deliberately ignore them for conciseness. To show $\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b}) = \boldsymbol{\theta}_h^{\mathsf{OP}*} + \mathcal{O}_p(\frac{\sigma}{n})$, by the definition of stochastic convergence, it is equivalent to show

$$\Pr_{\boldsymbol{b}}\left[\left\|\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b}) - \boldsymbol{\theta}_h^{\mathsf{OP}*}\right\|_2 \geq \epsilon\frac{\sigma}{n}\right] \leq C_\epsilon, \quad \forall \epsilon > 0, \tag{57}$$

with a constant C_ϵ that depends on ϵ. Therefore, we only need to find a proper C_ϵ. By a contrapositive statement of Lemma 9.21 in Wainwright (2019) (Lemma 10), i.e., if $\|\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b}) - \boldsymbol{\theta}_h^{\mathsf{OP}*}\|_2 \geq \epsilon\frac{\sigma}{n}$, then

$$\inf_{\boldsymbol{\delta}\in\partial\mathcal{B}(\epsilon\sigma/n)}\left(\mathcal{L}_h^{\#}(\boldsymbol{\theta}_h^{\mathsf{OP}*} + \boldsymbol{\delta}) + \frac{\boldsymbol{b}^\top(\boldsymbol{\theta}_h^{\mathsf{OP}*} + \boldsymbol{\delta})}{n}\right) - \left(\mathcal{L}_h^{\#}(\boldsymbol{\theta}_h^{\mathsf{OP}*}) + \frac{\boldsymbol{b}^\top\boldsymbol{\theta}_h^{\mathsf{OP}*}}{n}\right) \leq 0. \tag{58}$$

Thus, the probability on the l.h.s. of (57) can be upper bounded by the happening probability of (58). Working on the l.h.s. of (58), we have

$$\begin{aligned}
\text{l.h.s. of (58)} &= \inf_{\boldsymbol{\delta}\in\partial\mathcal{B}(\epsilon\sigma/n)} \mathcal{L}_h^{\#}(\boldsymbol{\theta}_h^{\mathsf{OP}*} + \boldsymbol{\delta}) - \mathcal{L}_h^{\#}(\boldsymbol{\theta}_h^{\mathsf{OP}*}) + \frac{\boldsymbol{b}^\top\boldsymbol{\delta}}{n} \\
&= \inf_{\boldsymbol{\delta}\in\partial\mathcal{B}(\epsilon\sigma/n)} \int_0^1 \nabla^\top \mathcal{L}_h^{\#}(\boldsymbol{\theta}_h^{\mathsf{OP}*} + t\boldsymbol{\delta})\, dt \cdot \boldsymbol{\delta} + \frac{\boldsymbol{b}^\top\boldsymbol{\delta}}{n}, \\
&= \inf_{\boldsymbol{\delta}\in\partial\mathcal{B}(\epsilon\sigma/n)} \frac{1}{2}\boldsymbol{\delta}^\top\boldsymbol{J}\boldsymbol{\delta} + \lambda\|\boldsymbol{\delta}\|_2^2 + \frac{\boldsymbol{b}^\top\boldsymbol{\delta}}{n} && \text{(by (54), Lemma 9, and Taylor expansion)} \\
&> \lambda_{\min}\cdot\frac{\epsilon^2\sigma^2}{2n^2} - \frac{\|\boldsymbol{b}\|_2\,\epsilon\sigma}{n^2} && (\lambda_{\min} > 0 \text{ is the minimal eigenvalue of } \boldsymbol{J})
\end{aligned}$$

The above analysis implies

$$\begin{aligned}
\text{l.h.s. of (57)} < \Pr_{\boldsymbol{b}}\left[\lambda_{\min}\cdot\frac{\epsilon^2\sigma^2}{2n^2} - \frac{\|\boldsymbol{b}\|_2\,\epsilon\sigma}{n^2} \leq 0\right] &= \Pr_{\boldsymbol{b}}\left[\frac{\|\boldsymbol{b}\|_2}{\sigma} \geq \lambda_{\min}\frac{\epsilon}{2}\right] \\
&\leq 2\exp\left(-\lambda_{\min}^2\epsilon^2/(8d)\right) =: C_\epsilon, \quad \forall \epsilon > 0,
\end{aligned}$$

where the last inequality is from Gaussian vector's tail bound since $\boldsymbol{b} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I})$. The constant C_ϵ defined above validates (57), further confirming our conjecture $\boldsymbol{\theta}_h^{\mathsf{OP}*}(\boldsymbol{b}) = \boldsymbol{\theta}_h^{\mathsf{OP}*} + \mathcal{O}_p(\frac{\sigma}{n})$.

Combining above all analysis together, we get $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}} = \boldsymbol{\theta}^* - 2\lambda\boldsymbol{u} - \frac{1}{2}h^2\kappa_2\boldsymbol{v} + \mathcal{O}_p\left(\frac{1}{\sqrt{n}} + \frac{\sigma}{n}\right)$, which immediately results in the desired proposition.

□

C.3. Lemma 11 and Proof

Lemma 11 (Asymptotic Covariance Matrix of $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$). *Suppose Assumption 1 holds and $\langle\boldsymbol{\theta}^*, \boldsymbol{x}\rangle \geq 0$ for any $\boldsymbol{x}$. If we set $\lambda = o(1)$ and $h = o(1)$, then the variance of estimator obtained from Algorithm* OP *satisfies*

$$\mathsf{Var}_{\mathcal{D},\boldsymbol{b}}\left[\sqrt{n}\cdot\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right] \succcurlyeq \boldsymbol{J}^{-1}\boldsymbol{\Sigma}_r\boldsymbol{J}^{-1} - c_1\lambda\boldsymbol{J}^{-1}\boldsymbol{\Sigma}_r\boldsymbol{J}^{-1} - 2h\mathsf{K}\boldsymbol{J}^{-1} + \frac{\sigma^2}{n}\boldsymbol{J}^{-1}\boldsymbol{J}^{-1},$$

where $\boldsymbol{J} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0)\cdot\boldsymbol{x}\boldsymbol{x}^T\right] \succ \boldsymbol{0}$, $\boldsymbol{\Sigma}_r := r(1-r)\boldsymbol{\Sigma} \succ \boldsymbol{0}$, $\mathsf{K} := \int_{-\infty}^{\infty} uK(u)\cdot\int_{-\infty}^{u} K(v)\,dv\cdot du \in (0, \kappa_1)$, and $c_1 > 0$ is a constant. Higher-order infinitesimals of h and λ are omitted. The randomness is taken over both dataset $\mathcal{D}$ generating process and algorithm's randomness.

Proof of Lemma 11. Because the following four facts hold for any $h, \lambda > 0$:

(i) the loss function $\ell_h^{\mathrm{OP}}(\boldsymbol{\theta}; \boldsymbol{x}, y) := \ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) + \lambda \|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top \boldsymbol{\theta}}{n}$ is differentiable in $\boldsymbol{\theta}$ and has a finite expected gradient $\mathbb{E}\left[\|\nabla \ell_h^{\mathrm{OP}}(\boldsymbol{\theta}_h^{\mathrm{OP}*})\|_2^2\right] < \infty$ at optimal $\boldsymbol{\theta}_h^{\mathrm{OP}*}$ if $\sigma = \mathcal{O}(n)$. Furthermore, the function ℓ_h^{OP} is also $(\frac{\overline{K} B_x^2}{h} + 2\lambda)$-smooth, implying that

$$\left\|\nabla \ell_h^{\mathrm{OP}}(\boldsymbol{\theta}_1) - \nabla \ell_h^{\mathrm{OP}}(\boldsymbol{\theta}_2)\right\|_2 \le (\frac{\overline{K} B_x^2}{h} + 2\lambda) \cdot \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2, \quad \forall \boldsymbol{\theta}_1, \boldsymbol{\theta}_2;$$

(ii) the population-level function $\mathcal{L}_h^{\mathrm{OP}}(\boldsymbol{\theta})$'s Hessian matrix $\boldsymbol{H}_h^{\#}(\boldsymbol{\theta}) := \nabla^2 \mathcal{L}_h(\boldsymbol{\theta}) + 2\lambda \boldsymbol{I}$ is positive definite at optimal $\boldsymbol{\theta}_h^{\mathrm{OP}*}$ (or equivalently at $\boldsymbol{\theta}_h^{\#*}$, see Lemma 9), i.e., $\boldsymbol{H}_h^{\#} := \boldsymbol{H}_h^{\#}(\boldsymbol{\theta}_h^{\#*}) = \mathbb{E}_{\boldsymbol{x},y}\left[K_h(y - \boldsymbol{\theta}_h^{\#*\top} \boldsymbol{x}) \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] + 2\lambda \boldsymbol{I} \succ \boldsymbol{0}$;

(iii) the empirical gradient satisfies $\nabla \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}) = \boldsymbol{0}$;

(iv) the private estimator $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} \xrightarrow{\mathbb{P}^n \times \mathbb{Q}} \boldsymbol{\theta}_h^{\mathrm{OP}*} = \boldsymbol{\theta}_h^{\#*}$,

by Van der Vaart (2000, Theorem 5.23), the private estimator $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}$ admits a Bahadur representation:

$$\sqrt{n}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} - \boldsymbol{\theta}_h^{\mathrm{OP}*}) = -\boldsymbol{H}_h^{\#-1} \cdot \sqrt{n} \nabla \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\boldsymbol{\theta}_h^{\mathrm{OP}*}) + o_p(1), \tag{59}$$

where the stochastic convergence $o_p(1)$ is with respect to both data sampling and algorithm's internal randomness. To obtain (59), we need to check the validity of preceding four facts. The first three facts are easy to check. The fourth fact is a corollary of Lemma 9 and is an intermediate result in the proof of Proposition 2.

Hence, by (59), the asymptotic covariance matrix of $\sqrt{n}\, \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}$ is

$$\mathsf{Var}_{\mathcal{D},\boldsymbol{b}}\left[\sqrt{n}\, \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}\right] = \boldsymbol{H}_h^{\#-1} \cdot \mathsf{Var}_{\mathcal{D},\boldsymbol{b}}\left[\sqrt{n} \nabla \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\boldsymbol{\theta}_h^{\mathrm{OP}*})\right] \cdot \boldsymbol{H}_h^{\#-1}. \tag{60}$$

Since the middle term of the preceding sandwich expression is

$$\begin{aligned}
\mathsf{Var}_{\mathcal{D},\boldsymbol{b}}\left[\sqrt{n} \nabla \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\boldsymbol{\theta}_h^{\mathrm{OP}*})\right] &= \mathsf{Var}_{\mathcal{D},\boldsymbol{b}}\left[\sqrt{n} \nabla \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\boldsymbol{\theta}_h^{\#*})\right] \\
&= n \cdot \mathsf{Var}_{\mathcal{D},\boldsymbol{b}}\left[\frac{1}{n}\sum_{i=1}^n \nabla \ell_h(\boldsymbol{\theta}_h^{\#*}; \boldsymbol{x}_i, y_i) + 2\lambda \boldsymbol{\theta}_h^{\#*} + \frac{\boldsymbol{b}}{n}\right] \\
&= \mathsf{Var}_{(\boldsymbol{x},y)}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right] + \frac{\sigma^2}{n}\boldsymbol{I},
\end{aligned}$$

it remains to figure out $\boldsymbol{H}_h^{\#-1}$ and $\mathsf{Var}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right]$. We will show later that

$$\boldsymbol{H}_h^{\#-1} = \boldsymbol{J}^{-1} - 2\lambda \cdot \boldsymbol{J}^{-1}\left(\boldsymbol{I} - \boldsymbol{V}'\right)\boldsymbol{J}^{-1} + o(\lambda) + o(h) \succ \boldsymbol{0}; \tag{61}$$

$$\mathsf{Var}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right] = \boldsymbol{\Sigma}_r + \lambda \cdot (4r - 2)\boldsymbol{V} - h \cdot 2\mathsf{K}\boldsymbol{J} + o(\lambda) + o(h) \succ \boldsymbol{0}, \tag{62}$$

where matrices $\boldsymbol{J} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0) \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \succ \boldsymbol{0}$, $\boldsymbol{V} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \succ \boldsymbol{0}$, $\boldsymbol{V}' := \mathbb{E}_{\boldsymbol{x}}\left[f'_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} \cdot \boldsymbol{x}\boldsymbol{x}^\top\right]$, and $\boldsymbol{\Sigma}_r := r(1-r)\boldsymbol{\Sigma} \succ \boldsymbol{0}$; constant $\mathsf{K} := \int_{-\infty}^{\infty} uK(u) \cdot \int_{-\infty}^{u} K(v)\, dv \cdot du \in (0, \kappa_1)$ is derived from kernel function K and independent of h by definition. Substituting preceding two expressions into (60), we obtain by dropping dominated terms,

$$\begin{aligned}
\mathsf{Var}_{\mathcal{D},\boldsymbol{b}}\left[\sqrt{n}\, \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}\right] &= \boldsymbol{J}^{-1}\boldsymbol{\Sigma}_r\boldsymbol{J}^{-1} - \lambda \cdot \boldsymbol{J}^{-1}\left[(4r-2)\boldsymbol{V} + 4(\boldsymbol{I} - \boldsymbol{V}')\boldsymbol{J}^{-1}\boldsymbol{\Sigma}_r\right]\boldsymbol{J}^{-1} \\
&\quad - h \cdot 2\mathsf{K}\boldsymbol{J}^{-1} + \frac{\sigma^2}{n} \cdot \boldsymbol{J}^{-1}\boldsymbol{J}^{-1} + o(\lambda) + o(h) \\
&\succcurlyeq \boldsymbol{J}^{-1}\boldsymbol{\Sigma}_r\boldsymbol{J}^{-1} - \lambda \cdot c_1\boldsymbol{J}^{-1}\boldsymbol{\Sigma}_r\boldsymbol{J}^{-1} - h \cdot 2\mathsf{K}\boldsymbol{J}^{-1} + \frac{\sigma^2}{n}\boldsymbol{J}^{-1}\boldsymbol{J}^{-1} + o(\lambda) + o(h), \quad \text{as } \lambda, h \to 0,
\end{aligned}$$

for some positive constant c_1. Thus, we get the desired statement.

To complete the proof, we only need to derive (61) and (62). Denote $\boldsymbol{\delta}_h^{\#*} := \boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}^*$.

- We first derive (62). By noticing that

$$\mathsf{Var}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right] = \mathbb{E}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*}) \cdot \nabla \ell_h(\boldsymbol{\theta}_h^{\#*})^\top\right] - \mathbb{E}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right] \cdot \mathbb{E}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right]^\top, \tag{63}$$

we can separately address the two terms. Following (54), we have

$$\begin{aligned}
\mathbb{E}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right] &= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot \int_0^{\boldsymbol{\delta}_h^{\#*\top}\boldsymbol{x} - uh} f_{\epsilon|\boldsymbol{x}}(t)\, dt\, du \cdot \boldsymbol{x}\right] && (\boldsymbol{\delta}_h^{\#*} := \boldsymbol{\theta}_h^{\#*} - \boldsymbol{\theta}^*) \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot \int_0^{-2\lambda(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} - uh + o(\lambda) + o(h^2)} f_{\epsilon|\boldsymbol{x}}(t)\, dt\, du \cdot \boldsymbol{x}\right] && \text{(by Lemma 9)} \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u) \cdot \left[f_{\epsilon|\boldsymbol{x}}(0)(-2\lambda(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x}) + o(\lambda) + o(h^2)\right] du \cdot \boldsymbol{x}\right] && \text{(Taylor expansion)} \qquad (64) \\
&= \mathbb{E}_{\boldsymbol{x}}\left[-2\lambda f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} \cdot \boldsymbol{x}\right] + o(\lambda) + o(h^2) \\
&= -2\lambda \boldsymbol{\theta}^* + o(\lambda) + o(h^2). && (65)
\end{aligned}$$

We then come to address $\mathbb{E}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*}) \cdot \nabla \ell_h(\boldsymbol{\theta}_h^{\#*})^\top\right]$, which is equal to

$$\mathbb{E}\left[(\mathcal{K}_h(\boldsymbol{\theta}_h^{\#*\top}\boldsymbol{x} - y) - r)^2 \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] = \mathbb{E}_{\boldsymbol{x}}\left[\left(\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[\mathcal{K}_h^2(\boldsymbol{\theta}_h^{\#*\top}\boldsymbol{x} - y) - 2r\mathcal{K}_h(\boldsymbol{\theta}_h^{\#*\top}\boldsymbol{x} - y)\right] + r^2\right) \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \tag{66}$$

We focus on the inner expectation term $\mathbb{E}_{\epsilon|\boldsymbol{x}}[\cdot]$ conditional on $\boldsymbol{x}$ on the r.h.s. of (66). Following a similar idea for (64), we can show that,

$$\begin{aligned}
\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[\mathcal{K}_h^2(\boldsymbol{\theta}_h^{\#*\top}\boldsymbol{x} - y)\right] &= \int_{-\infty}^{\infty} F_{\epsilon|\boldsymbol{x}}(t) \cdot 2\mathcal{K}_h(\boldsymbol{\delta}_h^{\#*\top}\boldsymbol{x} - t) K_h(\boldsymbol{\delta}_h^{\#*\top}\boldsymbol{x} - t)\, dt \quad \text{(integration by parts)} \\
&= \int_{-\infty}^{\infty} F_{\epsilon|\boldsymbol{x}}(\boldsymbol{\delta}_h^{\#*\top}\boldsymbol{x} - uh) \cdot 2\mathcal{K}_h(uh) K(u)\, du \\
&= \int_{-\infty}^{\infty} \left[\int_0^{\boldsymbol{\delta}_h^{\#*\top}\boldsymbol{x} - uh} f_{\epsilon|\boldsymbol{x}}(t)\, dt\right] \cdot 2\mathcal{K}_h(uh) K(u)\, du + r \underbrace{\int_{-\infty}^{\infty} 2\mathcal{K}_h(uh) K(u)\, du}_{=1} \\
&= r + \int_{-\infty}^{\infty} \left(-2\lambda f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} - f_{\epsilon|\boldsymbol{x}}(0) uh + o(\lambda) + o(h^2)\right) \cdot 2\mathcal{K}_h(uh) K(u)\, du \\
&= r - 2\lambda f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} - 2h f_{\epsilon|\boldsymbol{x}}(0)\mathsf{K} + o(\lambda) + o(h^2), && (67)
\end{aligned}$$

where $\mathsf{K} = \int_{-\infty}^{\infty} u\mathcal{K}_h(uh) K(u)\, du \in (0, \kappa_1)$ is a real value derived from kernel function K and independent of h by definition. Additionally, following the analysis on $\mathbb{E}\left[\nabla \ell_h(\boldsymbol{\theta}_h^{\#*})\right]$ and (64), we have

$$\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[\mathcal{K}_h(\boldsymbol{\theta}_h^{\#*\top}\boldsymbol{x} - y)\right] = r - 2\lambda f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} + o(\lambda) + o(h^2). \tag{68}$$

Plugging (67) and (68) back into (66) gives

$$(66) = \boldsymbol{\Sigma}_r + 2\lambda(2r-1)\boldsymbol{V} - 2h\mathsf{K}\boldsymbol{J} + o(\lambda) + o(h) \succcurlyeq \boldsymbol{0} \tag{69}$$

where $\boldsymbol{\Sigma}_r := r(1-r)\boldsymbol{\Sigma} \succ \boldsymbol{0}$, and $\boldsymbol{V} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top \boldsymbol{x} \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \succ \boldsymbol{0}$ as $\boldsymbol{\theta}^{*\top}\boldsymbol{x}$ is always positive by assumption. Lastly, inserting (65) and (69) into (63) gives (62).

- We next analyze $\boldsymbol{H}_h^{\#-1}$:

$$\begin{aligned}
\boldsymbol{H}_h^{\#} - 2\lambda \boldsymbol{I} &= \mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[K_h(y - \boldsymbol{\theta}_h^{\#*\top}\boldsymbol{x})\right]\cdot \boldsymbol{x}\boldsymbol{x}^\top\right] \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u)\cdot f_{\epsilon|\boldsymbol{x}}(\boldsymbol{\delta}_h^{\#*\top}\boldsymbol{x} + uh)\,du \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] && \text{(by (55))} \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u)\left[f_{\epsilon|\boldsymbol{x}}(\boldsymbol{\delta}_h^{\#*\top}\boldsymbol{x} + uh) - f_{\epsilon|\boldsymbol{x}}(0)\right]du \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] + \boldsymbol{J} \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\int_{-\infty}^{\infty} K(u)\left[-2\lambda f'_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top\boldsymbol{x} + uh\right]du \cdot \boldsymbol{x}\boldsymbol{x}^\top\right] + \boldsymbol{J} + o(\lambda) + o(h^2) \\
&= -2\lambda \boldsymbol{V}' + \boldsymbol{J} + o(\lambda) + o(h^2),
\end{aligned}$$

where $\boldsymbol{V}' := \mathbb{E}_{\boldsymbol{x}}\left[f'_{\epsilon|\boldsymbol{x}}(0)(\boldsymbol{J}^{-1}\boldsymbol{\theta}^*)^\top\boldsymbol{x}\cdot\boldsymbol{x}\boldsymbol{x}^\top\right]$. Then, applying second-order approximation of an inverse matrix to $\boldsymbol{H}_h^{\#}$ leads to (61).

□

C.4. Proof of Proposition 3

Proof. According to Lemma 11, the conditional variance of non-private and private order quantities are

$$\begin{aligned}
\mathsf{Var}\left[\widehat{q}(\boldsymbol{x})|\boldsymbol{x}\right] &= \frac{\left\|\boldsymbol{\Sigma}_r^{\frac{1}{2}}\boldsymbol{J}^{-1}\boldsymbol{x}\right\|_2^2}{n} && \forall \boldsymbol{x}\in\mathcal{X}; \\
\mathsf{Var}\left[\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x})|\boldsymbol{x}\right] &\geq \frac{\left\|\boldsymbol{\Sigma}_r^{\frac{1}{2}}\boldsymbol{J}^{-1}\boldsymbol{x}\right\|_2^2}{n} - c_1\lambda\frac{\left\|\boldsymbol{\Sigma}_r^{\frac{1}{2}}\boldsymbol{J}^{-1}\boldsymbol{x}\right\|_2^2}{n} - 2h\mathsf{K}\frac{\left\|\boldsymbol{J}^{-\frac{1}{2}}\boldsymbol{x}\right\|_2^2}{n} + \frac{\sigma^2\left\|\boldsymbol{J}^{-1}\boldsymbol{x}\right\|_2^2}{n^2}, && \forall \boldsymbol{x}\in\mathcal{X}
\end{aligned}$$

where $\boldsymbol{J} := \mathbb{E}_{\boldsymbol{x}}\left[f_{\epsilon|\boldsymbol{x}}(0)\cdot\boldsymbol{x}\boldsymbol{x}^T\right] \succ \boldsymbol{0}$, and $\boldsymbol{\Sigma}_r := r(1-r)\boldsymbol{\Sigma} \succ \boldsymbol{0}$, $\mathsf{K} := \int_{-\infty}^{\infty} uK(u)\cdot\int_{-\infty}^{u}K(v)\,dv\cdot du \in (0,\kappa_1)$, $c_1 > 0$ is a constant, and $\sigma \asymp \bar{r}\sqrt{\ln(1/\delta)}/\varepsilon$. Hence, the increase in conditional variance due to DP algorithms $\Delta V(\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x}), \widehat{q}(\boldsymbol{x})) := \mathsf{Var}\left[\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x})|\boldsymbol{x}\right] - \mathsf{Var}\left[\widehat{q}(\boldsymbol{x})|\boldsymbol{x}\right]$ is lower bounded as

$$\begin{aligned}
\Delta V(\widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x}), \widehat{q}(\boldsymbol{x})) &\geq \frac{(-c_1\lambda - c_2 h)\cdot r(1-r)}{n}\left\|\boldsymbol{\Sigma}^{\frac{1}{2}}\boldsymbol{J}^{-1}\boldsymbol{x}\right\|_2^2 + \frac{\sigma^2}{n^2}\left\|\boldsymbol{J}^{-1}\boldsymbol{x}\right\|_2^2, \quad \forall \boldsymbol{x}\in\mathcal{X} \\
&= \Omega\left(r(1-r)\frac{-\lambda - h}{n} + \frac{\sigma^2}{n^2}\right) \\
&= \Omega\left(r(1-r)\frac{-\lambda - h}{n} + \max\{r, 1-r\}\frac{\ln(1/\delta)}{n^2\varepsilon^2}\right).
\end{aligned}$$

The upper bound is a corollary of Proposition 2 as the increased deviation is of order σ/n.

□

C.5. Formal Statement of Proposition 4 and Proof

Proposition 6 (Supplier Gains a Lower Profit Share; formal). *Consider a newsvendor-supplier channel under the wholesale price contract with production cost $v > 0$, wholesale price $w > v$, and retail price $p > w$. Assume demand $y = \langle\boldsymbol{\theta}^*, \boldsymbol{x}\rangle + \mathcal{N}(0,1)$, and Assumption 1 holds. Further assume that the optimal order quantity $q^*(\boldsymbol{x}) := \langle\boldsymbol{\theta}^*, \boldsymbol{x}\rangle$ is always positive for any $\boldsymbol{x}\in\mathcal{X}$, where $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d}\mathcal{L}(\boldsymbol{\theta})$. For any algorithm $\mathcal{A}$ associated with an estimator $\widehat{\boldsymbol{\theta}}^{\mathcal{A}}$, let the order quantity be $\widehat{q}^{\mathcal{A}}(\boldsymbol{x}) := \langle\widehat{\boldsymbol{\theta}}^{\mathcal{A}}, \boldsymbol{x}\rangle, \forall\boldsymbol{x}\in\mathcal{X}$, let the expected profit of the newsvendor under policy $\mathcal{A}$ be $\widehat{\Pi}^N(\boldsymbol{x}) = \mathbb{E}_{y|\boldsymbol{x}}\left[p\min\{y, \widehat{q}^{\mathcal{A}}(\boldsymbol{x})\} - w\widehat{q}^{\mathcal{A}}(\boldsymbol{x})\right], \forall\boldsymbol{x}\in\mathcal{X}$, and let the profit of the supplier be $\widehat{\Pi}^S(\boldsymbol{x}) = (w-v)\widehat{q}^{\mathcal{A}}(\boldsymbol{x})$. Then, with high probability, adopting the private algorithm* OP *would raise up newsvendor's profit share (correspondingly, pull down supplier's share):*

$$\frac{\widehat{\Pi}^N(\boldsymbol{x})}{\widehat{\Pi}^N(\boldsymbol{x}) + \widehat{\Pi}^S(\boldsymbol{x})} = \frac{\mathsf{Rev}^N(\boldsymbol{x}) - C - U}{\mathsf{Rev}(\boldsymbol{x}) - C - U - \widehat{P}(\boldsymbol{x}, \mathcal{D})}, \quad \forall\boldsymbol{x}\in\mathcal{X}, \tag{70}$$

by setting a positive value of $\widehat{P}(\boldsymbol{x},\mathcal{D}) := (w-v)(q^*(\boldsymbol{x}) - \widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x}))$ *in* (70)*. The value* $\widehat{P}(\boldsymbol{x},\mathcal{D})$ *is the supplier's revenue loss due to newsvendor's private learning process. Other values on the right hand side of* (70) *are constants:* $\mathsf{Rev}^N(\boldsymbol{x}) := (p-w)q^*(\boldsymbol{x})$ *is newsvendor's revenue at optimal order quantity;* $\mathsf{Rev}(\boldsymbol{x}) := (p-v)q^*(\boldsymbol{x})$ *is the whole chain's optimal revenue;* $C := \min_{\boldsymbol{\theta}} \mathbb{E}_{y,\boldsymbol{x}}\left[(p-w)(y-\boldsymbol{\theta}^\top\boldsymbol{x})^+ + w(\boldsymbol{\theta}^\top\boldsymbol{x}-y)^+\right]$ *is newsvendor's minimal cost;* $U := (p-w)\Phi^{-1}(\frac{p-w}{p})$ *is the revenue loss due to demand uncertainty. Infinitesimals of* $\widehat{P}(\boldsymbol{x},\mathcal{D})$ *appearing both in the numerator and denominator of the right-hand side term of* (70) *are omitted.*

Proof. Let $C(\boldsymbol{\theta}) := \mathbb{E}_{y,\boldsymbol{x}}\left[(p-w)(y-\boldsymbol{\theta}^\top\boldsymbol{x})^+ + w(\boldsymbol{\theta}^\top\boldsymbol{x}-y)^+\right]$ be the newsvendor's cost at $\boldsymbol{\theta}$. By Lemma 4, we know that the minimizer $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}} C(\boldsymbol{\theta})$ takes the form $\boldsymbol{\theta}^* := \boldsymbol{\theta}^\star + [\Phi^{-1}(\frac{p-w}{p}), 0, \ldots, 0]$, and the data generating process can be reformulated as $y = \boldsymbol{\theta}^{*\top}\boldsymbol{x} + \mathcal{N}(-\Phi^{-1}(\frac{p-w}{p}), 1)$.

Given a dataset $\mathcal{D}$, let $\widehat{\boldsymbol{\theta}}^{\mathcal{A}}$ be an estimator to $\boldsymbol{\theta}^*$. Let $q^*(\boldsymbol{x}) := \boldsymbol{\theta}^{*\top}\boldsymbol{x}$ and $\widehat{q}^{\mathcal{A}}(\boldsymbol{x}) := \widehat{\boldsymbol{\theta}}^{\mathcal{A}\top}\boldsymbol{x}$ be the order quantities under context $\boldsymbol{x}$. Recall that, under context $\boldsymbol{x}$, the newsvendor's expected profit with an estimator $\widehat{\boldsymbol{\theta}}^{\mathcal{A}}$ is

$$\begin{aligned}
\widehat{\Pi}^N(\boldsymbol{x}) &= \mathbb{E}_{y|\boldsymbol{x}}\left[p\min\{y, \widehat{\boldsymbol{\theta}}^{\mathcal{A}\top}\boldsymbol{x}\} - w\widehat{\boldsymbol{\theta}}^{\mathcal{A}\top}\boldsymbol{x}\right] \\
&= (p-w)q^*(\boldsymbol{x}) - (p-w)\Phi^{-1}\left(\frac{p-w}{p}\right) - C(\widehat{\boldsymbol{\theta}}) \\
&= (p-w)q^*(\boldsymbol{x}) - (p-w)\Phi^{-1}\left(\frac{p-w}{p}\right) - C(\boldsymbol{\theta}^*) - \mathcal{O}(\|\boldsymbol{\delta}\|_2^2),
\end{aligned}$$

where $\boldsymbol{\delta} := \widehat{\boldsymbol{\theta}}^{\mathcal{A}} - \boldsymbol{\theta}^*$, and the last equality follows from analysis by mean-value theorem:

$$\begin{aligned}
C(\widehat{\boldsymbol{\theta}}) - C(\boldsymbol{\theta}^*) &= \boldsymbol{\delta}^\top \cdot \int_0^1 \nabla C(\boldsymbol{\theta}^* + t\boldsymbol{\delta})\, dt \\
&= \boldsymbol{\delta}^\top \cdot \int_0^1 \mathbb{E}_{\boldsymbol{x}}\left[\left(\Phi\left(t\boldsymbol{\delta}^\top\boldsymbol{x} + \Phi^{-1}(\frac{p-w}{p})\right) - \frac{p-w}{p}\right)\cdot\boldsymbol{x}\right] dt \\
&= \mathcal{O}(\|\boldsymbol{\delta}\|_2^2) > 0.
\end{aligned}$$

Similarly, under context $\boldsymbol{x}$, the supplier's profit is

$$\widehat{\Pi}^S(\boldsymbol{x}) = (w-v)\widehat{q}^{\mathcal{A}}(\boldsymbol{x}).$$

Therefore, when the newsvendor adopts Algorithm OP, the profit share of the newsvendor is:

$$\begin{aligned}
\frac{\widehat{\Pi}^N(\boldsymbol{x})}{\widehat{\Pi}^N(\boldsymbol{x}) + \widehat{\Pi}^S(\boldsymbol{x})} &= \frac{(p-w)q^*(\boldsymbol{x}) - (p-w)\Phi^{-1}\left(\frac{p-w}{p}\right) - C(\boldsymbol{\theta}^*) - \mathcal{O}(\|\boldsymbol{\delta}\|_2^2)}{(p-w)q^*(\boldsymbol{x}) - (p-w)\Phi^{-1}\left(\frac{p-w}{p}\right) - C(\boldsymbol{\theta}^*) - \mathcal{O}(\|\boldsymbol{\delta}\|_2^2) + (w-v)\widehat{q}(\boldsymbol{x})} \\
&= \frac{\mathsf{Rev}^N(\boldsymbol{x}) - U - C - \mathcal{O}(\|\boldsymbol{\delta}\|_2^2)}{\mathsf{Rev}(\boldsymbol{x}) - U - C - \widehat{P}(\boldsymbol{x},\mathcal{D}) + \mathcal{O}(\|\boldsymbol{\delta}\|_2^2)} \\
&= \frac{\mathsf{Rev}^N(\boldsymbol{x}) - U - C - o(\widehat{P}(\boldsymbol{x},\mathcal{D}))}{\mathsf{Rev}(\boldsymbol{x}) - U - C - \widehat{P}(\boldsymbol{x},\mathcal{D}) - o(\widehat{P}(\boldsymbol{x},\mathcal{D}))}.
\end{aligned}$$

The second line basically only involves notation changes: $\mathsf{Rev}^N(\boldsymbol{x}) := (p-w)q^*(\boldsymbol{x})$, $\widehat{P}(\boldsymbol{x},\mathcal{D}) := (w-v)(q^*(\boldsymbol{x}) - \widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x}))$; $U := (p-w)\Phi^{-1}(\frac{p-w}{p})$; $C := C(\boldsymbol{\theta}^*) = \min_{\boldsymbol{\theta}} \mathbb{E}_{y,\boldsymbol{x}}\left[(p-w)(y-\boldsymbol{\theta}^\top\boldsymbol{x})^+ + w(\boldsymbol{\theta}^\top\boldsymbol{x}-y)^+\right]$; and $\mathsf{Rev}(\boldsymbol{x}) := (p-v)q^*(\boldsymbol{x})$. In the last line, we replace $\mathcal{O}(\|\boldsymbol{\delta}\|_2^2)$ with $o(\widehat{P}(\boldsymbol{x},\mathcal{D}))$ since $\widehat{P}(\boldsymbol{x},\mathcal{D}) = \mathcal{O}(\|\boldsymbol{\delta}\|_2)$.

To show the profit share of the supplier is decreased, it suffices to show $\widehat{P}(\boldsymbol{x},\mathcal{D}) := (w-v)(q^*(\boldsymbol{x}) - \widehat{q}_h^{\mathsf{OP}}(\boldsymbol{x}))$ is positive with high probability. Recall Proposition 2, we know that

$$\widehat{P}(\boldsymbol{x},\mathcal{D}) = (w-v)\left[2\lambda\boldsymbol{u}^\top\boldsymbol{x} + \frac{1}{2}h^2\kappa_2\boldsymbol{v}^\top\boldsymbol{x}\right] + \mathcal{O}_p\left(\frac{1}{\sqrt{n}} + \frac{\sigma}{n}\right),$$

with $\boldsymbol{u} = \boldsymbol{J}^{-1}\boldsymbol{\theta}^*$, $\boldsymbol{v} = \boldsymbol{J}^{-1}\mathbb{E}_{\boldsymbol{x}}\left[f'_\epsilon(0)\boldsymbol{x}\right]$, and $\boldsymbol{J} := \mathbb{E}_{\boldsymbol{x}}\left[f_\epsilon(0)\cdot\boldsymbol{x}\boldsymbol{x}^T\right] \succ \boldsymbol{0}$. As we assume $q^*(\boldsymbol{x}) := \boldsymbol{\theta}^{*\top}\boldsymbol{x}$ is always positive, the value $\boldsymbol{u}^\top\boldsymbol{x}$ is therefore always positive. Additionally, under our assumptions, we have $\boldsymbol{v}^\top\boldsymbol{x} = f'_\epsilon(0)/f_\epsilon(0)$. Since $\epsilon \sim \mathcal{N}(-\Phi^{-1}(\frac{p-w}{p}), 1)$:

- when $(p-w)/p \le 1/2$ (i.e., $w \ge p/2$), we have $f'_\epsilon(0) \ge 0$, which ensures $\boldsymbol{v}^\top\boldsymbol{x}$ is always non-negative, and further ensures $\widehat{P}(\boldsymbol{x}, \mathcal{D})$ is positive w.h.p;
- when $(p-w)/p > 1/2$ (i.e., $w < p/2$), $\boldsymbol{v}^\top\boldsymbol{x}$ is negative. Nevertheless, recall that when running OP, we require $\lambda \gtrsim h^2$ (this can be verified by the values of λ, h, ε, and analysis around Eq. (24)). Hence, the positive value of $\boldsymbol{u}^\top\boldsymbol{x}$ will finally dominates the negative value of $\boldsymbol{v}^\top\boldsymbol{x}$, resulting in a positive $\widehat{P}(\boldsymbol{x}, \mathcal{D})$ w.h.p.

Combining both cases completes the proof.

□

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